Statistical Inference for Linear Stochastic Approximation with Richardson-Romberg Extrapolation

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Abstract

This thesis studies statistical inference for constant-stepsize linear stochastic approximation with Markovian noise. The main object is the Polyak–Ruppert averaged Richardson–Romberg estimator formed from two coupled step sizes, α and 2α. We prove a stationary augmented-chain Berry–Esseen assembly for scalar RR statistics, identify the asymptotic covariance target Σ_∞, and then transfer the result to deterministic starts under mixing-scale burn-in conditions with logarithmic factors. At the balanced scale $\alpha = cn^{-1/2}$, the final burned-in statistic has a non-asymptotic normal approximation with $n^{-1/4}$ polynomial rate up to logarithmic factors.

1. Introduction

Stochastic approximation (SA) algorithms are a cornerstone of modern computational statistics, optimization, and reinforcement learning. Introduced by Robbins and Monro (1951), these iterative procedures provide a principled way to find roots of equations or optimize objectives when only noisy observations are available. A particularly important subclass is the *linear stochastic approximation* (LSA) algorithm, which arises naturally in temporal-difference (TD) learning, policy evaluation, and stochastic gradient descent for linear models.

In this work, we study the LSA recursion with a *constant step size* $\alpha > 0$:

$$\theta_k^{(\alpha)} = \theta_{k-1}^{(\alpha)} - \alpha\Big\{A(Z_k)\theta_{k-1}^{(\alpha)} - b(Z_k)\Big\}, \quad k \geq 1, \tag{1}$$

where $\{Z_k\}_{k\in\mathbb{N}}$ is a time-homogeneous Markov chain on a measurable space $(Z, \mathcal{Z})$ with transition kernel Q and unique invariant distribution π. The mappings $A : Z \to \mathbb{R}^{d\times d}$ and $b : Z \to \mathbb{R}^d$ are measurable functions satisfying $\overline{A} := \int_Z A(z)d\pi(z)$ and $\overline{b} := \int_Z b(z)d\pi(z)$. We use the stability convention that $-\overline{A}$ is Hurwitz, equivalently all eigenvalues of $\overline{A}$ have strictly positive real parts. Then the target parameter $\theta^* = \overline{A}^{-1}\overline{b}$ is uniquely defined.

The *Polyak–Ruppert averaging* procedure (Polyak, 1990; Ruppert, 1988) provides an effective variance reduction technique. Given a burn-in period $n_0 \geq 0$, the averaged iterate is defined as

$$\overline{\theta}_n^{(\alpha)} = \frac{1}{n - n_0}\sum_{k=n_0}^{n-1}\theta_k^{(\alpha)}. \tag{2}$$

1.1. Bias of constant step-size iterates

The use of a constant step size $\alpha > 0$ offers several practical advantages: it enables geometrically fast forgetting of the initial condition (Dieuleveut, Durmus, and Bach, 2020) and simplifies hyperparameter tuning compared to diminishing step-size schedules. However, unlike the classical regime $\alpha_k \to 0$ with $\sum \alpha_k = \infty$ and $\sum \alpha_k^2 < \infty$, a constant step size produces iterates that converge only *in distribution* to a stationary measure Π_α, rather than almost surely to θ^*. The stationary expectation $\mathbb{E}\left[\theta_\infty^{(\alpha)}\right]$ is generally *biased* with respect to θ^*, and this bias cannot be eliminated by Polyak–Ruppert averaging alone.

As shown in Levin, Naumov, and Samsonov (2025), the stationary bias has a leading linear term in α:

$$\lim_{n\to\infty} \mathbb{E}\left[\theta_n^{(\alpha)}\right] = \theta^* + \alpha\Delta + O\left(\alpha^{3/2}\right), \tag{3}$$

where $\Delta = \overline{A}^{-1} \sum_{k=1}^{\infty} \mathbb{E}\left[\left\{Q^k \tilde{A}(Z_\infty)\right\}\epsilon(Z_\infty)\right]$ depends on the correlation structure of the Markov chain, and $\tilde{A}(z) = A(z) - \overline{A}$ is the centered matrix-valued function. Under stronger expansion assumptions, the power-series approach of Huo, Chen, and Xie (2023) gives higher-order bias expansions in integer powers of α; in the Levin decomposition, the first misadjustment bias component itself has an $O(\alpha^2)$ remainder after the leading $\alpha\Delta$ term.

1.2. Richardson–Romberg extrapolation

To eliminate the leading $O(\alpha)$ bias term, we employ the *Richardson–Romberg* (RR) *extrapolation* procedure. Two LSA sequences are run *on the same Markov chain trajectory* $\{Z_k\}$ with step sizes α and 2α, and the RR iterate is formed as

$$\overline{\theta}_n^{(\alpha,\ \mathrm{RR})} = 2\overline{\theta}_n^{(\alpha)} - \overline{\theta}_n^{(2\alpha)}. \tag{4}$$

Since both sequences share the same noise realization, the leading bias term $\alpha\Delta$ cancels, leaving a residual bias of order $O\left(\alpha^{3/2}\right)$ or higher (Levin et al., 2025).

More generally, one can consider the multi-level extrapolation with M step sizes $\mathcal{A} = \{\alpha_1, ..., \alpha_M\}$ and coefficients $\{h_m\}$ determined by the Vandermonde system (Huo et al., 2023):

$$\sum_{m=1}^{M} h_m = 1, \quad \sum_{m=1}^{M} h_m \alpha_m^l = 0, \quad l = 1, ..., M-1, \tag{5}$$

which cancels successive powers in settings where such a power-series expansion is available.

1.3. Problem statement and goals

The high-order moment bounds for the PR-averaged RR iterate $\overline{\theta}_n^{(\alpha,\ \mathrm{RR})}$ established in Levin et al. (2025) show that the leading error term scales as $\sqrt{\mathrm{Tr}\ \Sigma_\epsilon^{(\mathrm{M})}} \cdot n^{-1/2}$, where $\Sigma_\epsilon^{(\mathrm{M})}$ is the Markovian noise covariance defined below in the key quantities subsection, matching the minimax optimal rate. Berry–Esseen type bounds and bootstrap inference procedures for the *standard* Polyak–Ruppert average $\overline{\theta}_n$ (without extrapolation) under Markovian noise have been obtained in Samsonov, Sheshukova, Moulines, and Naumov (2025).

These results do not directly give the distributional approximation needed for the PR-averaged Richardson–Romberg statistic under Markovian noise. Two distinctions are essential here. First, the stationary theorem is proved for an augmented-chain scalar statistic, where the martingale noise, Poisson boundary term, and Richardson–Romberg misadjustment are controlled separately. Second, the deterministic-start estimator requires an additional burn-in transfer, because the deterministic transient and startup remainders are absent from the stationary augmented-chain theorem.

The main goal of this work is therefore to build a non-asymptotic distributional approximation for scalar projections of the PR-averaged Richardson–Romberg statistic. The stationary result should be read as an $n_0 = 0$ augmented-chain theorem, not as a fixed-α central limit theorem centered exactly at θ^*. Its final deterministic-start consequence is obtained at the balanced triangular-array scale $\alpha_n = cn^{-1/2}$, where the residual RR terms are absorbed into explicit remainders and the covariance target is Σ_∞.

Concretely, we establish:

1. A stationary full-window augmented-chain Berry–Esseen assembly for scalar RR statistics, including the martingale approximation, predictable-variance comparison, Poisson remainder, and stationary RR misadjustment.
2. A balanced triangular-array specialization, in particular for $\alpha_n = cn^{-1/2}$, which identifies Σ_∞ as the covariance target and gives the resulting stationary $n_0 = 0$ CLT interpretation.
3. A deterministic-start transfer theorem under mixing-scale burn-in conditions with logarithmic factors, yielding the corresponding balanced-scale bound for the main burned-in statistic.

1.4. Contribution map and notation guide

The proof combines existing non-asymptotic tools with RR-specific algebra and the deterministic-start transfer developed here. The main proof blocks are:

Block	**Role in the proof**	**Status**
RR weight algebra	Controls the deterministic kernels $\mathcal{Q}_l^{\mathrm{RR}}$ and $Q_{l;n_0,n}^{\mathrm{RR}}$ and their variation.	Derived in Sections Section 2 and Section 4.
Poisson and martingale reduction	Turns the Markovian depth-zero weighted sum into a martingale plus a bounded Poisson remainder.	Adapted from the Samsonov et al. framework and reproved for RR weights.
Stationary misadjustment	Controls the non-martingale RR remainder $J^{(1)} + J^{(2)} + H^{(2)}$ under the stationary augmented chain.	Uses Levin et al. inputs; the stationary RR assembly is carried out here.
Burn-in transfer	Transfers the stationary theorem to deterministic starts by controlling the deterministic transient, random initial product, and startup discrepancy.	Developed in Section Section 5.

The following notation is used throughout the later chapters:

Notation	**Meaning**
$\alpha, 2\alpha$	The two constant step sizes used by the two-level RR estimator.
n_0 and $m = n - n_0$	Burn-in length and effective averaging window.
$B_w = I - w\bar{A}$	Deterministic linearized contraction at step size w.
$\mathcal{Q}_l^{\mathrm{RR}}$	Full-window RR PR weight in the stationary $n_0 = 0$ chapter.
$Q_{l;n_0,n}^{\mathrm{RR}}$ or Q_l^{bRR}	Burned-in RR PR weight for the window $k = n_0, ..., n-1$.
$S_{n,\,\mathrm{stat}}^{\mathrm{RR}}(u)$	Stationary augmented-chain scalar statistic.
$T_{n,n_0}^{\mathrm{RR}}(u)$	Finite-start burned-in scalar statistic with $\sqrt{m}$ normalization.
$\Xi_{n,n_0}^{\mathrm{bRR}}(u)$	Finite-window normalized burned-in statistic.
$\sigma(u)^2 = u^\top \Sigma_\infty u$	Asymptotic scalar variance target.

1.5. Setting and assumptions

We now formalize the setting and state the assumptions that will be used throughout this work.

Let $\{Z_k\}_{k\in\mathbb{N}}$ be a Markov chain on a complete separable metric space $(Z, \mathcal{Z})$ with transition kernel Q.

Assumption 1 (Uniform geometric ergodicity). The kernel Q admits a unique invariant distribution π and is *uniformly geometrically ergodic*: there exists $t_{\mathrm{mix}} \in \mathbb{N}^*$ such that for all $k \in \mathbb{N}^*$,

$$\Delta\big(Q^k\big) \coloneqq \sup_{z,z'\in Z} \frac{1}{2}\, \|Q^k(z,\cdot) - Q^k(z',\cdot)\|_{\mathrm{TV}} \le (1/4)^{\lfloor k/t_{\mathrm{mix}} \rfloor}. \tag{6}$$

Equivalently, there exist constants $\zeta > 0$ and $\rho \in (0,1)$ such that $\sup_z \|Q^k(z,\cdot) - \pi\|_{\mathrm{TV}} \le \zeta\rho^k$ for all $k \ge 1$.

Assumption 2 (Hurwitz condition and boundedness). The matrix $-\overline{A}$ is Hurwitz; equivalently, all eigenvalues of $\overline{A}$ have strictly positive real parts. Moreover,

$$C_A := \max\left(\sup_{z\in Z}\|A(z)\|, \sup_{z\in Z}\|\tilde{A}(z)\|\right) < \infty, \tag{7}$$

where $\tilde{A}(z) := A(z) - \overline{A}$.

Assumption 3 (Noise regularity). The noise function $\epsilon(z) = \tilde{A}(z)\theta^* - \tilde{b}(z)$, where $\tilde{b}(z) = b(z) - \overline{b}$, satisfies

$$\|\epsilon\|_\infty := \sup_{z\in Z}\|\epsilon(z)\| < +\infty. \tag{8}$$

By construction, $\pi\left(\tilde{A}\right) = 0$, $\pi\left(\tilde{b}\right) = 0$, and hence $\pi(\epsilon) = 0$.

Under Assumptions 1–3, the error $\theta_k^{(\alpha)} - \theta^*$ satisfies the recursion

$$\theta_k^{(\alpha)} - \theta^* = (I - \alpha A(Z_k))\left(\theta_{k-1}^{(\alpha)} - \theta^*\right) - \alpha\epsilon(Z_k). \tag{9}$$

1.6. Key quantities

The *Markovian noise covariance matrix* captures both the marginal variance and the temporal correlations of the noise:

$$\Sigma_\epsilon^{(\mathrm{M})} = \mathbb{E}_\pi\left[\epsilon(Z_0)\epsilon(Z_0)^\top\right] + \sum_{\ell=1}^{\infty}\left(\mathbb{E}_\pi\left[\epsilon(Z_0)\epsilon(Z_\ell)^\top\right] + \mathbb{E}_\pi\left[\epsilon(Z_\ell)\epsilon(Z_0)^\top\right]\right). \tag{10}$$

This matrix is the limiting covariance in the Markov chain CLT for the partial sums $n^{-1/2}\sum_{t=0}^{n-1}\epsilon(Z_t)$ (cf. Douc et al., 2018, Theorem 21.2.10).

The *asymptotically optimal covariance matrix* is given by

$$\Sigma_\infty = \overline{A}^{-1}\Sigma_\epsilon^{(\mathrm{M})}\left(\overline{A}^{-1}\right)^\top. \tag{11}$$

This is the covariance target attained by the averaged linearized recursion. We call it optimal in the usual averaged-SA sense; a full Hájek–Le Cam optimality statement would require an additional local-asymptotic experiment argument, which is not part of this thesis.

The *Lyapunov equation* plays a central role in the contraction analysis. For any $P = P^\top \succ 0$, there exists a unique $Q = Q^\top \succ 0$ satisfying $\overline{A}^\top Q + Q\overline{A} = P$. Defining $a = \lambda_{\min}(P)/(2\,\|Q\|)$ and $\kappa_Q = \lambda_{\max}(Q)/\lambda_{\min}(Q)$, the key contraction property holds: for all $\alpha \in [0, \alpha_\infty]$,

$$\|I - \alpha\overline{A}\|_Q^2 \le 1 - \alpha a. \tag{12}$$

Since the iterates $\theta_k^{(\alpha)}$ alone are generally not Markovian (due to the Markovian noise), we consider the *joint process* $\left(\theta_k^{(\alpha)}, Z_{k+1}\right)$ with kernel

$$\overline{P}_\alpha f(\theta, z) = \int_Z Q(z, dz')f(F_{z'}(\theta), z'), \tag{13}$$

where $F_z(\theta) = (I - \alpha A(z))\theta + \alpha b(z)$. Under Assumptions 1–3, this joint chain admits a unique invariant distribution Π_α for sufficiently small $\alpha > 0$ (Levin et al., 2025).

2. Zeroth-Order Richardson–Romberg Difference

2.1. LSA Error Decomposition

We consider the recursion

$$\theta_k = \theta_{k-1} - \alpha_k(A(Z_k)\theta_{k-1} - b(Z_k)), \quad \alpha_k = \alpha = \text{const.} \tag{14}$$

Define the transition products

$$\Gamma_{m:k} = \prod_{l=m}^{k}(I - \alpha A(Z_l)). \tag{15}$$

Write $B_\alpha := I - \alpha\overline{A}$ and introduce the deterministic-product linearized term

$$J_k^{(0,\alpha)} = -\alpha \sum_{l=1}^{k} B_\alpha^{k-l}\,\epsilon(Z_l), \quad J_0^{(0,\alpha)} = 0. \tag{16}$$

The difference between the exact random-product expansion and this deterministic-product term is denoted by $R_k^{(\alpha)}$:

$$\theta_k^{(\alpha)} - \theta^* = J_k^{(0,\alpha)} + B_\alpha^k(\theta_0 - \theta^*) + R_k^{(\alpha)}. \tag{17}$$

This convention matches the weight decomposition in Section 4.1.

A standard PR-averaged decomposition (cf. Chapter 4 for the full derivation) yields

$$\sqrt{n}\left(\overline{\theta}_n^{(\alpha)} - \theta^*\right) = W + D_1, \tag{18}$$

where the leading martingale-like term is

$$W = -\frac{1}{\sqrt{n}}\sum_{l=1}^{n-1} Q_l\,\epsilon(Z_l), \quad Q_l = \alpha\sum_{k=l}^{n-1} B_\alpha^{k-l}, \tag{19}$$

and the residual term is

$$D_1 = \frac{1}{\sqrt{n}}\sum_{k=0}^{n-1} B_\alpha^k(\theta_0 - \theta^*) + \frac{1}{\sqrt{n}}\sum_{k=1}^{n-1} R_k^{(\alpha)}. \tag{20}$$

2.2. Last-Iterate RR Combination and the $\tilde{J}_{n,\,\text{last}}^{(0,\alpha)}$ Term

Applying the deterministic-product decomposition of the previous subsection separately to step sizes α and 2α, and writing $B_{2\alpha} := I - 2\alpha\overline{A}$, define the chapter-local last-iterate Richardson–Romberg object

$$\theta_{n,\,\text{last}}^{(\text{RR},\alpha)} := 2\theta_n^{(\alpha)} - \theta_n^{(2\alpha)}. \tag{21}$$

It is not the PR-averaged RR estimator $\overline{\theta}_n^{(\text{RR},\alpha)}$ studied in Chapters 4–5. Its deterministic-product decomposition has the form

$$\begin{aligned}\theta_{n,\,\text{last}}^{(\text{RR},\alpha)} - \theta^* &= [2B_\alpha^n - B_{2\alpha}^n](\theta_0 - \theta^*)\\ &\quad + \left[2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)}\right] + \left[2R_n^{(\alpha)} - R_n^{(2\alpha)}\right].\end{aligned} \tag{22}$$

The first bracket is the deterministic transient in this convention. The second bracket is the linearized stochastic RR difference, and the last one is the higher-order random-product remainder. In this subsection we focus on the zeroth-order RR difference, which we denote

$$\tilde{J}^{(0,\alpha)}_{n,\,\text{last}} = 2J^{(0,\alpha)}_n - J^{(0,2\alpha)}_n, \tag{23}$$

where, by definition,

$$J^{(0,\alpha)}_n = -\alpha \sum_{j=1}^{n} \left(I - \alpha\overline{A}\right)^{n-j} \epsilon\left(Z_j\right). \tag{24}$$

Substituting and using the elementary identity $X^m - Y^m = (X - Y)\sum_{i=1}^{m} X^{i-1}Y^{m-i}$ with $X = I - \alpha\overline{A}$, $Y = I - 2\alpha\overline{A}$, $X - Y = \alpha\overline{A}$, we get

$$\begin{aligned}
\tilde{J}^{(0,\alpha)}_{n,\,\text{last}} &= -2\alpha \sum_{j=1}^{n} \left[\left(I - \alpha\overline{A}\right)^{n-j} - \left(I - 2\alpha\overline{A}\right)^{n-j}\right]\epsilon\left(Z_j\right) \\
&= -2\alpha^2\overline{A} \sum_{j=1}^{n} \underbrace{\sum_{i=1}^{n-j} \left(I - \alpha\overline{A}\right)^{i-1} \left(I - 2\alpha\overline{A}\right)^{n-j-i}}_{=:H^{(n)}_j} \epsilon\left(Z_j\right) \\
&= -2\alpha^2\overline{A} \sum_{j=1}^{n-1} H^{(n)}_j \, \epsilon\left(Z_j\right),
\end{aligned} \tag{25}$$

where the last equality uses $H^{(n)}_n = 0$ because the inner sum is empty. The left-factored matrix-power identity is legitimate here because $I - \alpha\overline{A}$, $I - 2\alpha\overline{A}$, and $\overline{A}$ are polynomials in the same matrix $\overline{A}$ and therefore commute. The extra factor $\alpha\overline{A}$ pulled out front is the source of the additional α-decay of the RR difference compared to a single LSA trajectory.

2.3. Norm Estimate for $H^{(n)}_j$

To bound $\|H^{(n)}_j\|$ we use the following standard result.

Lemma 1. Let $-\overline{A}$ be a Hurwitz matrix. Then for any $P = P^\top \succ 0$ there exists a unique $Q = Q^\top \succ 0$ satisfying

$$\overline{A}^\top Q + Q\overline{A} = P. \tag{26}$$

Moreover, letting

$$a := \frac{\lambda_{\min}(P)}{2\,\|Q\|}, \quad \alpha_\infty := \min\left(\frac{\lambda_{\min}(P)}{2\kappa_Q\,\|\overline{A}\|_Q^2}, \frac{\|Q\|}{\lambda_{\min}(P)}\right), \tag{27}$$

with $\kappa_Q := \lambda_{\max}(Q)/\lambda_{\min}(Q)$, one has for all $\alpha \in [0, \alpha_\infty]$:

$$\alpha a \le 1/2, \quad \|I - \alpha\overline{A}\|_Q^2 \le 1 - \alpha a. \tag{28}$$

For $1 \le j \le n-1$, we estimate $\|H^{(n)}_j\|$ by combining the triangle inequality, submultiplicativity, the equivalence $\|\ X\ \| \le \kappa_Q^{1/2}\ \|\ X\ \|_Q$ (applied to **each** operator-norm factor, hence two factors of $\kappa_Q^{1/2}$ multiplying to κ_Q), and the Lyapunov contraction at step sizes α and 2α:

$$
\begin{aligned}
\|H_j^{(n)}\| &\le \sum_{i=1}^{n-j} \|I-\alpha\overline{A}\|^{i-1}\,\|I-2\alpha\overline{A}\|^{n-j-i} && \text{(triangle + submult.)}\\
&\le \kappa_Q \sum_{i=1}^{n-j} (1-\alpha a)^{(i-1)/2}(1-2\alpha a)^{(n-j-i)/2} && \text{(equiv. + Lyapunov)}\\
&= \kappa_Q (1-\alpha a)^{(n-j-1)/2} \sum_{k=0}^{n-j-1} \left(\frac{1-2\alpha a}{1-\alpha a}\right)^{k/2} && \text{(reindex } k=n-j-i\text{)}\\
&\le \kappa_Q (1-\alpha a)^{(n-j-1)/2} \frac{1}{1-\sqrt{\frac{1-2\alpha a}{1-\alpha a}}} && \text{(geometric series).}
\end{aligned}
\tag{29}
$$

For $\alpha a \le 1/2$ the geometric rate is bounded below by an **elementary** inequality: combining $\sqrt{(1-2\alpha a)/(1-\alpha a)} \le \sqrt{1-\alpha a}$ (since $(1-2\alpha a) \le (1-\alpha a)^2$) with $1-\sqrt{1-x} \ge x/2$ on $[0,1]$ gives

$$
1-\sqrt{\frac{1-2\alpha a}{1-\alpha a}} \ge 1-\sqrt{1-\alpha a} \ge \frac{\alpha a}{2}, \quad \text{i.e.} \quad \frac{1}{1-\sqrt{\frac{1-2\alpha a}{1-\alpha a}}} \le \frac{2}{\alpha a}. \tag{30}
$$

Defining the chapter-local norm-equivalence constant

$$
K_Q := \kappa_Q \tag{31}
$$

(distinct from the assumption-2 sup-norm constant C_A, which enters separately via $\| \overline{A} \|$ in the next subsection), we arrive at the final estimate

$$
\|H_j^{(n)}\| \le K_Q\,(1-\alpha a)^{(n-j-1)/2}\,\frac{2}{\alpha a}. \tag{32}
$$

The kernel decays geometrically in $n-j$ at rate $\sqrt{1-\alpha a}$ but its summed weight is of order $1/(\alpha a)$. The next subsection keeps the remaining powers of a explicit when this kernel is multiplied by the prefactor α^2 in $\tilde{J}_{n,\,\text{last}}^{(0,\alpha)}$.

2.4. Scalar L^p Bound for the Zeroth-Order Term

The expression

$$
\tilde{J}_{n,\,\text{last}}^{(0,\alpha)} = -\sum_{j=1}^{n-1} 2\alpha^2 \overline{A} H_j^{(n)}\,\epsilon\big(Z_j\big) \tag{33}
$$

is a weighted sum of values of the centered noise function ϵ along the Markov chain. In the Berry–Esseen argument below only fixed scalar projections are used, so we state the concentration input in scalar form.

Lemma 2. Assume **UGE 1**. Let $\{g_i\}_{i=1}^n$ be a family of measurable functions $g_i : Z \to \mathbb{R}$ such that

$$c_i = \|g_i\|_\infty < \infty \quad \text{for all } i \geq 1, \quad \pi(g_i) = 0 \quad \text{for all } i \in \{1, ..., n\}. \tag{34}$$

Then, for any initial distribution ξ on $(Z, \mathcal{Z})$, any $n \in \mathbb{N}$ and any $t \geq 0$,

$$\mathbb{P}_\xi \left(\left| \sum_{i=1}^n g_i(Z_i) \right| \geq t \right) \leq 2 \exp\left(-\frac{t^2}{2v_n^2} \right), \tag{35}$$

where

$$v_n = 8 \left(\sum_{i=1}^n c_i^2 \right)^{1/2} \sqrt{t_{\text{mix}}}. \tag{36}$$

Proof. This is the scalar specialization of Levin et al. (2025, Lemma 11). Their lemma gives the same sub-Gaussian tail bound for vector-valued time-inhomogeneous functions under **UGE 1**, assumes the centering condition $\pi(g_i) = 0$, and is stated for an arbitrary initial law ξ. Thus the imported estimate is a tail bound around zero, not merely around $\mathbb{E}_\xi \sum_i g_{i(Z_i)}$, and no additional initial-bias term is introduced here.

Fix a deterministic direction $u \in \mathbb{R}^d$. We apply the lemma with $g_j^{u(z)} = -2\alpha^2 u^\top \overline{A} H_j^{(n)} \epsilon(z)$ for $1 \leq j \leq n-1$ and $g_n^u = 0$. Each g_j^u is centered under π since $\pi(\epsilon) = 0$, so the centering hypothesis holds. Combining the bound $\| \overline{A} \| \leq C_A$ (Assumption 2) with the previous subsection, define

$$K_{A,Q} := C_A \, K_Q = C_A \, \kappa_Q. \tag{37}$$

Then the per-summand bound is

$$\begin{aligned} \|g_j^u\|_\infty &\leq 2\alpha^2 \, \|u\| \, K_{A,Q} (1 - \alpha a)^{(n-j-1)/2} \frac{2}{\alpha a} \, \|\epsilon\|_\infty \\ &= \frac{4\alpha \, \|u\| \, K_{A,Q} \, \|\epsilon\|_\infty}{a} (1 - \alpha a)^{(n-j-1)/2}, \quad 1 \leq j \leq n-1. \end{aligned} \tag{38}$$

Note the prefactor α^2 collapsing to α/a: the $1/(\alpha a)$ blow-up in $H_j^{(n)}$ is absorbed only by one factor of α. Squaring and summing over j:

$$\begin{aligned} \sum_{j=1}^n \|g_j^u\|_\infty^2 &= \sum_{j=1}^{n-1} \|g_j^u\|_\infty^2 \\ &\leq \frac{16\alpha^2 \, \|u\|^2 \, K_{A,Q}^2 \, \|\epsilon\|_\infty^2}{a^2} \sum_{j=1}^{n-1} (1 - \alpha a)^{n-j-1} \\ &\leq \frac{16\alpha^2 \, \|u\|^2 \, K_{A,Q}^2 \, \|\epsilon\|_\infty^2}{a^2} \frac{1}{\alpha a} \\ &= \frac{16\alpha \, \|u\|^2 \, K_{A,Q}^2 \, \|\epsilon\|_\infty^2}{a^3}. \end{aligned} \tag{39}$$

Plugging this into the variance proxy $v_n^2 = 64 \, t_{\text{mix}} \sum_{j=1}^n \|g_j^u\|_\infty^2$ from the lemma and defining

$$K_{\text{last},0} := 32 \, K_{A,Q} \, \|\epsilon\|_\infty \, \frac{\sqrt{t_{\text{mix}}}}{a^{3/2}}, \tag{40}$$

gives

$$v_n^2 \leq \|u\|^2 \, K_{\text{last},0}^2 \, \alpha. \tag{41}$$

To convert the sub-Gaussian tail into a moment bound, we use the following standard fact.

Lemma 3. Let X be an $\mathbb{R}$-valued random variable satisfying

$$\mathbb{P}(|X| \geq t) \leq 2\exp\left(-\frac{t^2}{2\sigma^2}\right) \quad \text{for all } t \geq 0, \tag{42}$$

for some $\sigma^2 > 0$. Then, for any $p \geq 2$, it holds that

$$\mathbb{E}[\![X|^p] \leq 2\, p^{p/2}\, \sigma^p. \tag{43}$$

Applying the lemma with $X = u^\top \tilde{J}^{(0,\alpha)}_{n,\,\text{last}}$ and $\sigma^2 = v_n^2 \leq \|u\|^2\ K^2_{\text{last},0}\alpha$ gives, for any $p \geq 2$,

$$\mathbb{E}^{1/p}\left[\!\!\left[u^\top \tilde{J}^{(0,\alpha)}_{n,\,\text{last}}|^p\right]\right. \leq 2^{1/p}\sqrt{p}\,\|u\|\, K_{\text{last},0}\,\sqrt{\alpha} \leq 2\sqrt{p}\,\|u\|\, K_{\text{last},0}\,\sqrt{\alpha}, \tag{44}$$

or equivalently

$$\frac{1}{\sqrt{\alpha}}\,\|u^\top \tilde{J}^{(0,\alpha)}_{n,\,\text{last}}\|_{L_p} \leq 2\sqrt{p}\,\|u\|\, K_{\text{last},0}. \tag{45}$$

Thus every fixed scalar projection of the last-iterate zeroth-order RR difference is $O(\sqrt{\alpha})$ in L^p, uniformly in n. A Euclidean-norm bound can be recovered in fixed dimension by applying the scalar estimate to a coordinate basis, but no dimension-free vector concentration is claimed here.

3. Last Iterate Analysis

3.1. Centered Bound for the Shifted First-Order Perturbation

The purpose of this section is to isolate the last-iteration weighted term which appears in the analysis of $J_n^{(1,\alpha)}$, and to record a clean L_p bound for its centered part. The proof uses the deterministic-product perturbation expansion from Samsonov et al. (2025, Proposition 9), specialized to the constant-stepsize setting, and isolates the future-centered bilinear concentration input used in that argument.

Let

$$B = I - \alpha\overline{A}. \tag{46}$$

Throughout this section we assume $0 < \alpha \le \alpha_\infty$, so that

$$\|B^m\|_Q \le (1-\alpha a)^{m/2}, \quad m \ge 0. \tag{47}$$

When the shifted estimate is transferred back from $T_n^{(1,\alpha)}$ to $J_n^{(1,\alpha)}$, we also use the elementary inverse-admissibility ceiling

$$\alpha_{\text{inv}} := \frac{1}{2 \parallel \overline{A} \parallel}. \tag{48}$$

Indeed, if $w \le \alpha_{\text{inv}}$, then the Neumann series gives $\parallel \left(I - w\overline{A}\right)^{-1} \parallel \le 2$. The single-stepsize estimates below are stated for α. Whenever the Richardson–Romberg combination is formed, the same estimates are applied at both $w = \alpha$ and $w = 2\alpha$; the local assumptions are then $2\alpha \le \alpha_\infty$ and $2\alpha \le \alpha_{\text{inv}}$. All constants denoted by C may depend on C_A, κ_Q, and norm equivalence constants, but not on α, n, p, or t_{mix}.

The deterministic-product components are defined by

$$J_n^{(0,\alpha)} = BJ_{n-1}^{(0,\alpha)} - \alpha\epsilon(Z_n), \quad J_0^{(0,\alpha)} = 0, \tag{49}$$

and

$$J_n^{(1,\alpha)} = BJ_{n-1}^{(1,\alpha)} - \alpha\tilde{A}(Z_n)J_{n-1}^{(0,\alpha)}, \quad J_0^{(1,\alpha)} = 0. \tag{50}$$

Hence

$$J_n^{(0,\alpha)} = -\alpha\sum_{k=1}^{n} B^{n-k}\epsilon(Z_k), \tag{51}$$

and

$$\begin{aligned} J_n^{(1,\alpha)} &= -\alpha\sum_{j=1}^{n} B^{n-j}\tilde{A}\big(Z_j\big)J_{j-1}^{(0,\alpha)} \\ &= \alpha^2 \sum_{1\le k<j\le n} B^{n-j}\tilde{A}\big(Z_j\big)B^{j-1-k}\epsilon(Z_k) \\ &= \alpha^2\sum_{k=1}^{n-1}\sum_{r=1}^{n-k} B^{n-k-r}\tilde{A}\big(Z_{k+r}\big)B^{r-1}\epsilon(Z_k). \end{aligned} \tag{52}$$

The sign in the second line is positive because $J_{j-1}^{(0,\alpha)}$ already contains the minus sign.

The last display is for $J_n^{(1,\alpha)}$ itself. The last-iteration estimate below operates on a **shifted** version, obtained by inserting one additional left factor B into the recursion:

$$S_n = \sum_{t=0}^{n-1} B^{n-t} \tilde{A}(Z_{t+1}) J_t^{(0,\alpha)}. \tag{53}$$

Reindexing $j = t+1$ rewrites S_n in the same form as $J_n^{(1,\alpha)}$ but with one extra power of B:

$$S_n = \sum_{j=1}^{n} B^{n-j+1} \tilde{A}\left(Z_j\right) J_{j-1}^{(0,\alpha)} = -\frac{1}{\alpha} B\, J_n^{(1,\alpha)}. \tag{54}$$

Thus the corresponding shifted first-order contribution is

$$T_n^{(1,\alpha)} = -\alpha S_n = B\, J_n^{(1,\alpha)}, \tag{55}$$

i.e. $T_n^{(1,\alpha)}$ is exactly $J_n^{(1,\alpha)}$ pre-multiplied by one additional B. Transferring a bound from $T_n^{(1,\alpha)}$ back to $J_n^{(1,\alpha)}$ therefore requires a local inverse bound on $B^{-1} = \left(I - \alpha\overline{A}\right)^{-1}$; this inverse-bound step is used explicitly when the shifted estimate is applied below.

Lemma 4. (Imported future-centered bilinear estimate.) Assume UGE 1 and stationarity. Let $\mathcal{F}_k = \sigma(Z_1, ..., Z_k)$. For $1 \leq k \leq n-1$ and $1 \leq l \leq n-k$, let $g_{k,l} : \mathcal{Z} \to \mathbb{R}^d$ be deterministic functions satisfying $\pi\left(g_{k,l}\right) = 0$ and $\|g_{k,l}\|_\infty \leq \beta_{k,l}$. Let ξ_k be $\mathcal{F}_k$-measurable vectors with $\sup_k \|\xi_k\|_\infty \leq M_\xi$. Then, for every $p \geq 2$,

$$\begin{aligned} &\| \sum_{k=1}^{n-1} \left\{ \sum_{l=1}^{n-k} g_{k,l}(Z_{k+l})^\top \xi_k - \mathbb{E}\left[\sum_{l=1}^{n-k} g_{k,l}(Z_{k+l})^\top \xi_k \mid \mathcal{F}_k \right] \right\} \|_{L_p} \\ &\leq C p^{3/2} t_{\mathrm{mix}}^{1/2} M_\xi \left(\sum_{k=1}^{n-1} \sum_{l=1}^{n-k} \beta_{k,l}^2 \right)^{1/2}. \end{aligned} \tag{56}$$

This is the scalar, constant-stepsize specialization of the block-decomposition and Berbee-coupling estimate used in Samsonov et al. (2025, Appendix D.2, Proposition 9; see in particular their Lemma 11 and the treatment of the coupled term $T_{21} + T_{22}$). The centering in the display is the conditional centering with respect to $\mathcal{F}_k$; no stationary-centering inequality is applied to a future chain started from Z_k.

Lemma 5. Assume the Markov chain is started from stationarity, that is, the law of Z_1 is π, and $\pi\left(\tilde{A}\right) = 0$. For every deterministic direction $u \in \mathbb{R}^d$ and every $p \geq 2$,

$$\|u^\top (S_n - \mathbb{E} S_n)\|_{L_p} \leq C\ \|u\|\ \|\epsilon\|_\infty \left(p^{3/2} t_{\mathrm{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\mathrm{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right). \tag{57}$$

Consequently,

$$\|u^\top \left(T_n^{(1,\alpha)} - \mathbb{E} T_n^{(1,\alpha)}\right)\|_{L_p} \leq C\alpha\ \|u\|\ \|\epsilon\|_\infty \left(p^{3/2} t_{\mathrm{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\mathrm{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right). \tag{58}$$

Proof. Since $J_0^{(0,\alpha)} = 0$, the term $t = 0$ in S_n vanishes. Substituting the explicit formula for $J_t^{(0,\alpha)}$ gives

$$\begin{aligned} S_n &= -\alpha \sum_{t=1}^{n-1} \sum_{k=1}^{t} B^{n-t} \tilde{A}(Z_{t+1}) B^{t-k} \epsilon(Z_k) \\ &= -\alpha \sum_{k=1}^{n-1} H_{k+1}^{(w)} \epsilon(Z_k), \end{aligned} \tag{59}$$

where, after the change of summation index $l = t - k + 1$,

$$H_{k+1}^{(w)} = \sum_{l=1}^{n-k} B^{n-k-l+1} \tilde{A}(Z_{k+l}) B^{l-1}. \tag{60}$$

The kernel $H_{k+1}^{(w)}$ acts on the past noise $\epsilon(Z_k)$ through future states Z_{k+l}, so $H_{k+1}^{(w)}\epsilon(Z_k)$ is a future-weighted bilinear functional of the trajectory.

Define

$$\mu_k^{(w)} = \mathbb{E}_\pi\left[H_{k+1}^{(w)}\epsilon(Z_k)\right]. \tag{61}$$

Then

$$S_n - \mathbb{E}S_n = -\alpha \sum_{k=1}^{n-1}\left\{H_{k+1}^{(w)}\epsilon(Z_k) - \mu_k^{(w)}\right\}. \tag{62}$$

Let $\mathcal{F}_k = \sigma(Z_1, ..., Z_k)$. The Markov property gives

$$\mathbb{E}\left[H_{k+1}^{(w)}\epsilon(Z_k) \mid \mathcal{F}_k\right] = v_k^{(w,\varepsilon)}(Z_k), \tag{63}$$

where

$$v_k^{(w)}(z) = \sum_{l=1}^{n-k} B^{n-k-l+1}\left(Q^l\tilde{A}\right)(z)B^{l-1}, \quad v_k^{(w,\varepsilon)}(z) = v_k^{(w)}(z)\epsilon(z). \tag{64}$$

Here Q denotes the one-step Markov transition kernel of $(Z_k)_{k\geq 1}$, acting on bounded matrix-valued functions by integration against the conditional law:

$$\left(Q\tilde{A}\right)(z) = \int \tilde{A}(u)\, Q(z, du) = \mathbb{E}\left[\tilde{A}(Z_{k+1}) \mid Z_k = z\right], \tag{65}$$

and Q^l is its l-fold iterate, the l-step kernel $Q^l(z, du) = \mathbb{P}(Z_{k+l} \in du \mid Z_k = z)$, so that

$$\left(Q^l\tilde{A}\right)(z) = \int \tilde{A}(u)\, Q^l(z, du) = \mathbb{E}\left[\tilde{A}(Z_{k+l}) \mid Z_k = z\right]. \tag{66}$$

In particular $\left(Q^l\tilde{A}\right)(z) \to \pi\left(\tilde{A}\right) = 0$ at the geometric rate dictated by UGE, which is the only fact about Q^l used below.

Under stationarity, $\mu_k^{(w)} = \pi\left(v_k^{(w,\varepsilon)}\right)$. Therefore

$$S_n - \mathbb{E}S_n = -\alpha(U_M + U_R), \tag{67}$$

with

$$U_M = \sum_{k=1}^{n-1}\left\{H_{k+1}^{(w)}\epsilon(Z_k) - v_k^{(w,\varepsilon)}(Z_k)\right\} \tag{68}$$

and

$$U_R = \sum_{k=1}^{n-1}\left\{v_k^{(w,\varepsilon)}(Z_k) - \pi\left(v_k^{(w,\varepsilon)}\right)\right\}. \tag{69}$$

This decomposition splits the centered statistic into two structurally different parts: U_R is an ordinary centered additive functional of the original Markov chain, while U_M is a future-centered bilinear term — the conditional expectation given $\mathcal{F}_k$ vanishes summand-wise, but the summands are not forward martingale differences. We bound each piece separately.

Fix a deterministic direction u. The case $u = 0$ is trivial. By homogeneity, it is enough to prove the estimate for $\|u\| = 1$; the general statement follows by multiplying the right-hand side by $\|u\|$.

Step 1: bound on $u^\top U_R$. Because $\pi\left(\tilde{A}\right) = 0$, applying Q^l followed by integration against π replaces $\tilde{A}$ by its l-step propagation away from stationarity. UGE then gives the geometric Dobrushin bound

$$\left\|\left(Q^l\tilde{A}\right)(z)\right\| = \left\|\int \tilde{A}(y)\left\{Q^l(z,dy) - \pi(dy)\right\}\right\| \le 2C_A\Delta\left(Q^l\right) \le 2C_A(1/4)^{\lfloor l/t_{\mathrm{mix}}\rfloor}. \tag{70}$$

Inserting this into $v_k^{(w)}$ and factoring out the slow rate $(1-\alpha a)^{(n-k)/2}$ from the two B-powers, the inner sum becomes a fast-decaying l-series:

$$\begin{aligned}
\|v_k^{(w)}\|_\infty &\le C\sum_{l=1}^{n-k}(1-\alpha a)^{(n-k-l+1)/2}\Delta\left(Q^l\right)(1-\alpha a)^{(l-1)/2} && \text{(triangle + Lyapunov)}\\
&\le C(1-\alpha a)^{(n-k)/2}\sum_{l=1}^{\infty}(1/4)^{\lfloor l/t_{\mathrm{mix}}\rfloor} && \text{(extend to } l=\infty\text{)}\\
&\le Ct_{\mathrm{mix}}(1-\alpha a)^{(n-k)/2} && \text{(geometric block sum).}
\end{aligned} \tag{71}$$

Multiplying by $\|\epsilon\|_\infty$ gives

$$\|v_k^{(w,\varepsilon)}\|_\infty \le Ct_{\mathrm{mix}}\ \|\epsilon\|_\infty\ (1-\alpha a)^{(n-k)/2}. \tag{72}$$

The functions $v_k^{(w,\varepsilon)}(Z_k) - \pi\left(v_k^{(w,\varepsilon)}\right)$ are centered under π and uniformly bounded by the previous display, so the weighted Markov concentration/Rosenthal bound for centered time-dependent functions yields

$$\begin{aligned}
\|u^\top U_R\|_{L_p} &\le Cp^{1/2}t_{\mathrm{mix}}^{1/2}\left(\sum_{k=1}^{n-1}\|v_k^{(w,\varepsilon)}\|_\infty^2\right)^{1/2} && \text{(Rosenthal)}\\
&\le Cp^{1/2}t_{\mathrm{mix}}^{3/2}\ \|\epsilon\|_\infty\ \left(\sum_{k=1}^{n-1}(1-\alpha a)^{n-k}\right)^{1/2} && \text{(plug previous bound)}\\
&\le Cp^{1/2}t_{\mathrm{mix}}^{3/2}\ \|\epsilon\|_\infty\ \frac{1}{\sqrt{\alpha a}} && \text{(geometric series).}
\end{aligned} \tag{73}$$

Step 2: bound on $u^\top U_M$. Unfold the projected matrix kernel through $\left(H_{k+1}^{(w)}\right)^\top u$:

$$\left(H_{k+1}^{(w)}\right)^\top u = \sum_{l=1}^{n-k} g_{k,l}(Z_{k+l}), \quad g_{k,l}(z) = \left(B^{l-1}\right)^\top \tilde{A}(z)^\top \left(B^{n-k-l+1}\right)^\top u. \tag{74}$$

Each $g_{k,l}$ is centered under π (since $\pi\left(\tilde{A}\right) = 0$), and the two B-powers give the uniform bound

$$\|g_{k,l}\|_\infty \le C(1-\alpha a)^{(n-k)/2}, \tag{75}$$

which is independent of l. Squaring and summing over l produces only linear growth in $n-k$:

$$\sum_{l=1}^{n-k}\|g_{k,l}\|_\infty^2 \le C(n-k)(1-\alpha a)^{n-k}. \tag{76}$$

Note the contrast with the U_R analysis: there, UGE folded the l-sum into a single t_{mix} factor, whereas here the same l-independent bound is applied $(n-k)$ times — the price of conditional centering being weaker than π-centering.

The conditional expectation in the imported future-centered estimate is exactly

$$\mathbb{E}\left[\sum_{l=1}^{n-k} g_{k,l}(Z_{k+l})^\top \epsilon(Z_k) \mid \mathcal{F}_k\right] = u^\top v_k^{(w,\varepsilon)}(Z_k). \tag{77}$$

Therefore the whole future-centered sum $u^\top U_M$ can be bounded in one step by applying the imported estimate with $\xi_k = \epsilon(Z_k)$ and $\beta_{k,l} = C(1-\alpha a)^{(n-k)/2}$:

$$\begin{aligned}
\|u^\top U_M\|_{L_p} &\le C p^{3/2} t_{\mathrm{mix}}^{1/2}\, \|\epsilon\|_\infty \left(\sum_{k=1}^{n-1} \sum_{l=1}^{n-k} (1-\alpha a)^{n-k} \right)^{1/2} \\
&= C p^{3/2} t_{\mathrm{mix}}^{1/2}\, \|\epsilon\|_\infty \left(\sum_{k=1}^{n-1} (n-k)(1-\alpha a)^{n-k} \right)^{1/2} \\
&\le C p^{3/2} t_{\mathrm{mix}}^{1/2}\, \|\epsilon\|_\infty \, \frac{1}{\alpha a} \qquad\qquad (\text{use} \sum_m m\, r^m \le C(1-r)^{-2}, r = 1-\alpha a\ ).
\end{aligned} \tag{78}$$

The extra factor $1/\sqrt{\alpha a}$ relative to U_R is precisely the cost of using conditional rather than stationary centering.

Step 3: assembly. Combining the two pieces via $S_n - \mathbb{E}S_n = -\alpha(U_M + U_R)$ and the triangle inequality,

$$\begin{aligned}
\|u^\top (S_n - \mathbb{E}S_n)\|_{L_p} &\le \alpha \left(\|u^\top U_M\|_{L_p} + \|u^\top U_R\|_{L_p} \right) \\
&\le C\, \|\epsilon\|_\infty \left(p^{3/2} t_{\mathrm{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\mathrm{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right).
\end{aligned} \tag{79}$$

The first term (from U_M) is the leading contribution for small α; the second (from U_R) carries the heavier t_{mix} dependence but vanishes as $\alpha \to 0$. Restoring the factor $\|u\|$ and multiplying by α gives the asserted bound for $T_n^{(1,\alpha)} = -\alpha S_n$. □

3.2. A Depth-One RR Misadjustment Bound and Its Limitation

This subsection records the natural depth-one attempt and explains why it is not used in the final Berry–Esseen assembly. The actual stationary and burned-in theorems use the depth-two Levin transfer developed in the next chapter.

The PR-averaged Richardson–Romberg expansion produces, after Step (S8) of the Samsonov scheme applied separately at step sizes α and 2α, a depth-one "misadjustment" remainder

$$D_1^{\text{mis, RR}} = \frac{\sqrt{n}}{n - n_0} \sum_{k=n_0}^{n-1} \left(2 J_k^{(1,\alpha)} - J_k^{(1,2\alpha)} \right), \tag{80}$$

whose centered part must be controlled to feed into a Berry–Esseen statement by this route. We write

$$D_{1,\text{ c}}^{\text{mis, RR}} := D_1^{\text{mis, RR}} - \mathbb{E} D_1^{\text{mis, RR}} \tag{81}$$

for this centered statistic.

In this subsection assume explicitly that $2\alpha \le \alpha_\infty$ and $2\alpha \le \alpha_{\mathrm{inv}}$. The first condition makes the one-stepsize last-iterate lemma admissible at both RR levels; the second makes the shifted-to-unshifted transfer $T_k^{(1,w)} = \left(I - w\overline{A}\right) J_k^{(1,w)}$ uniformly invertible for $w \in \{\alpha, 2\alpha\}$.

The stationary bias is smaller than the fluctuation term. By Levin et al. (2025, Proposition 2),

$$\mathbb{E}_\pi \left[J_\infty^{(1,\alpha)} \right] = \alpha \Delta + O(\alpha^2), \tag{82}$$

so the linear term $\alpha\Delta$ cancels in the RR-combination and the per-iterate stationary RR bias is $O(\alpha^2)$. Therefore the PR-scaled stationary bias satisfies

$$\|\mathbb{E} D_1^{\text{mis, RR}}\| \le C \sqrt{n}\, \alpha^2. \tag{83}$$

What remains is the centered fluctuation.

Define

$$\Phi(p,\alpha) := p^{3/2}\,\frac{t_{\mathrm{mix}}^{1/2}}{a} + p^{1/2}\,t_{\mathrm{mix}}^{3/2}\sqrt{\alpha/a}. \tag{84}$$

Fix a deterministic direction u. The lemma applied at α and at 2α gives, separately,

$$\|u^\top\Big(T_n^{(1,\alpha)} - \mathbb{E}T_n^{(1,\alpha)}\Big)\|_{L_p} \le C\ \|u\|\,\alpha\,\Phi(p,\alpha), \quad \|u^\top\Big(T_n^{(1,2\alpha)} - \mathbb{E}T_n^{(1,2\alpha)}\Big)\|_{L_p} \le C\ \|u\|\,\alpha\,\Phi(p,2\alpha) \le C'\ \|u\|\,\alpha\,\Phi(p,\alpha), \tag{85}$$

where $C' = \sqrt{2}C$ absorbs the $\sqrt{2}$-factor coming from the 2α scaling. Combining the two by the triangle inequality and using the index-shift identity $T_k^{(1,w)} = \Big(I - w\overline{A}\Big)\,J_k^{(1,w)}$ gives

$$u^\top J_k^{(1,w)} = \Big(\Big(I - w\overline{A}\Big)^{-\top} u\Big)^\top T_k^{(1,w)}. \tag{86}$$

The local inverse bound $\|\ \Big(I - w\overline{A}\Big)^{-1}\ \| \le 2$ therefore holds for $w \in \{\alpha, 2\alpha\}$ and yields

$$\|u^\top\Big(2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)}\Big) - \mathbb{E}\Big(2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)}\Big)\|_{L_p} \le C\ \|u\|\,\alpha\,\Phi(p,\alpha), \tag{87}$$

uniformly in k. PR-averaging through $\frac{\sqrt{n}}{n-n_0}$ and absorbing the constant therefore yields

$$\|u^\top D_{1,\,\mathrm{c}}^{\mathrm{mis,\,RR}}\|_{L_p} \le C\ \|u\|\,\sqrt{n}\,\alpha\,\Phi(p,\alpha) = O\big(\sqrt{n}\,\alpha\big). \tag{88}$$

Together with the bias estimate,

$$\|u^\top D_1^{\mathrm{mis,\,RR}}\|_{L_p} \le C\ \|u\|\,\sqrt{n}\,\alpha\,\Phi(p,\alpha) + C\ \|u\|\,\sqrt{n}\,\alpha^2. \tag{89}$$

At the optimal scale $\alpha \asymp n^{-1/2}$ the centered-fluctuation term is $O(1)$, whereas the stationary bias is $O\big(n^{-1/2}\big)$. Hence this depth-one route still does not yield a useful Berry–Esseen remainder of order $n^{-1/4}$: the centered misadjustment must be controlled more sharply to be subleading.

4. Richardson–Romberg PR Weight Bounds

Throughout this chapter, unnamed constants C may change from line to line and are independent of n, α, p, and q. Named constants display the dependencies that matter for the rate; fixed problem parameters may be absorbed only when they are not being tracked explicitly in the surrounding display.

4.1. Imported Inputs and Admissibility Thresholds

The proof below is a self-contained assembly of RR weights, Poisson decomposition, smoothing, and burn-in transfer. It uses the following external non-asymptotic inputs in the displayed forms.

Input A: Markov concentration for centered inhomogeneous sums. Under **UGE 1**, there is a universal constant C_{MC} such that, for any bounded measurable functions g_i with $\pi(g_i) = 0$ and $\|g_i\|_\infty \le c_i$, every initial distribution ξ, and every $p \ge 2$,

$$\| \sum_{i=1}^{N} g_{i(Z_i)} \|_{L_{p(\xi)}} \le C_{\mathrm{MC}} \sqrt{p\, t_{\mathrm{mix}} \sum_{i=1}^{N} c_i^2}. \tag{90}$$

This is the scalar time-inhomogeneous concentration consequence of Levin et al. (2025, Lemma 11) used in the predictable-variation estimates and in the preliminary last-iterate bounds.

Input B: Bolthausen–Fan martingale Berry–Esseen. For scalar martingale differences X_l with $|X_l| \le \kappa$, partial sum $S_N = \sum_l X_l$, predictable variation $V_N^2 = \sum_l \mathbb{E}[X_l^2 \mid \mathcal{F}_{l-1}]$, and deterministic scale $s_N^2 > 0$, Samsonov et al. (2025, Lemma 21) gives, for every $p \ge 1$,

$$\begin{aligned} d_{K(S_N/s_N, \mathcal{N}(0,1))} \le L_{B(\kappa)} &\frac{(2N+1)\log(2N+1)}{s_N^3} \\ &+ C_1 \sqrt{p} s_N^{-2p/(2p+1)} \left(\mathbb{E}|V_N^2 - s_N^2|^p\right)^{1/(2p+1)} \\ &+ C_2 p s_N^{-2p/(2p+1)} \kappa^{2p/(2p+1)}. \end{aligned} \tag{91}$$

Here $L_{B(\kappa)} < \infty$ and C_1, C_2 are the constants in that martingale theorem.

Input C: Levin stationary depth-two inputs. There exists a step-size ceiling $\alpha_*(q, t_{\mathrm{mix}}) > 0$ such that, for $q \ge 2$, $2 \le p \le q/2$, and $w \le \alpha_*(q, t_{\mathrm{mix}})$, the stationary bias, centered bilinear, and depth-two moment estimates of Levin et al. (2025) hold in the working forms stated later as Lemma 10, Lemma 11, and Lemma 12. Their constants may depend on the fixed problem parameters displayed in those lemmas, but not on n, p, or w beyond the explicit factors. The cited papers sometimes use the plus-form SA convention; throughout this thesis we use $\theta_{k+1} = \theta_k - w(A(Z_{k+1})\theta_k - b(Z_{k+1}))$, so their stability assumptions are read after this sign conversion.

Input D: startup and random-product stability. There exists a positive threshold $\alpha_{\mathrm{st}}(p)$ such that the product-stability and full-state startup contractions used in the burn-in chapter hold for $2\alpha \le \alpha_{\mathrm{st}}(p)$; their working forms are Lemma 19 and Lemma 25. This input is not used in the stationary theorem, only in the deterministic-start transfer.

For the shifted-to-unshifted first-order transfer we also use the local inverse ceiling

$$\alpha_{\mathrm{inv}} := \frac{1}{2 \| \overline{A} \|}. \tag{92}$$

If $w \le \alpha_{\mathrm{inv}}$, the Neumann series yields $\| \left(I - w\overline{A}\right)^{-1} \| \le 2$.

4.2. PR-Averaged Error Decomposition and the RR Weight

The deterministic weight estimates start from the finite-start representation of $\sqrt{n}\left(\bar{\theta}_n^{(\mathrm{RR},\alpha)} - \theta^*\right)$ as a noise-weighted sum with a deterministic kernel. The stationary $n_0 = 0$ Berry–Esseen bound proved below is stated for the augmented-chain assembly built from the same full-window weights.

Depth-one expansion. Unfolding the error recursion of Chapter 1 gives, for every $k \geq 1$,

$$\theta_k^{(\alpha)} - \theta^* = -\alpha \sum_{l=1}^{k} \Gamma_{l+1:k}^{(\alpha)} \epsilon(Z_l) + \Gamma_{1:k}^{(\alpha)}(\theta_0 - \theta^*), \quad \Gamma_{l+1:k}^{(\alpha)} := \prod_{j=l+1}^{k} \left(I - \alpha A(Z_j)\right). \tag{93}$$

Replacing the random products in the noise sum by their deterministic counterparts $B_\alpha^{k-l} := \left(I - \alpha \bar{A}\right)^{k-l}$ and keeping the initial-product discrepancy separate yields the **depth-one decomposition** (Samsonov et al., 2025, Proposition 9):

$$R_{k,\,\mathrm{init}}^{(\alpha)} := \left(\Gamma_{1:k}^{(\alpha)} - B_\alpha^k\right)(\theta_0 - \theta^*), \tag{94}$$

$$\theta_k^{(\alpha)} - \theta^* = -\alpha \sum_{l=1}^{k} B_\alpha^{k-l} \epsilon(Z_l) + B_\alpha^k(\theta_0 - \theta^*) + R_{k,\,\mathrm{init}}^{(\alpha)} + R_k^{(\alpha)}, \tag{95}$$

where $R_k^{(\alpha)} := J_k^{(1,\alpha)} + H_k^{(1,\alpha)}$ is the first noise-driven misadjustment remainder. The term $R_{k,\,\mathrm{init}}^{(\alpha)}$ vanishes when $\theta_0 = \theta^*$; otherwise it is a finite-start transient and is handled in the burn-in transfer theorem, not by the stationary augmented-chain misadjustment bound. The leading component $J_k^{(1,\alpha)}$ has a stationary bias of order α, so it is not treated as an $\alpha^{3/2}$ term. The Berry–Esseen proof below controls this noise-driven remainder by refining it to depth two, where $J^{(2)} + H^{(2)}$ carries the $\alpha^{3/2}$ moment scale. The first sum is the **depth-zero** term and is the only piece that carries the limiting Gaussian.

PR averaging produces $Q_l^{(\alpha)}$. Recall the PR average $\bar{\theta}_n^{(\alpha)} = (n - n_0)^{-1} \sum_{k=n_0}^{n-1} \theta_k^{(\alpha)}$. For notational clarity, and for the stationary result proved in this chapter, we set $n_0 = 0$ from this point on. A burned-in average has a different deterministic weight,

$$Q_{l,n_0}^{(\alpha)} = \frac{n}{n - n_0} \alpha \sum_{k=\max(n_0,l)}^{n-1} B_\alpha^{k-l}, \tag{96}$$

when the statistic is normalized as $-n^{-1/2} \sum_l Q_{l,n_0}^{(\alpha)} \epsilon(Z_l)$. The burned-in non-stationary theorem therefore requires separate weight, Poisson, and variance-comparison arguments. Subtracting θ^*, substituting the depth-one decomposition above, and **exchanging the order of summation** in the depth-zero piece,

$$\sum_{k=0}^{n-1} \sum_{l=1}^{k} B_\alpha^{k-l} \epsilon(Z_l) = \sum_{l=1}^{n-1} \left(\sum_{k=l}^{n-1} B_\alpha^{k-l} \right) \epsilon(Z_l), \tag{97}$$

isolates each noise sample $\epsilon(Z_l)$ together with the deterministic **PR weight**

$$Q_l^{(\alpha)} := \alpha \sum_{k=l}^{n-1} B_\alpha^{k-l} = \alpha \sum_{j=0}^{n-l-1} B_\alpha^j. \tag{98}$$

Multiplying the resulting identity by $\sqrt{n}$ gives the PR-averaged decomposition

$$\sqrt{n}\left(\bar{\theta}_n^{(\alpha)} - \theta^*\right) = W^{(\alpha)} + D^{(\alpha)}, \tag{99}$$

where the **leading martingale-like sum** is

$$W^{(\alpha)} := -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l^{(\alpha)} \, \epsilon(Z_l), \tag{100}$$

and the remainder $D^{(\alpha)}$ contains the deterministic transient $D_{\mathrm{tr}}^{(\alpha)}$, the random initial-product transient $D_{\mathrm{init}}^{(\alpha)}$, and the higher-order noise-driven stochastic part $D_R^{(\alpha)}$:

$$D^{(\alpha)} := D_{\mathrm{tr}}^{(\alpha)} + D_{\mathrm{init}}^{(\alpha)} + D_R^{(\alpha)}, \quad D_{\mathrm{tr}}^{(\alpha)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} B_\alpha^k (\theta_0 - \theta^*), \quad D_{\mathrm{init}}^{(\alpha)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} R_{k,\,\mathrm{init}}^{(\alpha)}, \quad D_R^{(\alpha)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} R_k^{(\alpha)} \tag{101}$$

The first sum is the deterministic initial-condition transient. Under the full-average convention it is only $O\Big(\big(\sqrt{n}\alpha a\big)^{-1}\Big)$ in general, so it must either be retained explicitly or removed by a centered initialization $\theta_0 = \theta^*$. The second sum is stochastic but still purely finite-start: it is controlled after burn-in by random-product stability. The third is the source of the leading non-Gaussian correction in the Berry–Esseen rate. Its RR-combination $2D_R^{(\alpha)} - D_R^{(2\alpha)}$ is the **misadjustment** $D_1^{\mathrm{mis,\,RR}}$ controlled below by the Levin depth-two transfer.

RR combination produces $\mathcal{Q}_l^{\mathrm{RR}}$. The Richardson–Romberg iterate $\overline{\theta}_n^{(\mathrm{RR},\alpha)} := 2\overline{\theta}_n^{(\alpha)} - \overline{\theta}_n^{(2\alpha)}$ inherits the PR decomposition **by linearity**: applying the previous display at step sizes α and 2α separately and combining,

$$\sqrt{n}\left(\overline{\theta}_n^{(\mathrm{RR},\alpha)} - \theta^*\right) = W^{\mathrm{RR}} + D^{\mathrm{RR}}, \tag{102}$$

$$W^{\mathrm{RR}} := 2W^{(\alpha)} - W^{(2\alpha)} = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} \mathcal{Q}_l^{\mathrm{RR}} \, \epsilon(Z_l), \tag{103}$$

where the **RR weight** is

$$\mathcal{Q}_l^{\mathrm{RR}} := 2Q_l^{(\alpha)} - Q_l^{(2\alpha)}. \tag{104}$$

Since both PR averages share the **same** noise realization $\{Z_k\}$, the bracket $\mathcal{Q}_l^{\mathrm{RR}}$ is a single deterministic matrix kernel: all the stochastic content of W^{RR} now lives in $\epsilon(Z_l)$ alone.

Two deterministic estimates. The rest of the martingale approximation uses two quantities determined by $\mathcal{Q}_l^{\mathrm{RR}}$:

1. **Variance comparison.** The target CLT covariance of W^{RR} is $\Sigma_\infty = \overline{A}^{-1}\Sigma_\epsilon^{(\mathrm{M})}\overline{A}^{-\top}$ (the Markov-chain CLT covariance). Replacing each $\mathcal{Q}_l^{\mathrm{RR}}$ by the asymptotic weight $\overline{A}^{-1}$ gives this covariance, so the finite-n deviation is controlled by $\sum_l \|\mathcal{Q}_l^{\mathrm{RR}} - \overline{A}^{-1}\|^2$ — see Section 4.5.
2. **Poisson-equation / Abel-summation remainder.** The standard Berry–Esseen route for Markov-chain noise solves the Poisson equation $\hat{\epsilon} - \mathbf{Q}\hat{\epsilon} = \epsilon$, replaces $\epsilon(Z_l)$ by $\hat{\epsilon}(Z_l) - \hat{\epsilon}(Z_{l+1})$ (up to a martingale increment), and Abel-sums against the weight sequence $\mathcal{Q}_l^{\mathrm{RR}}$. The resulting remainder has norm bounded by the **total variation** $\sum_l \|\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}}\|$.

Both quantities are estimated in Sections 4.3–4.4 below. The closed-form identities of Section 4.2 reduce them to elementary operator-norm calculations on $B_\alpha^m - B_{2\alpha}^m$.

4.3. Closed-Form Identities

Throughout this chapter write

$$B_\alpha := I - \alpha\overline{A}, \quad B_{2\alpha} := I - 2\alpha\overline{A}, \quad k := n - l. \tag{105}$$

We assume $\alpha, 2\alpha \in (0, \alpha_\infty]$, so that the Lyapunov contraction $\| \, B_\alpha^m \, \|_Q^2 \le (1-\alpha a)^m$ and $\| \, B_{2\alpha}^m \, \|_Q^2 \le (1-2\alpha a)^m$ hold (and we use freely $1 - 2\alpha a \le 1 - \alpha a$).

The geometric-series identity $\sum_{j=0}^{m-1} B_\alpha^j = \left(\alpha\overline{A}\right)^{-1}(I - B_\alpha^m)$ converts Eq. 98 into the closed form

$$Q_l^{(\alpha)} = \alpha \sum_{j=0}^{n-l-1} B_\alpha^j = \alpha\left(\alpha\overline{A}\right)^{-1}\left(I - B_\alpha^{n-l}\right) = \overline{A}^{-1}\left(I - B_\alpha^{n-l}\right). \tag{106}$$

This makes the convergence $Q_l^{(\alpha)} \to \overline{A}^{-1}$ as $k = n - l \to \infty$ explicit: the only obstruction is the geometric Lyapunov tail B_α^k. Two immediate consequences are

$$Q_l^{(\alpha)} - \overline{A}^{-1} = -\overline{A}^{-1}B_\alpha^{n-l}, \quad Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)} = -\alpha B_\alpha^{n-l-1}. \tag{107}$$

The first is the **asymptotic-weight error** and decays geometrically in k. The second is the **discrete derivative** used in Abel summation; note the explicit factor α in front.

Applying both identities at α and 2α and combining yields the basic **RR identities**:

$$\mathcal{Q}_l^{\mathrm{RR}} - \overline{A}^{-1} = -\overline{A}^{-1}\left(2B_\alpha^k - B_{2\alpha}^k\right), \tag{108}$$

$$\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}} = -2\alpha\left(B_\alpha^{k-1} - B_{2\alpha}^{k-1}\right). \tag{109}$$

These are the exact expressions bounded in the next subsection. Note the structural asymmetry between them:

- The asymptotic-weight error $2B_\alpha^k - B_{2\alpha}^k$ is at the **single-trajectory** rate $(1-\alpha a)^{k/2}$ — the triangle inequality $|2X - Y| \le 2\,|X| + |Y|$ does not see the RR coupling, and indeed the leading $-\overline{A}^{-1}$ in $\mathcal{Q}_l^{\mathrm{RR}}$ is the same as in the single-step weight $Q_l^{(\alpha)}$.
- The discrete derivative $B_\alpha^{k-1} - B_{2\alpha}^{k-1}$, however, is a **true** difference of contractions evaluated at the same exponent. The elementary identity $X^m - Y^m = (X - Y)\sum_{i=1}^m X^{i-1}Y^{m-i}$ with $X = B_\alpha$, $Y = B_{2\alpha}$, $X - Y = \alpha\overline{A}$ extracts an extra factor of α, gaining one full power of α over the single-step case $Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)} = -\alpha B_\alpha^{k-1}$. The left-factored form is valid because B_α and $B_{2\alpha}$ are polynomials in the same matrix $\overline{A}$, hence commute with each other and with $\overline{A}$.

This local α-gain in the discrete derivative is the **only** manifestation of Richardson–Romberg cancellation visible at the level of the PR weights.

4.4. Pointwise Bounds for the RR Weights

Lemma 6. Let $\alpha, 2\alpha \in (0, \alpha_\infty]$, set $C_Q := \kappa_Q^{1/2} \,\|\, \overline{A}^{-1} \,\|$ and $\tilde{C}_A := \kappa_Q \,\|\, \overline{A} \,\|$, and write $k = n - l$.

(i) For every $1 \le l \le n-1$,

$$\|\, \mathcal{Q}_l^{\mathrm{RR}} - \overline{A}^{-1} \,\| \le 3C_Q(1-\alpha a)^{k/2}. \tag{110}$$

(ii) For every $1 \le l \le n-2$,

$$\|\, \mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}} \,\| \le 2\tilde{C}_A\,\alpha^2\,(k-1)\,(1-\alpha a)^{(k-2)/2}. \tag{111}$$

Proof of (i). Take norms in the basic RR identity above. Using the equivalence $\|\cdot\| \le \kappa_Q^{1/2} \,\|\cdot\|_Q$ and the Lyapunov contraction at both step sizes,

$$\|\, 2B_\alpha^k - B_{2\alpha}^k \,\|_Q \le 2\,\|\, B_\alpha^k \,\|_Q + \|\, B_{2\alpha}^k \,\|_Q \le 2(1-\alpha a)^{k/2} + (1-2\alpha a)^{k/2} \le 3(1-\alpha a)^{k/2}. \tag{112}$$

Submultiplicativity in $\|\cdot\|_Q$ followed by norm equivalence then gives

$$\|\, \mathcal{Q}_l^{\mathrm{RR}} - \overline{A}^{-1} \,\| \le \kappa_Q^{1/2} \,\|\, \overline{A}^{-1} \,\| \cdot \|\, 2B_\alpha^k - B_{2\alpha}^k \,\|_Q \le 3C_Q(1-\alpha a)^{k/2}. \tag{113}$$

Proof of (ii). Apply the elementary identity $X^m - Y^m = (X - Y)\sum_{i=1}^{m} X^{i-1}Y^{m-i}$ with $X = B_\alpha$, $Y = B_{2\alpha}$, $X - Y = \alpha\overline{A}$, and $m = k - 1$. The matrices commute because they are polynomials in $\overline{A}$, so the factor $\alpha\overline{A}$ may be placed on the left:

$$B_\alpha^{k-1} - B_{2\alpha}^{k-1} = \alpha\overline{A}\sum_{i=1}^{k-1} B_\alpha^{i-1}\, B_{2\alpha}^{k-1-i}. \tag{114}$$

Each summand obeys, by submultiplicativity in $\|\cdot\|_Q$ and Lyapunov contraction,

$$\begin{aligned} \|\, B_\alpha^{i-1}\, B_{2\alpha}^{k-1-i}\,\| &\le \kappa_Q^{1/2}\,(1-\alpha a)^{(i-1)/2}(1-2\alpha a)^{(k-1-i)/2} \\ &\le \kappa_Q^{1/2}\,(1-\alpha a)^{(k-2)/2}, \end{aligned} \tag{115}$$

where in the last step we used $1 - 2\alpha a \le 1 - \alpha a$. Summing over $i \in \{1, ..., k-1\}$ and absorbing constants,

$$\|\, B_\alpha^{k-1} - B_{2\alpha}^{k-1}\,\| \le \alpha\kappa_Q \,\|\, \overline{A}\,\|\, (k-1)(1-\alpha a)^{(k-2)/2}. \tag{116}$$

Inserting this bound into the discrete-difference identity finishes the proof. □

4.5. Summed Bounds and Comparison with the Single-Step Case

Corollary 1. Under the assumptions of the previous lemma, uniformly in $n \ge 2$,

$$\sum_{l=1}^{n-1}\|\, Q_l^{\mathrm{RR}} - \overline{A}^{-1}\,\|^2 \le \frac{C_1}{\alpha a}, \quad \sum_{l=1}^{n-2}\|\, Q_{l+1}^{\mathrm{RR}} - Q_l^{\mathrm{RR}}\,\| \le \frac{C_2}{a^2}, \tag{117}$$

with $C_1 = 9C_Q^2$ and $C_2 = 32\tilde{C}_A$.

Proof. For the first sum apply (i) and the geometric series $\sum_{k\ge1}(1-\alpha a)^k \le \frac{1}{\alpha a}$. For the second, apply (ii) and

$$\sum_{k\ge2}(k-1)\,(1-\alpha a)^{(k-2)/2} = \sum_{m\ge0}(m+1)(1-\alpha a)^{m/2} \le \frac{1}{\left(1-\sqrt{1-\alpha a}\right)^2} \le \frac{4}{(\alpha a)^2}, \tag{118}$$

where the last step uses $1 - \sqrt{1-\alpha a} \ge \alpha\frac{a}{2}$ for $\alpha a \le 1/2$. Multiplying by $2\tilde{C}_A\alpha^2$ gives the claim, up to the stated universal constant. □

4.6. Variance Comparison

After the Poisson decomposition the $l = 1$ noise sample is absorbed into the Poisson boundary remainder, while the martingale increments are indexed by $l \in \{2, ..., n-1\}$. The deterministic variance proxy used in the martingale Berry–Esseen step must therefore use the same index set:

$$\Sigma_n^{\mathrm{RR}} := \frac{1}{n}\sum_{l=2}^{n-1} Q_l^{\mathrm{RR}}\,\Sigma_\epsilon^{(\mathrm{M})}\left(Q_l^{\mathrm{RR}}\right)^\top, \quad \sigma_n^{2,\,\mathrm{RR}}(u) := u^\top\Sigma_n^{\mathrm{RR}}u, \tag{119}$$

with $\Sigma_\epsilon^{(\mathrm{M})}$ the symmetric long-run noise covariance of the Markov chain,

$$\Sigma_\epsilon^{(\mathrm{M})} = \mathbb{E}_\pi\left[\epsilon(Z_0)\epsilon(Z_0)^\top\right] + \sum_{j\ge1}\left(\mathbb{E}_\pi\left[\epsilon(Z_0)\epsilon(Z_j)^\top\right] + \mathbb{E}_\pi\left[\epsilon(Z_j)\epsilon(Z_0)^\top\right]\right). \tag{120}$$

The asymptotic covariance is

$$\Sigma_\infty = \overline{A}^{-1}\Sigma_\epsilon^{(\mathrm{M})}\overline{A}^{-\top}, \quad \sigma^2(u) = u^\top\Sigma_\infty u. \tag{121}$$

The bounds of the previous section give a quantitative comparison between Σ_n^{RR} and Σ_∞.

Lemma 7. Let $\alpha, 2\alpha \in (0, \alpha_\infty]$, set $\Sigma := \Sigma_\epsilon^{(\mathrm{M})}$, and assume $\| \Sigma \| < \infty$. Then

$$\| \Sigma_n^{\mathrm{RR}} - \Sigma_\infty \| \le \frac{C_3}{n\,\alpha a}, \tag{122}$$

with $C_3 = 12\, C_Q \| \overline{A}^{-1} \| \| \Sigma \| + 9\, C_Q^2 \| \Sigma \| + 2 \| \Sigma_\infty \|$. Consequently, for every $u \in \mathbb{R}^d$,

$$| \sigma_n^{2,\,\mathrm{RR}}(u) - \sigma^2(u) | \le \frac{C_3 \| u \|^2}{n\,\alpha a}. \tag{123}$$

At the working scale $\alpha = c\,n^{-1/2}$ this is $O\big(n^{-1/2}\big)$.

Proof. Write $\Delta_l := \mathcal{Q}_l^{\mathrm{RR}} - \overline{A}^{-1}$ and expand

$$\mathcal{Q}_l^{\mathrm{RR}}\, \Sigma \left(\mathcal{Q}_l^{\mathrm{RR}}\right)^\top - \Sigma_\infty = \underbrace{\Delta_l\, \Sigma\, \overline{A}^{-\top} + \overline{A}^{-1}\, \Sigma\, \Delta_l^\top}_{R_{1,l}} + \underbrace{\Delta_l\, \Sigma\, \Delta_l^\top}_{R_{2,l}}. \tag{124}$$

Submultiplicativity of the operator norm yields the pointwise bounds

$$\| R_{1,l} \| \le 2 \| \overline{A}^{-1} \| \| \Sigma \| \| \Delta_l \|, \quad \| R_{2,l} \| \le \| \Sigma \| \| \Delta_l \|^2. \tag{125}$$

Summing the linear part with the bound from part (i) of the previous lemma and the geometric series $\sum_{k\ge 1} (1-\alpha a)^{k/2} \le \frac{1}{1-\sqrt{1-\alpha a}} \le \frac{2}{\alpha a}$ (valid for $\alpha a \le 1/2$),

$$\sum_{l=1}^{n-1} \| \Delta_l \| \le 3 C_Q \sum_{k=1}^{n-1} (1-\alpha a)^{k/2} \le \frac{6 C_Q}{\alpha a}. \tag{126}$$

The quadratic part is bounded directly by the previous corollary,

$$\sum_{l=1}^{n-1} \| \Delta_l \|^2 \le \frac{9 C_Q^2}{\alpha a}. \tag{127}$$

Since the sum starts at $l = 2$, replacing every weight by $\overline{A}^{-1}$ gives $((n-2)/n)\Sigma_\infty$, not Σ_∞. Hence there is an additional deterministic finite-sum boundary term $2\Sigma_\infty/n$. Combining,

$$\begin{aligned} \| \Sigma_n^{\mathrm{RR}} - \Sigma_\infty \| &\le \frac{2 \| \Sigma_\infty \|}{n} + \frac{1}{n} \sum_{l=2}^{n-1} \big(\| R_{1,l} \| + \| R_{2,l} \|\big) \\ &\le \frac{1}{n} \left(2 \| \overline{A}^{-1} \| \| \Sigma \| \cdot \frac{6 C_Q}{\alpha a} + \| \Sigma \| \cdot \frac{9 C_Q^2}{\alpha a} + 2 \| \Sigma_\infty \| / (\alpha a) \right) = \frac{C_3}{n\,\alpha a}, \end{aligned} \tag{128}$$

where we used $\alpha a \le 1$. This proves the operator-norm bound. The scalar bound on $| \sigma_n^{2,\,\mathrm{RR}}(u) - \sigma^2(u) | = | u^\top (\Sigma_n^{\mathrm{RR}} - \Sigma_\infty) u |$ follows from the Cauchy–Schwarz inequality. □

4.7. Poisson Martingale Approximation

The variance comparison of the previous section identifies the limiting covariance of W^{RR}, but it does not by itself produce a martingale. This subsection converts W^{RR} into a martingale plus a quantitatively small remainder via the Poisson equation for the Markov chain $\{Z_l\}$. The deterministic kernel bounds of Sections 4.3–4.4 enter once and for all in the boundary/Abel control of the remainder.

Poisson kernel. Let Q denote the one-step Markov transition kernel of $(Z_k)_{k\ge 1}$, acting on bounded measurable functions $f : Z \to \mathbb{R}^d$ by

$$(Qf)(z) = \mathbb{E}[f(Z_{k+1}) \mid Z_k = z]. \tag{129}$$

Under UGE 1 with mixing time t_{mix}, the geometric Dobrushin bound $\| Q^k \epsilon \|_\infty \le 2 \| \epsilon \|_\infty (1/4)^{\lfloor k/t_{\text{mix}} \rfloor}$ (valid for centered ϵ, $\pi(\epsilon) = 0$) makes the Poisson series

$$\hat{\epsilon} := \sum_{k=0}^{\infty} Q^k \epsilon \tag{130}$$

absolutely convergent in sup-norm, with

$$\| \hat{\epsilon} \|_\infty \le 2 \| \epsilon \|_\infty \sum_{k=0}^{\infty} (1/4)^{\lfloor k/t_{\text{mix}} \rfloor} \le 3\, t_{\text{mix}} \| \epsilon \|_\infty, \quad \| Q\hat{\epsilon} \|_\infty \le \| \hat{\epsilon} \|_\infty, \tag{131}$$

the second inequality because Q is a Markov kernel and contracts the sup-norm. By construction $\hat{\epsilon}$ solves the **Poisson equation**

$$\hat{\epsilon} - Q\hat{\epsilon} = \epsilon. \tag{132}$$

Conditional centering. Let $\mathcal{F}_l := \sigma(Z_1, ..., Z_l)$. Substituting Eq. 132 into the noise sample and adding and subtracting the conditional mean,

$$\epsilon(Z_l) = \underbrace{[\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})]}_{\text{martingale increment}} + \underbrace{[Q\hat{\epsilon}(Z_{l-1}) - Q\hat{\epsilon}(Z_l)]}_{\text{telescope}}, \quad l \ge 2, \tag{133}$$

where the first bracket is centered conditionally on $\mathcal{F}_{l-1}$ since the Markov property gives $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$. The $l = 1$ term has no past state to condition on; we treat it directly via Eq. 132 as $\epsilon(Z_1) = \hat{\epsilon}(Z_1) - Q\hat{\epsilon}(Z_1)$. The decomposition therefore separates W^{RR} into a martingale piece and a deterministic-coefficient telescope; Abel summation against the kernel sequence $\{\mathcal{Q}_l^{\text{RR}}\}$ converts the telescope into boundary terms plus a sum against the discrete derivative $\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}}$, which is exactly the object bounded in the Corollary of Section 4.4.

Lemma 8. Assume **UGE 1** and $\pi(\epsilon) = 0$. Set

$$\Delta M_l^{\text{RR}} := \mathcal{Q}_l^{\text{RR}} \left(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})\right), \quad 2 \le l \le n-1, \tag{134}$$

and let $M_n^{\text{RR}} := \sum_{l=2}^{n-1} \Delta M_l^{\text{RR}}$. Then $\{\Delta M_l^{\text{RR}}\}_{l=2}^{n-1}$ is a sequence of $\mathcal{F}_l$-martingale differences, and

$$W^{\text{RR}} = -\frac{1}{\sqrt{n}} M_n^{\text{RR}} + D_{2,n}^{\text{RR}}, \tag{135}$$

with the **Poisson boundary/Abel remainder**

$$D_{2,n}^{\text{RR}} := -\frac{1}{\sqrt{n}} \left[\mathcal{Q}_1^{\text{RR}}\, \hat{\epsilon}(Z_1) + \sum_{l=1}^{n-2} \left(\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}}\right) Q\hat{\epsilon}(Z_l) \right]. \tag{136}$$

(The right boundary $\mathcal{Q}_{n-1}^{\text{RR}}\, Q\hat{\epsilon}(Z_{n-1})$ vanishes identically because $\mathcal{Q}_{n-1}^{\text{RR}} = 0$.) Moreover, with $C_\mathcal{Q} := \| \overline{A}^{-1} \| + 3C_Q$ a uniform bound on $\| \mathcal{Q}_l^{\text{RR}} \|$, and C_2 the constant from the Corollary of Section 4.4,

$$\| D_{2,n}^{\text{RR}} \|_\infty \le \frac{3\, t_{\text{mix}} \| \epsilon \|_\infty}{\sqrt{n}} \left(C_\mathcal{Q} + \frac{C_2}{a^2} \right). \tag{137}$$

Consequently, for every $p \ge 1$,

$$\| D_{2,n}^{\text{RR}} \|_{L_p} \le \frac{C\, t_{\text{mix}} \| \epsilon \|_\infty}{a^2 \sqrt{n}}, \tag{138}$$

with a constant C depending only on $\| \overline{A}^{-1} \|$, κ_Q, and $\| \overline{A} \|$.

Proof. The increments ΔM_l^{RR} are $\mathcal{F}_l$-martingale differences because $\mathcal{Q}_l^{\mathrm{RR}}$ is deterministic and the Markov property gives $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$.

For the decomposition, substitute Eq. 132 into the definition in Eq. 103:

$$W^{\mathrm{RR}} = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} \mathcal{Q}_l^{\mathrm{RR}} \left(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_l)\right). \tag{139}$$

Peel off $l = 1$ and apply the conditional-centering identity above to the remaining $l \in \{2, ..., n-1\}$:

$$W^{\mathrm{RR}} = -\frac{1}{\sqrt{n}} \mathcal{Q}_1^{\mathrm{RR}}\, \hat{\epsilon}(Z_1) + \frac{1}{\sqrt{n}} \mathcal{Q}_1^{\mathrm{RR}}\, Q\hat{\epsilon}(Z_1) - \frac{1}{\sqrt{n}} M_n^{\mathrm{RR}} - \frac{1}{\sqrt{n}} \sum_{l=2}^{n-1} \mathcal{Q}_l^{\mathrm{RR}} \left(Q\hat{\epsilon}(Z_{l-1}) - Q\hat{\epsilon}(Z_l)\right). \tag{140}$$

Set $g_l \coloneqq Q\hat{\epsilon}(Z_l)$ and Abel-sum the last sum:

$$\sum_{l=2}^{n-1} \mathcal{Q}_l^{\mathrm{RR}} (g_{l-1} - g_l) = \mathcal{Q}_2^{\mathrm{RR}}\, g_1 - \mathcal{Q}_{n-1}^{\mathrm{RR}}\, g_{n-1} + \sum_{l=2}^{n-2} \left(\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}}\right) g_l. \tag{141}$$

Adding the contribution $\mathcal{Q}_1^{\mathrm{RR}}\, g_1$ from the peeled $l = 1$ term combines with $-\mathcal{Q}_2^{\mathrm{RR}}\, g_1$ to yield $-\left(\mathcal{Q}_2^{\mathrm{RR}} - \mathcal{Q}_1^{\mathrm{RR}}\right) g_1$, which folds the boundary $l = 1$ summand back into the discrete-derivative sum:

$$\begin{aligned} &-\mathcal{Q}_1^{\mathrm{RR}}\, \hat{\epsilon}(Z_1) + \mathcal{Q}_1^{\mathrm{RR}}\, g_1 - \mathcal{Q}_2^{\mathrm{RR}}\, g_1 + \mathcal{Q}_{n-1}^{\mathrm{RR}}\, g_{n-1} - \sum_{l=2}^{n-2} \left(\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}}\right) g_l \\ &= -\mathcal{Q}_1^{\mathrm{RR}}\, \hat{\epsilon}(Z_1) - \sum_{l=1}^{n-2} \left(\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}}\right) g_l + \mathcal{Q}_{n-1}^{\mathrm{RR}}\, g_{n-1}. \end{aligned} \tag{142}$$

The rightmost boundary $\mathcal{Q}_{n-1}^{\mathrm{RR}}\, g_{n-1}$ vanishes because $\mathcal{Q}_{n-1}^{\mathrm{RR}} = 2\alpha I - 2\alpha I = 0$ identically (from $Q_{n-1}^{(\alpha)} = \alpha I$ and $Q_{n-1}^{(2\alpha)} = 2\alpha I$). Multiplying by $1/\sqrt{n}$ and absorbing the sign into $D_{2,n}^{\mathrm{RR}}$ produces exactly the stated formula.

For the sup-norm bound Eq. 137, use $\| g_l \|_\infty \le \| \hat{\epsilon} \|_\infty \le 3t_{\mathrm{mix}} \| \epsilon \|_\infty$, the uniform bound $\| \mathcal{Q}_1^{\mathrm{RR}} \| \le C_{\mathcal{Q}}$ on the left boundary, and the summed-total-variation bound $\sum_{l=1}^{n-2} \| \mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}} \| \le C_2/a^2$ on the Abel sum:

$$\| \sqrt{n}\, D_{2,n}^{\mathrm{RR}} \| \le 3t_{\mathrm{mix}} \| \epsilon \|_\infty \left(C_{\mathcal{Q}} + C_2/a^2\right). \tag{143}$$

The L_p bound is immediate from the deterministic sup-norm bound. □

4.8. Predictable Quadratic Variation Concentration

The Berry–Esseen step for the martingale M_n^{RR} requires a quantitative control of its predictable quadratic variation $\langle M^{\mathrm{RR}} \rangle_n$ around its deterministic asymptotic counterpart $n\, \sigma_n^{2,\,\mathrm{RR}}(u)$. This subsection produces such a control by applying the Markov concentration result of Section 2.2 to a suitable centered quadratic functional of the chain. The result is the RR analogue of Lemmas 22–23 of Samsonov et al. (2025).

Predictable quadratic variation. By the Poisson Martingale Approximation lemma above, the increments $\Delta M_l^{\mathrm{RR}} = \mathcal{Q}_l^{\mathrm{RR}} \left(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})\right)$ are $\mathcal{F}_l$-martingale differences. The Markov property $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$ gives, by direct computation,

$$\begin{aligned} \mathbb{E}\Big[\Delta M_l^{\mathrm{RR}} \left(\Delta M_l^{\mathrm{RR}}\right)^\top \mid \mathcal{F}_{l-1}\Big] &= \mathcal{Q}_l^{\mathrm{RR}} \mathbb{E}\Big[\hat{\epsilon}(Z_l)\, \hat{\epsilon}(Z_l)^\top \mid \mathcal{F}_{l-1}\Big] \left(\mathcal{Q}_l^{\mathrm{RR}}\right)^\top \\ &\quad - \mathcal{Q}_l^{\mathrm{RR}}\, Q\hat{\epsilon}(Z_{l-1})\, (Q\hat{\epsilon})(Z_{l-1})^\top \left(\mathcal{Q}_l^{\mathrm{RR}}\right)^\top \\ &= \mathcal{Q}_l^{\mathrm{RR}}\, \mathcal{V}_\epsilon(Z_{l-1}) \left(\mathcal{Q}_l^{\mathrm{RR}}\right)^\top, \end{aligned} \tag{144}$$

where the cross-terms in the expansion of $(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1}))(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1}))^\top$ cancel by the Markov property and we have set

$$\mathcal{V}_\epsilon(z) := Q(\hat\epsilon\hat\epsilon^\top)(z) - (Q\hat\epsilon)(z)\,(Q\hat\epsilon)(z)^\top. \tag{145}$$

This is a matrix-valued conditional covariance of the Poisson martingale increment; the vector noise remains denoted by ϵ. Summing over $l \in \{2, ..., n-1\}$ and tracking that $\langle M^{\mathrm{RR}}\rangle_n := \sum_{l=2}^{n-1} \mathbb{E}\left[\Delta M_l^{\mathrm{RR}}\left(\Delta M_l^{\mathrm{RR}}\right)^\top \mid \mathcal{F}_{l-1}\right]$,

$$\langle M^{\mathrm{RR}}\rangle_n = \sum_{l=2}^{n-1} \mathcal{Q}_l^{\mathrm{RR}}\,\mathcal{V}_\epsilon(Z_{l-1})\left(\mathcal{Q}_l^{\mathrm{RR}}\right)^\top. \tag{146}$$

The conditional covariance function $\mathcal{V}_\epsilon$ has stationary mean equal to the long-run noise covariance,

$$\pi(\mathcal{V}_\epsilon) = \Sigma_\epsilon^{(\mathrm{M})} =: \Sigma, \tag{147}$$

which is a standard identity for the Poisson solution of a Markov chain (Samsonov et al. 2025, Eq. (10); see also Douc–Moulines–Priouret–Soulier 2018, Theorem 21.2.5).

Sup-norm bound on $\mathcal{V}_\epsilon$. Since $\|\hat\epsilon\|_\infty \le 3\,t_{\mathrm{mix}}\,\|\epsilon\|_\infty$ and Q is a Markov kernel (so Qf inherits the sup-norm of f),

$$\begin{aligned}\|\mathcal{V}_\epsilon\|_\infty &\le \|Q(\hat\epsilon\hat\epsilon^\top)\|_\infty + \|(Q\hat\epsilon)(Q\hat\epsilon)^\top\|_\infty\\ &\le 2\,\|\hat\epsilon\|_\infty^2 \le 18\,t_{\mathrm{mix}}^2\,\|\epsilon\|_\infty^2.\end{aligned} \tag{148}$$

Lemma 9. Assume **UGE 1** and $\pi(\epsilon) = 0$, with $\|\epsilon\|_\infty < \infty$. Let $C_{\mathcal{Q}}$ be the uniform bound on $\|\mathcal{Q}_l^{\mathrm{RR}}\|$ from the previous lemma. There exists a universal constant $C_4 > 0$ such that, for every $u \in \mathbb{R}^d$, every $p \ge 2$, every initial distribution ξ, and every $n \ge 2$,

$$\mathbb{E}_\xi^{1/p}\left[\,\left|\,u^\top\langle M^{\mathrm{RR}}\rangle_n u - n\,\sigma_n^{2,\,\mathrm{RR}}(u)\,\right|^p\right] \le C_4\,C_{\mathcal{Q}}^2\,\|u\|^2\,\|\epsilon\|_\infty^2\,t_{\mathrm{mix}}^{5/2}\sqrt{p\,n}. \tag{149}$$

Proof. Write $h_{l(z)} := u^\top\mathcal{Q}_l^{\mathrm{RR}}\,\mathcal{V}_\epsilon(z)\left(\mathcal{Q}_l^{\mathrm{RR}}\right)^\top u$ and $g_{l(z)} := h_{l(z)} - \pi(h_l)$. Submultiplicativity of the operator norm and Eq. 148 gives

$$|h_{l(z)}| \le C_{\mathcal{Q}}^2\,\|u\|^2\,\|\mathcal{V}_\epsilon\|_\infty \le 18\,C_{\mathcal{Q}}^2\,\|u\|^2\,t_{\mathrm{mix}}^2\,\|\epsilon\|_\infty^2, \quad \|g_l\|_\infty \le 2\,|h_{l(z)}| \le 36\,C_{\mathcal{Q}}^2\,\|u\|^2\,t_{\mathrm{mix}}^2\,\|\epsilon\|_\infty^2. \tag{150}$$

By Eq. 146 and the variance definition $n\,\sigma_n^{2,\,\mathrm{RR}}(u) = \sum_{l=2}^{n-1}\pi(h_l)$,

$$u^\top\langle M^{\mathrm{RR}}\rangle_n u - n\,\sigma_n^{2,\,\mathrm{RR}}(u) = \sum_{l=2}^{n-1} g_{l(Z_{l-1})}. \tag{151}$$

For the centered sum, set $\tilde g_i := g_{i+1}$ for $i \in \{1, ..., n-2\}$. By construction $\pi(\tilde g_i) = 0$ and $\|\tilde g_i\|_\infty \le c := 36\,C_{\mathcal{Q}}^2\,\|u\|^2\,t_{\mathrm{mix}}^2\,\|\epsilon\|_\infty^2$. The scalar Markov concentration lemma of Section 2.2, imported from Levin et al. (2025, Lemma 11) and valid for arbitrary initial law, applied to $\sum_{i=1}^{n-2}\tilde g_{i(Z_i)}$ yields, for every $t \ge 0$,

$$\mathbb{P}_\xi\left(\left|\sum_{i=1}^{n-2}\tilde g_{i(Z_i)}\right| \ge t\right) \le 2\exp\left(-\frac{t^2}{2\,u_n^2}\right), \quad u_n^2 \le 64\,t_{\mathrm{mix}}\,(n-2)\,c^2. \tag{152}$$

The sub-Gaussian-to-moment lemma of Section 2.2 then gives, for $p \ge 2$,

$$\mathbb{E}_\xi^{1/p}\left[\left|\sum_{l=2}^{n-1} g_{l(Z_{l-1})}\right|^p\right] \le 2^{1/p}\sqrt{p}\,u_n \le 2\sqrt{p}\,u_n \le 16\sqrt{2}\,\sqrt{p(n-2)}\,c\,\sqrt{t_{\mathrm{mix}}}. \tag{153}$$

Substituting c and bounding $\sqrt{n-2} \le \sqrt{n}$,

$$\mathbb{E}_\xi^{1/p}\left[\left|\sum_{l=2}^{n-1} g_{l(Z_{l-1})}\right|^p\right] \le 16\sqrt{2}\cdot 36\,C_{\mathcal{Q}}^2\,\|u\|^2\,\|\epsilon\|_\infty^2\,t_{\mathrm{mix}}^{5/2}\sqrt{p\,n} = 576\sqrt{2}\,C_{\mathcal{Q}}^2\,\|u\|^2\,\|\epsilon\|_\infty^2\,t_{\mathrm{mix}}^{5/2}\sqrt{p\,n}. \tag{154}$$

Rounding the resulting prefactor to $C_4 := 850$ gives the stated bound. □

Corollary 2. Under the assumptions of the previous lemma, for every $u \in \mathbb{R}^d$, every $p \geq 2$, and every $n \geq 2$,

$$\mathbb{E}_\xi^{1/p}[| \, u^\top \langle M^{\mathrm{RR}} \rangle_n u - n\,\sigma^2(u) \, |^p] \leq C_4\, C_Q^2 \parallel u \parallel^2 \parallel \epsilon \parallel_\infty^2 t_{\mathrm{mix}}^{5/2} \sqrt{p\, n} + \frac{C_3 \parallel u \parallel^2}{\alpha\, a}, \tag{155}$$

with C_3 the variance-comparison constant of Section 4.5.

Proof. The triangle inequality and the variance-comparison lemma give

$$| \, u^\top \langle M^{\mathrm{RR}} \rangle_n u - n\,\sigma^2(u) \, | \leq | \, u^\top \langle M^{\mathrm{RR}} \rangle_n u - n\,\sigma_n^{2,\,\mathrm{RR}}(u) \, | + n \, | \, \sigma_n^{2,\,\mathrm{RR}}(u) - \sigma^2(u) \, |. \tag{156}$$

The first piece is bounded in L_p by Eq. 149; the second is the deterministic bound $n\, C_3 \parallel u \parallel^2 /(n\,\alpha\, a) = C_3 \parallel u \parallel^2 /(\alpha a)$. □

4.9. Martingale Berry–Esseen Step

The Poisson decomposition (Section 4.6) writes $W^{\mathrm{RR}} = -n^{-1/2} M_n^{\mathrm{RR}} + D_{2,n}^{\mathrm{RR}}$, with the remainder $D_{2,n}^{\mathrm{RR}}$ controlled in sup-norm at the order $n^{-1/2}$. The quadratic variation concentration (Section 4.7) bounds $|u^\top \langle M^{\mathrm{RR}} \rangle_n u - n\,\sigma_n^{2,\,\mathrm{RR}}(u)|$ in L_p by $C\sqrt{p\, n}$. This subsection assembles those two inputs into a Berry–Esseen rate for the **scalar** martingale $u^\top M_n^{\mathrm{RR}}$, the leading contribution to the final RR Berry–Esseen bound.

Bounded increments. Fix $u \in \mathbb{R}^d$ and set $X_l := u^\top \Delta M_l^{\mathrm{RR}}$ for $2 \leq l \leq n-1$. The X_l are $\mathcal{F}_l$-martingale differences (Section 4.6), and bounding the three factors $\parallel u \parallel$, $\parallel Q_l^{\mathrm{RR}} \parallel \leq C_Q$, and $\parallel \hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1}) \parallel \leq 2 \parallel \hat{\epsilon} \parallel_\infty \leq 6 t_{\mathrm{mix}} \parallel \epsilon \parallel_\infty$ yields the deterministic bound

$$|X_l| \leq \varkappa(u), \quad \varkappa(u) := 6\, t_{\mathrm{mix}}\, C_Q \parallel \epsilon \parallel_\infty \parallel u \parallel. \tag{157}$$

The conditional variances $\sigma_l^2 := \mathbb{E}\left[X_l^2 \mid \mathcal{F}_{l-1}\right]$ sum to the predictable quadratic variation along u,

$$V_n^2 := \sum_{l=2}^{n-1} \sigma_l^2 = u^\top \langle M^{\mathrm{RR}} \rangle_n u. \tag{158}$$

Variance lower bound. Set $s_n^2 := n\,\sigma_n^{2,\,\mathrm{RR}}(u)$. The variance comparison lemma (Section 4.5) and assumption $\sigma^2(u) > 0$ give $|\sigma_n^{2,\,\mathrm{RR}}(u) - \sigma^2(u)| \leq C_3 \parallel u \parallel^2 /(n\alpha a)$, so whenever

$$n\,\alpha\, a \geq \frac{2\, C_3 \parallel u \parallel^2}{\sigma^2(u)} \tag{159}$$

one has $\sigma_n^{2,\,\mathrm{RR}}(u) \geq \sigma^2(u)/2$, i.e. $s_n^2 \geq n\sigma^2(u)/2$. At the working scale $\alpha = c\, n^{-1/2}$, the variance lower-bound condition Eq. 159 is satisfied for $n \geq \left(2C_3 \parallel u \parallel^2 /(c\, a\, \sigma^2(u))\right)^2$. The trivial upper bound $\sigma_n^{2,\,\mathrm{RR}}(u) \leq C_Q^2 \parallel \Sigma_\epsilon^{(\mathrm{M})} \parallel \parallel u \parallel^2$ gives $s_n^2 \leq K^2(u)\, n$ with $K(u) := C_Q \parallel \Sigma_\epsilon^{(\mathrm{M})} \parallel^{1/2} \parallel u \parallel$.

Bolthausen–Fan inequality. We apply the martingale Berry–Esseen input Eq. 91 with $N = n$ and bounded-increment parameter $\varkappa(u)$.

Theorem 1. Assume **UGE 1**, $\pi(\epsilon) = 0$, $\parallel \epsilon \parallel_\infty < \infty$, $\sigma^2(u) > 0$, and $\alpha, 2\alpha \in (0, \alpha_\infty]$. There exist constants $C_{K,1}(u), C_{K,2}(u) > 0$ depending only on $\parallel u \parallel$, $\sigma(u)$, C_Q, t_{mix}, $\parallel \epsilon \parallel_\infty$, $\parallel \Sigma_\epsilon^{(\mathrm{M})} \parallel$, and the constants $L_{B(\varkappa(u))}, C_1, C_2$ of Eq. 91, such that for every $n \geq 3$ satisfying the variance lower-bound condition Eq. 159,

$$d_K\left(\frac{u^\top M_n^{\mathrm{RR}}}{\sqrt{n}\,\sigma_n^{\mathrm{RR}}(u)}, \mathcal{N}(0,1)\right) \leq \frac{C_{K,1}(u) \log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u) \log n}{\sqrt{n}}. \tag{160}$$

Proof. Apply Eq. 91 with $S_n = u^\top M_n^{\mathrm{RR}}$, the increment bound $\varkappa = \varkappa(u)$ from Eq. 157, $s_n^2 = n\,\sigma_n^{2,\,\mathrm{RR}}(u)$, and $p = \lceil \log n \rceil$ (so $\log n \leq p \leq \log n + 1$). Bound the three terms in turn, using $s_n^2 \in \left[n\,\sigma^2(u)/2, K^2(u)\, n\right]$ from the variance lower and upper bounds.

Term I (classical Bolthausen). From $s_n^3 \geq (n\,\sigma^2(u)/2)^{3/2}$ and $(2n+1)\log(2n+1) \leq 3\,n\log n$ for $n \geq 3$,

$$L_{B(\varkappa(u))}\,\frac{(2n+1)\log(2n+1)}{s_n^3} \leq \frac{6\sqrt{2}\,L_{B(\varkappa(u))}}{\sigma^3(u)}\,\frac{\log n}{\sqrt{n}} =: \frac{C^{(\mathrm{I})}(u)\log n}{\sqrt{n}}. \tag{161}$$

Term III (Lindeberg). Write $a_p := 2p/(2p+1) = 1 - 1/(2p+1)$. Using $s_n^{-a_p} = s_n^{-1}\,s_n^{1/(2p+1)}$ and $s_n \leq K(u)\sqrt{n}$,

$$s_n^{1/(2p+1)} \leq \max(1, K(u))^{1/(2p+1)}\,n^{1/(2(2p+1))}. \tag{162}$$

For $p \geq \log n$ one has $n^{1/(2(2p+1))} \leq n^{1/(4\log n)} = e^{1/4}$; similarly $\max(1, K(u))^{1/(2p+1)} \leq e^{1/2}$ once $n \geq \max(1, K(u))$, which is absorbed into $C^{(\mathrm{III})}(u)$. Bounding $\varkappa(u)^{a_p} \leq \max(1, \varkappa(u))$ and $s_n^{-1} \leq \sqrt{2}/(\sigma(u)\sqrt{n})$,

$$C_2\,s_n^{-a_p}\,p\,\varkappa(u)^{a_p} \leq \sqrt{2}\,e^{3/4}\,\frac{C_2\max(1, \varkappa(u))}{\sigma(u)}\,\frac{p}{\sqrt{n}} \leq \frac{C^{(\mathrm{III})}(u)\log n}{\sqrt{n}}, \tag{163}$$

with $C^{(\mathrm{III})}(u)$ depending on $\parallel u \parallel, \sigma(u), C_Q, \parallel \Sigma_\epsilon^{(\mathrm{M})} \parallel, t_{\mathrm{mix}}, \parallel \epsilon \parallel_\infty$ and on C_2.

Term II (conditional-variance concentration). By the bracket concentration bound above, for every $p \geq 2$,

$$\left(\mathbb{E} \mid V_n^2 - s_n^2 \mid^p\right)^{1/p} \leq B(u)\,\sqrt{p\,n}, \quad B(u) := C_4\,C_Q^2 \parallel u \parallel^2 \parallel \epsilon \parallel_\infty^2 t_{\mathrm{mix}}^{5/2}, \tag{164}$$

Using $s_n^{-a_p} = s_n^{-1} s_n^{1/(2p+1)}$, the variance lower bound $s_n^2 \geq n\sigma^2\frac{u}{2}$, and the upper bound $s_n \leq K(u)\sqrt{n}$, the second Bolthausen–Fan term is bounded by

$$\begin{aligned} &C_1\sqrt{p}\,s_n^{-a_p}\left(\mathbb{E} \mid V_n^2 - s_n^2 \mid^p\right)^{1/(2p+1)} \\ &\quad\leq C\,C_1\,\sigma^{-1}(u)\max(1, K(u))^{1/(2p+1)}\max(1, B(u))^{p/(2p+1)} \\ &\quad\quad\times p^{(3p+1)/(2(2p+1))} n^{-p/(2(2p+1))}. \end{aligned} \tag{165}$$

For $p = \lceil \log n \rceil$ and n large enough depending on u, $p^{(3p+1)/(2(2p+1))} \leq C\log^{3/4} n$ and $n^{-p/(2(2p+1))} \leq Cn^{-1/4}$; the remaining $K(u)$- and $B(u)$-factors are absorbed into the direction-dependent constant. Thus

$$\text{Term II} \leq \frac{C^{(\mathrm{II})}(u)\log^{3/4} n}{n^{1/4}}. \tag{166}$$

Adding the three term bounds and setting $C_{K,1}(u) := C^{(\mathrm{II})}(u)$, $C_{K,2}(u) := C^{(\mathrm{I})}(u) + C^{(\mathrm{III})}(u)$ proves the martingale Berry–Esseen bound. □

Corollary 3. Under the hypotheses of the previous theorem,

$$d_K\left(\frac{u^\top M_n^{\mathrm{RR}}}{\sqrt{n}\,\sigma(u)}, \mathcal{N}(0,1)\right) \leq \frac{C_{K,1}(u)\log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u)\log n}{\sqrt{n}} + \frac{C_3 \parallel u \parallel^2}{n\,\alpha\,a\,\sigma^2(u)}. \tag{167}$$

At $\alpha = c\,n^{-1/2}$ the last term is $O(n^{-1/2})$, hence absorbed into $C_{K,2}(u)\log n/\sqrt{n}$ up to a constant.

Proof. Set $r := \sigma_n^{\mathrm{RR}}(u)/\sigma(u)$ and $W := u^\top M_n^{\mathrm{RR}}/(\sqrt{n}\sigma_n^{\mathrm{RR}}(u))$. Then the statistic normalised by $\sigma(u)$ is Wr. Under the variance lower-bound condition Eq. 159, $r \geq 1/\sqrt{2}$, while the trivial upper bound on $\sigma_n^{2,\,\mathrm{RR}}(u)$ gives $r \leq r_{\max(u)} < \infty$. On this compact interval the standard normal cdf satisfies

$$\sup_x |\Phi(x/r) - \Phi(x)| \leq C_\Phi\,|r-1|, \quad C_\Phi := \sqrt{2}/\sqrt{\pi e}. \tag{168}$$

Therefore

$$d_{K(Wr,\mathcal{N}(0,1))} \le d_{K(W,\mathcal{N}(0,1))} + C_\Phi\,|r-1|. \tag{169}$$

The variance comparison of Section 4.5 gives

$$|r-1| = \frac{|\sigma_n^{2,\,\mathrm{RR}}(u) - \sigma^2(u)|}{\sigma(u)\,(\sigma_n^{\mathrm{RR}}(u) + \sigma(u))} \le \frac{C_3 \parallel u \parallel^2}{n\,\alpha\,a\,\sigma^2(u)}. \tag{170}$$

Adding this perturbation to the martingale Berry–Esseen bound proves the stated claim after absorbing C_Φ into the constants. □

This theorem concerns only the martingale term; the stationary assembly below adds the Poisson remainder and the Levin depth-two misadjustment.

4.10. Misadjustment via Levin Depth-Two

The Berry–Esseen assembly so far controls the leading martingale piece (Section 4.8) and the Poisson boundary remainder $D_{2,n}^{\mathrm{RR}}$ (Section 4.6). The remaining non-martingale contribution to $\sqrt{n}\,u^\top\left(\overline{\theta}_n^{(\mathrm{RR},\alpha)} - \theta^*\right)$ is the **PR-averaged misadjustment**

$$R_n^{\mathrm{mis,\,RR}} := \frac{1}{\sqrt{n}}\sum_{k=0}^{n-1}\Big(2R_k^{(\alpha)} - R_k^{(2\alpha)}\Big), \tag{171}$$

where $R_k^{(\alpha)} := J_k^{(1,\alpha)} + H_k^{(1,\alpha)}$ is the depth-one remainder of Eq. 95. A direct kernel-difference route only bounds $R_n^{\mathrm{mis,\,RR}}$ at order $O(\sqrt{n}\,\alpha) = O(1)$ at the working scale $\alpha \asymp n^{-1/2}$ — too crude for a Berry–Esseen rate $n^{-1/4}$. The depth-two route below transfers four statements of Levin et al. (2025) into the present notation and recovers the target $n^{-1/4}\,\mathrm{polylog}(n)$ rate.

Depth-two refinement. The deterministic-product expansion of Levin et al. extends the depth-one decomposition by one more level. For $\ell \ge 1$ define

$$J_n^{(\ell,\alpha)} := \left(I - \alpha\overline{A}\right) J_{n-1}^{(\ell,\alpha)} - \alpha\tilde{A}(Z_n)\,J_{n-1}^{(\ell-1,\alpha)}, \tag{172}$$

$$H_n^{(\ell,\alpha)} := \left(I - \alpha A(Z_n)\right) H_{n-1}^{(\ell,\alpha)} - \alpha\tilde{A}(Z_n)\,J_{n-1}^{(\ell,\alpha)}, \tag{173}$$

with $J_0^{(\ell,\alpha)} = H_0^{(\ell,\alpha)} = 0$ and the $\ell = 0$ processes as in Chapter Section 3. Substituting $A(Z_n) = \overline{A} + \tilde{A}(Z_n)$ and grouping terms gives, by induction on n, the recursive identity

$$H_n^{(\ell,\alpha)} = J_n^{(\ell+1,\alpha)} + H_n^{(\ell+1,\alpha)}, \quad \ell \ge 0. \tag{174}$$

Applying Eq. 174 with $\ell = 1$ refines $R_k^{(\alpha)}$ to

$$R_k^{(\alpha)} = J_k^{(1,\alpha)} + J_k^{(2,\alpha)} + H_k^{(2,\alpha)}, \tag{175}$$

which splits the misadjustment into three structurally different pieces:

$$R_n^{\mathrm{mis,\,RR}} = T_n^{(1)} + T_n^{(2)} + T_n^{(H)}, \tag{176}$$

$$T_n^{(j)} := \frac{1}{\sqrt{n}}\sum_{k=0}^{n-1}\Big(2J_k^{(j,\alpha)} - J_k^{(j,2\alpha)}\Big), \quad j \in \{1,2\}, \tag{177}$$

with $T_n^{(H)}$ defined identically with $H^{(2)}$ in place of $J^{(j)}$. Each piece is bounded separately below.

Cited Levin inputs. Input C from Section 4.1 consists of the following Levin et al. (2025) statements, specialized to the constant step-size LSA setting. They are stated here in the exact working forms used below; their proofs are in the cited paper.

Lemma 10. (Levin Proposition 2 — stationary bias of $J^{(1)}$.) Under stationarity for the augmented chain $\left(Z_{t+1}, J_t^{(0,\alpha)}, J_t^{(1,\alpha)}\right)$,

$$\mathbb{E}_\pi\left[J_\infty^{(1,\alpha)}\right] = \alpha\,\Delta + R(\alpha), \quad \| R(\alpha) \| \le 12 \, \| \overline{A}^{-1} \| \, C_A^2 \, t_{\mathrm{mix}}^2 \, \alpha^2 \, \| \epsilon \|_\infty, \tag{178}$$

with $\Delta := \overline{A}^{-1} \sum_{k\ge 1} \mathbb{E}_\pi\left[\left(Q^k \tilde{A}\right)(Z_0)\, \epsilon(Z_0)\right]$.

Lemma 11. (Levin Corollary 6 — centered bilinear L_p bound.) Define $\overline{\psi}_\alpha(j,z) := \tilde{A}(z)j - \mathbb{E}_{\Pi_{J^{(0)},\alpha}}\left[\tilde{A}(Z_1)\, J_0^{(0,\alpha)}\right]$. For every initial distribution, every $r \ge 1$, and every $p \ge 2$,

$$\left\| \sum_{t=0}^{r-1} \overline{\psi}_\alpha\left(J_t^{(0,\alpha)}, Z_{t+1}\right) \right\|_{L_p} \le c_{W,1}\, p^{3/2} \sqrt{\alpha r} + c_{W,2}\, p^3\, \alpha^{-1/2} \log^{1/p}(1/(\alpha a)), \tag{179}$$

with $c_{W,1}, c_{W,2}$ depending only on $C_A, \kappa_Q, t_{\mathrm{mix}}, \| \epsilon \|_\infty$. The precise logarithmic factor is not rate-critical below; it is absorbed into $\mathrm{polylog}(n)$ in the working-scale corollaries.

Lemma 12. (Levin Propositions 8 and 9 — high-order moment bounds.) For every $q \ge 2$ and every p satisfying $2 \le p \le q/2$, under the step-size restriction $\alpha \le \alpha_*(q, t_{\mathrm{mix}})$ of Levin et al. (2025),

$$\| J_n^{(2,\alpha)} \|_{L_p} \le D_J \, t_{\mathrm{mix}}^{5/2}\, p^{7/2}\, \alpha^{3/2} \log^{3/2}(1/(\alpha a)), \tag{180}$$

$$\| H_n^{(2,\alpha)} \|_{L_p} \le D_H \, d^{1/q}\, t_{\mathrm{mix}}^{5/2}\, p^{7/2}\, \alpha^{3/2} \log^{3/2}(1/(\alpha a)), \tag{181}$$

uniformly in n, with D_J, D_H depending only on $C_A, \kappa_Q, \| \overline{A}^{-1} \|, \| \epsilon \|_\infty$.

Stationary augmented-chain convention. The recursions above are displayed in finite-time notation with $J_0^{(\ell,\alpha)} = H_0^{(\ell,\alpha)} = 0$, while Levin Proposition 2 is a statement about the stationary augmented chain. The misadjustment estimates below use the stationary versions of $\left(Z_{t+1}, J_t^{(0,w)}, J_t^{(1,w)}, J_t^{(2,w)}, H_t^{(2,w)}\right)$, $w \in \{\alpha, 2\alpha\}$, as deterministic centers and should be read under this stationary augmented-chain convention. We keep the same symbols $J_t^{(\ell,w)}$ and $H_t^{(\ell,w)}$ to avoid duplicating notation. For a zero-start full average, the transfer from finite-start sums to stationary centered sums is not a single terminal ρ^n term: summing pointwise contractions over $k = 0, \ldots, n-1$ produces a startup contribution of order

$$\frac{1}{\sqrt{n}} \sum_{k\ge 0} \rho_\alpha^k \asymp \frac{1}{\sqrt{n}\,\alpha a} \tag{182}$$

when $\rho_\alpha \approx \exp(-c\alpha a)$. This term is not part of the theorem proved here. A practical arbitrary-start theorem requires either a genuine burn-in average with the burned-in weights $Q_{l,n_0}^{(\alpha)}$ of Section 4.1 or a separate transfer lemma with the correct accumulated startup dependence.

For the shifted-to-unshifted first-order transfer we use the local inverse ceiling α_{inv} defined in Eq. 92.

Lemma 13. (Stationary-limit transfer for the centered first-order iterate.) Fix $w \in (0, \alpha_\infty]$ and set $B_w := I - w\overline{A}$. Assume the stationary augmented-chain convention above, **UGE 1**, $\pi\left(\tilde{A}\right) = 0$, and $\| \epsilon \|_\infty < \infty$. Let $\left(J_t^{(0,w)}, J_t^{(1,w)}\right)_{t\in\mathbb{Z}}$ be the stationary two-sided solution of the first two perturbation recursions, and set $T_t^{(1,w)} := B_w\, J_t^{(1,w)}$. Then, with

$$\Phi_+(p,w) := 1 + p^{3/2}\, t_{\mathrm{mix}}^{1/2}/a + p^{1/2}\, t_{\mathrm{mix}}^{3/2}\sqrt{w/a}, \tag{183}$$

there exists a constant $C_{\mathrm{stat},1}$, depending only on the constants in the zero-start centered shifted first-order bound in Lemma 5, such that for every $p \geq 2$ and every $t \in \mathbb{Z}$,

$$\| T_t^{(1,w)} - \mathbb{E}_\pi T_t^{(1,w)} \|_{L_p} \leq C_{\mathrm{stat},1}\, w\, \Phi_+(p,w). \tag{184}$$

Consequently, whenever $w \leq \alpha_{\mathrm{inv}}$,

$$\| J_t^{(1,w)} - \mathbb{E}_\pi J_t^{(1,w)} \|_{L_p} \leq 2C_{\mathrm{stat},1}\, w\, \Phi_+(p,w). \tag{185}$$

Proof. For $m \geq 1$, run the same recursions on the stationary driving chain over the finite window $\{t - m + 1, ..., t\}$ with zero initial conditions at time $t - m$; denote the resulting variables by $J_{t,m}^{(0,w)}$ and $J_{t,m}^{(1,w)}$. Stationarity of the driving chain implies that $B_w J_{t,m}^{(1,w)}$ has the same law as the zero-start $T_m^{(1,w)}$ in Lemma 5. Therefore the centered last-iterate bound gives, uniformly in m and t,

$$\| B_w J_{t,m}^{(1,w)} - \mathbb{E} B_w J_{t,m}^{(1,w)} \|_{L_p} \leq C_{\mathrm{stat},1}\, w\, \Phi_+(p,w). \tag{186}$$

The deterministic-product expansions give

$$J_{t,m}^{(0,w)} = -w \sum_{r=0}^{m-1} B_w^r \epsilon(Z_{t-r}), \tag{187}$$

and express $J_{t,m}^{(1,w)}$ as a bounded double series whose summands are dominated by $Cw^2 s(1 - wa)^{s/2}\, \|\epsilon\|_\infty$ at total lag s. For the fixed admissible w in the lemma, $\sum_{s\geq 0} s(1 - wa)^{s/2} < \infty$, so the finite-past approximations are Cauchy in every L_p and converge to the stationary two-sided solution. This domination is not asserted uniformly as $w \to 0$; the later estimates retain the resulting step-size dependence through $\Phi_+(p,w)$. Passing to the limit in the preceding uniform bound yields the displayed estimate for $T_t^{(1,w)}$. The estimate for $J_t^{(1,w)}$ then follows from $J_t^{(1,w)} - \mathbb{E}_\pi J_t^{(1,w)} = B_w^{-1}\left(T_t^{(1,w)} - \mathbb{E}_\pi T_t^{(1,w)}\right)$. □

The lemma is therefore compatible with the triangular-array substitution $w = \alpha_n$: the finite-past domination is fixed-w, while every later bound keeps the displayed w-dependence through $\Phi_+(p,w)$ and uses constants independent of n.

Telescoping identity for $J^{(1)}$. Summing the recursion $J_k^{(1,\alpha)} = \left(I - \alpha\overline{A}\right) J_{k-1}^{(1,\alpha)} - \alpha\tilde{A}(Z_k)\, J_{k-1}^{(0,\alpha)}$ from $k = 1$ to n and rearranging gives, for an arbitrary initial value,

$$\overline{A} \sum_{k=0}^{n-1} J_k^{(1,\alpha)} = -\sum_{k=1}^{n} \tilde{A}(Z_k)\, J_{k-1}^{(0,\alpha)} + \frac{1}{\alpha}\left(J_0^{(1,\alpha)} - J_n^{(1,\alpha)}\right). \tag{188}$$

The stationary version of the same recursion yields $\mathbb{E}_\pi\left[\tilde{A}(Z_1)\, J_0^{(0,\alpha)}\right] = -\overline{A}\, \mathbb{E}_\pi\left[J_\infty^{(1,\alpha)}\right]$. Under the stationary augmented-chain convention, for every k the pair $\left(J_{k-1}^{(0,w)}, Z_k\right)$ has the same law as $\left(J_0^{(0,w)}, Z_1\right)$. Thus the centered function $\overline{\psi}_w$ in Lemma 11 is centered for each summand in the telescoping sum below; the index shift is only notational. Subtracting $n\, \mathbb{E}_\pi\left[J_\infty^{(1,\alpha)}\right]$ from both sides and applying $\overline{A}^{-1}$ gives the **centered telescoping identity**

$$\sum_{k=0}^{n-1}\Big(J_k^{(1,\alpha)}-\mathbb{E}_\pi\Big[J_\infty^{(1,\alpha)}\Big]\Big)=-\overline{A}^{-1}\sum_{k=1}^{n}\overline{\psi}_\alpha\Big(J_{k-1}^{(0,\alpha)},Z_k\Big)+\frac{1}{\alpha}\,\overline{A}^{-1}\Big(J_0^{(1,\alpha)}-J_n^{(1,\alpha)}\Big). \tag{189}$$

This identity is the bridge between the (vector-valued) PR-average of $J^{(1)}$ and the centered bilinear sum bounded in Levin Corollary 6.

Lemma 14. Assume the stationary augmented-chain convention above, **UGE 1**, $\pi\Big(\tilde{A}\Big)=0$, $\pi(\epsilon)=0$, $\|\,\epsilon\,\|_\infty<\infty$, and $\alpha,2\alpha\in(0,\alpha_\infty]$ and $2\alpha\le\alpha_{\mathrm{inv}}$. Set

$$\Phi_+(p,\alpha):=1+p^{3/2}\,t_{\mathrm{mix}}^{1/2}/a+p^{1/2}\,t_{\mathrm{mix}}^{3/2}\sqrt{\alpha/a}. \tag{190}$$

There exists a constant $C_{\mathrm{mis},1}$ depending on $\|\,\overline{A}\,\|,\|\,\overline{A}^{-1}\,\|,\kappa_Q,C_A,\|\,\epsilon\,\|_\infty,t_{\mathrm{mix}}$, the universals $c_{W,1},c_{W,2}$ of Levin Corollary 6, and the constant of the centered bound in Lemma 5, such that for every $p\ge2$ and every $n\ge2$,

$$\begin{aligned}\|\,T_n^{(1)}\,\|_{L_p}&\le C_{\mathrm{mis},1}\sqrt{n}\,\alpha^2+C_{\mathrm{mis},1}\,p^{3/2}\sqrt{\alpha}\\&\quad+C_{\mathrm{mis},1}\,p^3\,(\alpha n)^{-1/2}\log^{1/p}(1/(\alpha a))+C_{\mathrm{mis},1}\,\Phi_+(p,\alpha)\,n^{-1/2}.\end{aligned} \tag{191}$$

Proof. Decompose $T_n^{(1)}=T_n^{(1,\,\mathrm{b})}+T_n^{(1,\,\mathrm{c})}$ via the bias-fluctuation split

$$T_n^{(1,\,\mathrm{b})}:=\sqrt{n}\,\Big(2\,\mathbb{E}_\pi\Big[J_\infty^{(1,\alpha)}\Big]-\mathbb{E}_\pi\Big[J_\infty^{(1,2\alpha)}\Big]\Big), \tag{192}$$

$$T_n^{(1,\,\mathrm{c})}:=\frac{1}{\sqrt{n}}\sum_{k=0}^{n-1}\sum_{w\in\{\alpha,2\alpha\}}c_w\Big(J_k^{(1,w)}-\mathbb{E}_\pi\Big[J_\infty^{(1,w)}\Big]\Big),\quad c_\alpha=2,c_{2\alpha}=-1. \tag{193}$$

The bias is deterministic; the centered piece carries all the L_p fluctuation.

Bias. By Levin Proposition 2 applied at $w\in\{\alpha,2\alpha\}$, $\mathbb{E}_\pi\Big[J_\infty^{(1,w)}\Big]=w\,\Delta+R(w)$ with $\|\,R(w)\,\|\le 12\,\|\,\overline{A}^{-1}\,\|\,C_A^2\,t_{\mathrm{mix}}^2\,w^2\,\|\,\epsilon\,\|_\infty$. The leading $w\,\Delta$ cancels in the RR combination, $2\alpha\,\Delta-2\alpha\,\Delta=0$, leaving

$$\|\,2\,\mathbb{E}_\pi\Big[J_\infty^{(1,\alpha)}\Big]-\mathbb{E}_\pi\Big[J_\infty^{(1,2\alpha)}\Big]\,\|\le2\,\|\,R(\alpha)\,\|+\|\,R(2\alpha)\,\|\le72\,\|\,\overline{A}^{-1}\,\|\,C_A^2\,t_{\mathrm{mix}}^2\,\alpha^2\,\|\,\epsilon\,\|_\infty. \tag{194}$$

Multiplying by $\sqrt{n}$ produces the first term of the bound.

Centered. Apply Eq. 189 at $w\in\{\alpha,2\alpha\}$, take L_p norms, and combine via the triangle inequality:

$$\begin{aligned}\|\,T_n^{(1,\,\mathrm{c})}\,\|_{L_p}\le\frac{1}{\sqrt{n}}\sum_{w\in\{\alpha,2\alpha\}}|c_w|\,\Bigg(&\|\,\overline{A}^{-1}\,\|\cdot\|\sum_{k=1}^{n}\overline{\psi}_w\Big(J_{k-1}^{(0,w)},Z_k\Big)\,\|_{L_p}\\&+\frac{1}{w}\,\|\,\overline{A}^{-1}\,\|\cdot\|\,J_0^{(1,w)}-J_n^{(1,w)}\,\|_{L_p}\Bigg).\end{aligned} \tag{195}$$

For the bilinear sum apply Levin Corollary 6 with $r=n$,

$$\|\sum_{k=1}^{n}\overline{\psi}_w\,\|_{L_p}\le c_{W,1}\,p^{3/2}\sqrt{wn}+c_{W,2}\,p^3\,w^{-1/2}\log^{1/p}(1/(wa)), \tag{196}$$

and divide by $\sqrt{n}$, using $w\in[\alpha,2\alpha]$ to upper-bound by the α-form,

$$\frac{1}{\sqrt{n}}\,\|\sum\overline{\psi}_w\,\|_{L_p}\le\sqrt{2}c_{W,1}\,p^{3/2}\sqrt{\alpha}+\sqrt{2}c_{W,2}\,p^3\,(\alpha n)^{-1/2}\log^{1/p}(1/(\alpha a)). \tag{197}$$

For the boundary term, apply the stationary-limit transfer Lemma 13 to $T_n^{(1,w)}=\Big(I-w\overline{A}\Big)\,J_n^{(1,w)}$ with $w\le\alpha_{\mathrm{inv}}$. This gives

$$\| J_n^{(1,w)} - \mathbb{E}J_n^{(1,w)} \|_{L_p} \le Cw\,\Phi_+(p,w), \tag{198}$$

and $\| \mathbb{E}J_n^{(1,w)} \| \le w \| \Delta \| + \| R(w) \| \le Cw$ by Levin Proposition 2; combining,

$$\| J_n^{(1,w)} \|_{L_p} \le Cw\,\Phi_+(p,w). \tag{199}$$

The same bound holds for $J_0^{(1,w)}$ under stationarity, so $w^{-1} \| J_0^{(1,w)} - J_n^{(1,w)} \|_{L_p} \le C\,\Phi_+(p,w) \le \sqrt{2}\,C\,\Phi_+(p,\alpha)$ after absorbing the factor 2 into C. Dividing by $\sqrt{n}$ produces the last term. Summing the three RR-combined pieces and absorbing universal factors into a single $C_{\text{mis},1}$ completes the proof. □

Lemma 15. Under the assumptions of the previous lemma, for every $p \ge 2$ and every $q \ge 2$ satisfying $p \le q/2$ and $2\alpha \le \alpha_*(q, t_{\text{mix}})$,

$$\| T_n^{(2)} \|_{L_p} + \| T_n^{(H)} \|_{L_p} \le C_{\text{mis},2}\left(1 + d^{1/q}\right) p^{7/2}\, t_{\text{mix}}^{5/2}\, \sqrt{n}\,\alpha^{3/2} \log^{3/2}(1/(\alpha a)), \tag{200}$$

$$\text{with } C_{\text{mis},2} := 6\,\sqrt{2}^{3}\,(D_J + D_H).$$

Proof. Triangle inequality on the n summands of $T_n^{(2)}$, then Levin Propositions 8 and 9 applied uniformly in k and $w \in \{\alpha, 2\alpha\}$:

$$\begin{aligned} \| T_n^{(2)} \|_{L_p} &\le \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \sum_{w\in\{\alpha,2\alpha\}} |c_w| \, \| J_k^{(2,w)} \|_{L_p} \\ &\le 3\,\sqrt{n} \sup_{w\in\{\alpha,2\alpha\}} \| J_k^{(2,w)} \|_{L_p} \\ &\le 3\,D_J\,(2\alpha)^{3/2}\, t_{\text{mix}}^{5/2}\, p^{7/2}\, \sqrt{n} \log^{3/2}(1/(\alpha a)), \end{aligned} \tag{201}$$

where $|c_\alpha| + |c_{2\alpha}| = 3$ absorbs the RR combination and $(2\alpha)^{3/2}$ the worse bound at the larger step. Identical argument for $T_n^{(H)}$, with the additional $d^{1/q}$ factor of Levin Proposition 9. Adding the two bounds gives the lemma. □

Theorem 2. (Stationary PR-averaged RR misadjustment bound.) Assume **UGE 1**, $\pi\left(\tilde{A}\right) = 0$, $\pi(\epsilon) = 0$, $\| \epsilon \|_\infty < \infty$, $\alpha, 2\alpha \in (0, \alpha_\infty]$, $2\alpha \le \alpha_{\text{inv}}$, and the stationary augmented-chain convention above. There exists a constant C depending on the universal and problem constants of the previous two lemmas such that for every $p \ge 2$, every $q \ge 2$ satisfying $p \le q/2$, every $2\alpha \le \alpha_*(q, t_{\text{mix}})$, and every $n \ge 2$,

$$\begin{aligned} \| R_n^{\text{mis, RR}} \|_{L_p} &\le C\sqrt{n}\,\alpha^2 + C\left(1 + d^{1/q}\right) p^{7/2}\, t_{\text{mix}}^{5/2}\, \sqrt{n}\,\alpha^{3/2} \log^{3/2}(1/(\alpha a)) \\ &\quad + C\,p^{3/2}\sqrt{\alpha} + C\,p^3\,(\alpha n)^{-1/2} \log^{1/p}(1/(\alpha a)) + C\,\Phi_+(p,\alpha)\,n^{-1/2}. \end{aligned} \tag{202}$$

Proof. Triangle inequality on Eq. 176: $\| R_n^{\text{mis, RR}} \|_{L_p} \le \| T_n^{(1)} \|_{L_p} + \| T_n^{(2)} \|_{L_p} + \| T_n^{(H)} \|_{L_p}$. Combine the centered $T^{(1)}$ bound and the raw $T^{(2)} + T^{(H)}$ bound. □

Corollary 4. At the working scale $\alpha = c\,n^{-1/2}$, with $p = \max(2, \lceil \log n \rceil)$ and $q = \max(2p, \lceil \log(e\,d) \rceil, 2)$, and for n large enough that $2\alpha \le \alpha_{\text{inv}}$ and $2\alpha \le \alpha_*(q, t_{\text{mix}})$,

$$\| R_n^{\text{mis, RR}} \|_{L_p} \le C\,\text{polylog}(n)\,n^{-1/4}, \tag{203}$$

of the same polynomial order as the leading martingale Berry–Esseen rate, up to logarithmic factors.

Proof. At $\alpha = c\,n^{-1/2}$: $\sqrt{n}\alpha^2 = c^2\,n^{-1/2}$, $\sqrt{n}\alpha^{3/2} = c^{3/2}\,n^{-1/4}$, $\sqrt{\alpha} = c^{1/2}\,n^{-1/4}$, $(\alpha n)^{-1/2} = c^{-1/2}\,n^{-1/4}$, and $\Phi_+(p,\alpha)\,n^{-1/2} = O\left(p^{3/2}\,n^{-1/2}\right)$. With $p \asymp \log n$ and $q \ge \log(ed)$, the factor $d^{1/q}$ is bounded by a universal constant, and the dominant order in the stationary misadjustment bound is $\text{polylog}(n)\,n^{-1/4}$. □

The theorem just proved is only a stationary augmented-chain misadjustment bound at $n_0 = 0$. A finite-start theorem with burn-in requires the burned-in weights $Q_{l,n_0}^{(\alpha)}$ and is not used in the assembly below.

4.11. Smoothing Assembly

The previous sections produced the four ingredients of the stationary $n_0 = 0$ Berry–Esseen program for the PR-averaged Richardson–Romberg statistic:

1. the **Poisson decomposition** $W^{\mathrm{RR}} = -n^{-1/2} M_n^{\mathrm{RR}} + D_{2,n}^{\mathrm{RR}}$ with deterministic sup-norm control of $D_{2,n}^{\mathrm{RR}}$ at the order $t_{\mathrm{mix}}/(a^2\sqrt{n})$;
2. the **predictable quadratic variation concentration** of $u^\top \langle M^{\mathrm{RR}} \rangle_n u$ around $n\,\sigma_n^{2,\,\mathrm{RR}}(u)$ in L_p at the order $\sqrt{p\,n}$;
3. the **martingale Berry–Esseen** for $u^\top M_n^{\mathrm{RR}}/(\sqrt{n}\,\sigma_n^{\mathrm{RR}}(u))$ at rate $\log^{3/4}(n)\,n^{-1/4}$;
4. the **Levin depth-two misadjustment bound** on $R_n^{\mathrm{mis,\,RR}}$ controlling the non-martingale residual at the same $\log^{c(n)} n^{-1/4}$ rate.

The smoothing inequality of Bobkov–Götze (Samsonov et al. 2025, Proposition 12) assembles these four into a single Kolmogorov bound on the stationary $n_0 = 0$ augmented-chain RR statistic.

Stationary assembled statistic. Combining the depth-one identity in Section 4.1 with the Poisson decomposition of Section 4.6 gives a finite-start identity. Since the transfer from a zero-start recursion to the stationary augmented chain is not proved here, the bound below is stated for the stationary $n_0 = 0$ assembled scalar statistic

$$S_{n,\,\mathrm{stat}}^{\mathrm{RR}}(u) := -\frac{u^\top M_n^{\mathrm{RR}}}{\sqrt{n}} + u^\top \mathcal{R}_{n,\,\mathrm{stat}}^{\mathrm{RR}}, \tag{204}$$

$$\mathcal{R}_{n,\,\mathrm{stat}}^{\mathrm{RR}} := D_{2,n}^{\mathrm{RR}} + R_n^{\mathrm{mis,\,RR}}, \tag{205}$$

where $D_{2,n}^{\mathrm{RR}}$ is the Poisson boundary/Abel remainder of Section 4.6 and $R_n^{\mathrm{mis,\,RR}}$ is the Levin depth-two misadjustment defined in Eq. 171, both read under the stationary augmented-chain convention. The deterministic-start transient $D_{\mathrm{tr}}^{\mathrm{RR}} := 2\,D_{\mathrm{tr}}^{(\alpha)} - D_{\mathrm{tr}}^{(2\alpha)}$ and the random initial-product discrepancy $2D_{\mathrm{init}}^{(\alpha)} - D_{\mathrm{init}}^{(2\alpha)}$ are not part of this stationary result; handling them together with the accumulated startup error belongs to the finite-start/burn-in transfer theorem. Dividing by $\sigma_n^{\mathrm{RR}}(u)$,

$$\frac{S_{n,\,\mathrm{stat}}^{\mathrm{RR}}(u)}{\sigma_n^{\mathrm{RR}}(u)} = X_n + Y_n, \tag{206}$$

$$X_n := -\frac{u^\top M_n^{\mathrm{RR}}}{\sqrt{n}\,\sigma_n^{\mathrm{RR}}(u)}, \quad Y_n := \frac{u^\top \mathcal{R}_{n,\,\mathrm{stat}}^{\mathrm{RR}}}{\sigma_n^{\mathrm{RR}}(u)}. \tag{207}$$

The martingale Berry–Esseen bound controls $d_{K(X_n,\mathcal{N}(0,1))}$ by applying Theorem 1 to the signed direction $-u$. Equivalently, use the martingale differences $-u^\top \Delta M_l^{\mathrm{RR}}$; their bounded-increment constant and predictable variance are the same as for u, because $\varkappa(-u) = \varkappa(u)$ and $\sigma_n^{2,\,\mathrm{RR}}(-u) = \sigma_n^{2,\,\mathrm{RR}}(u)$.

Smoothing inequality. For random variables X, Y on the same probability space and any $t > 0$,

$$d_K(X+Y, \mathcal{N}(0,1)) \le d_K(X, \mathcal{N}(0,1)) + \mathbb{P}(|Y| > t) + \frac{t}{\sqrt{2\pi}}, \tag{208}$$

where the third term uses the $1/\sqrt{2\pi}$ Lipschitz constant of the standard normal cdf (Bobkov–Götze; see Samsonov et al. 2025, Proposition 12 for an LSA-tailored statement). Bounding $\mathbb{P}(|Y| > t) \le \|\, Y \,\|_{L_p}^p \,/t^p$ via Markov's inequality and choosing $t = e \,\|\, Y \,\|_{L_p}$ gives the **working form**

$$d_K(X+Y, \mathcal{N}(0,1)) \le d_K(X, \mathcal{N}(0,1)) + \frac{e \,\|\, Y \,\|_{L_p}}{\sqrt{2\pi}} + e^{-p}, \tag{209}$$

in which the trailing tail probability e^{-p} is absorbed into $O(1/n)$ as soon as $p \ge \log n$.

***L_p*-bound on the composite remainder.** The two pieces of $\mathcal{R}^{\mathrm{RR}}_{n,\,\mathrm{stat}}$ contribute additively. The finite-start deterministic transient displayed in Section 4.1 is deliberately excluded: setting $\theta_0 = \theta^*$ would remove only that deterministic term, but not the startup discrepancy between zero-start perturbation variables and the stationary augmented chain.

Lemma 16. Assume the hypotheses and step-size restrictions of the stationary PR-averaged RR misadjustment bound above. Then for every $u \in \mathbb{R}^d$, every $p \geq 2$, and every admissible $q \geq 2$,

$$\begin{aligned}\| \, u^\top \mathcal{R}^{\mathrm{RR}}_{n,\,\mathrm{stat}} \, \|_{L_p} \leq \frac{C_{\mathrm{D2}} \, \| \, u \, \|}{\sqrt{n}} + \| \, u \, \| \, C_{\mathrm{mis}} \Big(&\sqrt{n}\, \alpha^2 + \big(1 + d^{1/q}\big)\, p^{7/2}\, t_{\mathrm{mix}}^{5/2} \sqrt{n}\, \alpha^{3/2} \log^{3/2}(1/(\alpha a)) \\ &+ p^{3/2} \sqrt{\alpha} + p^3\, (\alpha n)^{-1/2} \log^{1/p}(1/(\alpha a)) + \Phi_+(p,\alpha)\, n^{-1/2} \Big),\end{aligned} \tag{210}$$

with constants

$$C_{\mathrm{D2}} \coloneqq 3\, t_{\mathrm{mix}} \, \| \, \epsilon \, \|_\infty \, (C_Q + C_2/a^2), \tag{211}$$

and C_{mis} the constant in the misadjustment theorem.

Proof. Triangle inequality on $u^\top \mathcal{R}^{\mathrm{RR}}_{n,\,\mathrm{stat}}$:

(a) The deterministic sup-norm bound for $D^{\mathrm{RR}}_{2,n}$ yields $\| \, u^\top D^{\mathrm{RR}}_{2,n} \, \|_{L_p} \leq \| \, u \, \| \, \| \, D^{\mathrm{RR}}_{2,n} \, \|_\infty \leq C_{\mathrm{D2}} \, \| \, u \, \| \, / \sqrt{n}$ for every $p \geq 1$.

(b) The misadjustment theorem gives the remaining summands directly. □

Theorem 3. (Stationary augmented-chain Berry–Esseen assembly for the RR comparison statistic.) Assume **UGE 1**, $\pi\big(\tilde{A}\big) = 0$, $\pi(\epsilon) = 0$, $\| \, \epsilon \, \|_\infty < \infty$, $\sigma^2(u) > 0$, $\alpha, 2\alpha \in (0, \alpha_\infty]$, and the stationary augmented-chain convention above. Use the imported inputs summarized in Section 4.1. Set $p = \max(2, \lceil \log n \rceil)$ and $q = \max(2p, \lceil \log(e\, d) \rceil, 2)$. If n is large enough that the variance lower-bound condition Eq. 159 holds and $2\alpha \leq \alpha_{\mathrm{inv}}$ and $2\alpha \leq \alpha_*(q, t_{\mathrm{mix}})$, then

$$\begin{aligned} d_K \left(\frac{S^{\mathrm{RR}}_{n,\,\mathrm{stat}}(u)}{\sigma^{\mathrm{RR}}_n(u)}, \mathcal{N}(0,1) \right) \leq \frac{C_{K,1}(u) \log^{3/4} n}{n^{1/4}} &+ \frac{C_{K,2}(u) \log n}{\sqrt{n}} \\ &+ \frac{e \, \| \, u^\top \mathcal{R}^{\mathrm{RR}}_{n,\,\mathrm{stat}} \, \|_{L_p}}{\sqrt{2\pi}\, \sigma^{\mathrm{RR}}_n(u)} + \frac{e}{n}, \end{aligned} \tag{212}$$

with $\| \, u^\top \mathcal{R}^{\mathrm{RR}}_{n,\,\mathrm{stat}} \, \|_{L_p}$ bounded by the preceding lemma, and $C_{K,1}(u), C_{K,2}(u)$ the constants of the martingale Berry–Esseen theorem.

Proof. Apply the smoothing inequality with $X = X_n$ and $Y = Y_n$ from the displayed decomposition above. The first two terms come from Theorem 1 applied to the signed scalar martingale with increments $-u^\top \Delta M^{\mathrm{RR}}_l$; the constants are unchanged since the theorem's direction-dependent quantities are invariant under $u \to -u$. The remainder term is $e \, \| \, Y_n \, \|_{L_p} / \sqrt{2\pi} = e \, \| \, u^\top \mathcal{R}^{\mathrm{RR}}_{n,\,\mathrm{stat}} \, \|_{L_p} / \big(\sqrt{2\pi}\, \sigma^{\mathrm{RR}}_n(u) \big)$. Since $p = \max(2, \lceil \log n \rceil) \geq \log n$, the trailing e^{-p} is at most n^{-1}. □

Corollary 5. Let $\alpha = \alpha_n$ be an admissible step-size sequence and set $p_n := \max(2, \lceil \log n \rceil)$, $q_n := \max(2p_n, \lceil \log(e\,d) \rceil, 2)$, and $\Lambda_n := \log(1/(\alpha_n a))$. Assume the step-size restrictions $2\alpha_n \le \alpha_\infty$, $2\alpha_n \le \alpha_{\mathrm{inv}}$, and $2\alpha_n \le \alpha_*(q_n, t_{\mathrm{mix}})$, and the variance lower-bound condition Eq. 159. If

$$p_n^3 (n\alpha_n)^{-1/2} \Lambda_n^{1/p_n} \to 0, \quad p_n^{7/2} \sqrt{n}\, \alpha_n^{3/2} \Lambda_n^{3/2} \to 0, \tag{213}$$

then

$$d_K\left(\frac{S_{n,\,\mathrm{stat}}^{\mathrm{RR}}(u)}{\sigma_n^{\mathrm{RR}}(u)}, \mathcal{N}(0,1) \right) \to 0. \tag{214}$$

In particular, for a power scale $\alpha_n = c\,n^{-\gamma}$, up to logarithmic factors, the admissible window is

$$\frac{1}{3} < \gamma < 1. \tag{215}$$

Proof. Apply the stationary $n_0 = 0$ Berry–Esseen bound and the preceding remainder lemma with $p = p_n$, $q = q_n$, and $\alpha = \alpha_n$. The Poisson boundary term is $O(n^{-1/2})$. The bias term $\sqrt{n}\alpha_n^2$ is smaller than $\sqrt{n}\alpha_n^{3/2}$ for all sufficiently large n, because the admissibility conditions imply $\alpha_n \to 0$. The terms $p_n^{3/2}\sqrt{\alpha_n}$ and $\Phi_+(p_n, \alpha_n) n^{-1/2}$ are also negligible under the second admissibility condition. The remaining nontrivial terms are exactly the two quantities displayed in the statement. The martingale Berry–Esseen terms $\log^{3/4}(n) n^{-1/4}$ and $\log(n) n^{-1/2}$ vanish independently of α_n. For $\alpha_n = cn^{-\gamma}$, the two displayed conditions reduce, modulo logarithms, to $n^{-(1-\gamma)/2} \to 0$ and $n^{1/2 - 3\gamma/2} \to 0$, i.e. $\gamma < 1$ and $\gamma > 1/3$. □

Corollary 6. At the balanced scale $\alpha = c\,n^{-1/2}$ with $c > 0$ such that $\alpha, 2\alpha \in (0, \alpha_\infty]$ and $2\alpha \le \alpha_{\mathrm{inv}}$, with $p = \max(2, \lceil \log n \rceil)$ and $q = \max(2p, \lceil \log(e\,d) \rceil, 2)$ satisfying $2\alpha \le \alpha_*(q, t_{\mathrm{mix}})$, the bound Eq. 212 reduces to

$$d_K\left(\frac{S_{n,\,\mathrm{stat}}^{\mathrm{RR}}(u)}{\sigma_n^{\mathrm{RR}}(u)}, \mathcal{N}(0,1) \right) \le \frac{C(u)\,\mathrm{polylog}(n)}{n^{1/4}}, \tag{216}$$

where $C(u)$ depends on $\| u \|$, $\sigma(u)$, C_Q, $\| \overline{A} \|$, $\| \overline{A}^{-1} \|$, κ_Q, C_A, $\| \epsilon \|_\infty$, $\| \Sigma_\epsilon^{(\mathrm{M})} \|$, t_{mix}, a, α_∞, c, and the universal constants of the smoothing, Bolthausen–Fan, and Levin Markov-concentration inequalities.

Proof. The scale $\alpha = c\,n^{-1/2}$ satisfies Eq. 213, but gives the sharper balanced rate. Indeed, $C_{\mathrm{D2}} \| u \| / \sqrt{n} = O(n^{-1/2})$; $\sqrt{n}\alpha^2 = c^2\,n^{-1/2}$; $p^{7/2}\sqrt{n}\alpha^{3/2} \log^{3/2}(1/(\alpha a)) = O(\mathrm{polylog}(n) n^{-1/4})$; $p^{3/2}\sqrt{\alpha} = O(\mathrm{polylog}(n) n^{-1/4})$; $p^3 (\alpha n)^{-1/2} \log^{1/p}(1/(\alpha a)) = O(\mathrm{polylog}(n) n^{-1/4})$; and $\Phi_+(p, \alpha) n^{-1/2} = O(\mathrm{polylog}(n) n^{-1/2})$. After division by $\sigma_n^{\mathrm{RR}}(u) \ge \sigma(u)/\sqrt{2}$, these terms are dominated by $\mathrm{polylog}(n) n^{-1/4}$. Combining with the martingale Berry–Esseen term proves the claim. □

Corollary 7. Under the hypotheses of the stationary $n_0 = 0$ Berry–Esseen bound above, the same finite-n bound with asymptotic normalisation contains one additional variance-comparison term:

$$\begin{aligned} d_K\left(\frac{S_{n,\,\mathrm{stat}}^{\mathrm{RR}}(u)}{\sigma(u)}, \mathcal{N}(0,1) \right) &\le \frac{C_{K,1}(u) \log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u) \log n}{\sqrt{n}} \\ &\quad + \frac{e \| u^\top \mathcal{R}_{n,\,\mathrm{stat}}^{\mathrm{RR}} \|_{L_p}}{\sqrt{2\pi}\,\sigma_n^{\mathrm{RR}}(u)} + \frac{e}{n} + \frac{C \| u \|^2}{n\,\alpha\,a\,\sigma^2(u)}. \end{aligned} \tag{217}$$

Consequently, under the balanced-scale hypotheses above,

$$d_K\left(\frac{S_{n,\,\mathrm{stat}}^{\mathrm{RR}}(u)}{\sigma(u)}, \mathcal{N}(0,1) \right) \le \frac{C'(u)\,\mathrm{polylog}(n)}{n^{1/4}}. \tag{218}$$

Proof. Set $r := \sigma_n^{\mathrm{RR}}(u)/\sigma(u)$ and write $W := S_{n,\,\mathrm{stat}}^{\mathrm{RR}}(u)/\sigma_n^{\mathrm{RR}}(u)$, so $Wr = S_{n,\,\mathrm{stat}}^{\mathrm{RR}}(u)/\sigma(u)$. Under the variance lower-bound condition Eq. 159, $r \in \left[1/\sqrt{2}, r_{\max}\right]$ for some finite $r_{\max}$ (from the trivial upper bound $\sigma_n^{2,\,\mathrm{RR}}(u) \leq C_{\mathcal{Q}}^2 \parallel \Sigma \parallel \parallel u \parallel^2$). For r in this compact interval,

$$\sup_x | \ \Phi(x/r) - \Phi(x) \ | \leq C_\Phi \, |r-1|, \quad C_\Phi := \sqrt{2}/\sqrt{\pi e}, \tag{219}$$

because $\sup_x |x \, \varphi(x)| = 1/\sqrt{2\pi e}$. Hence

$$d_K(Wr, \mathcal{N}(0,1)) \leq d_K(W, \mathcal{N}(0,1)) + C_\Phi \, |r-1|. \tag{220}$$

The variance comparison of Section 4.5 gives

$$|r-1| = \frac{|\sigma_n^{2,\,\mathrm{RR}}(u) - \sigma^2(u)|}{\sigma(u)\,(\sigma_n^{\mathrm{RR}}(u) + \sigma(u))} \leq \frac{C_3 \parallel u \parallel^2}{n \, \alpha \, a \, \sigma^2(u)}, \tag{221}$$

using $\sigma_n^{\mathrm{RR}}(u) + \sigma(u) \geq \sigma(u)$. This proves the displayed asymptotic-normalisation bound. At $\alpha = cn^{-1/2}$ the final term is $O\left(n^{-1/2}\right)$, hence it is absorbed into the balanced-scale bound of the preceding corollary. □

The bound above is therefore an $n_0 = 0$ stationary augmented-chain statement for $S_{n,\,\mathrm{stat}}^{\mathrm{RR}}(u)$, not the deterministic-start RR average itself. A deterministic-start theorem with mixing-scale burn-in and logarithmic factors requires a separate transfer theorem with $Q_{l,n_0}^{(\alpha)}$ and the corresponding Poisson, variance-comparison, and misadjustment bounds.

5. Burn-in Transfer for Deterministic Starts

The previous chapter proves a stationary $n_0 = 0$ Berry–Esseen bound for $S^{\mathrm{RR}}_{n,\,\mathrm{stat}}(u)$. This chapter transfers that bound to the deterministic-start Richardson–Romberg average after burn-in. Burn-in changes the deterministic PR kernels, so the result is not obtained by simply inserting $n_0 > 0$ into the stationary theorem.

Throughout this chapter, constants with a `"burn"` subscript are independent of n, n_0, m, α, p, and q unless these variables appear as arguments. Dependencies on fixed problem constants and imported Levin/startup constants are absorbed into named constants; powers of a, t_{mix}, p, q, and d are kept explicit when they affect the final rate.

5.1. Target Statistic and Normalization

Let $0 \le n_0 < n$ and set $m := n - n_0$. The burned-in PR average is

$$\overline{\theta}^{(\alpha)}_{n,n_0} := \frac{1}{m} \sum_{k=n_0}^{n-1} \theta^{(\alpha)}_k, \tag{222}$$

and the two-level RR average is

$$\overline{\theta}^{(\mathrm{RR},\alpha)}_{n,n_0} := 2\overline{\theta}^{(\alpha)}_{n,n_0} - \overline{\theta}^{(2\alpha)}_{n,n_0}. \tag{223}$$

Both stepsizes are run from the same deterministic initial point θ_0 and on the same Markov trajectory. This coupling is part of the RR statistic; if the two levels use different paths, the deterministic-weight decomposition below is a different object.

The finite-start burned-in vector statistic is

$$\mathcal{T}^{\mathrm{RR}}_{n,n_0} := \sqrt{m}\left(\overline{\theta}^{(\mathrm{RR},\alpha)}_{n,n_0} - \theta^*\right). \tag{224}$$

Its scalar projection in direction $u \in \mathbb{R}^d$ is

$$T^{\mathrm{RR}}_{n,n_0}(u) := u^\top \mathcal{T}^{\mathrm{RR}}_{n,n_0} = \sqrt{m}\, u^\top \left(\overline{\theta}^{(\mathrm{RR},\alpha)}_{n,n_0} - \theta^*\right). \tag{225}$$

The stationary augmented-chain statistic from the previous chapter is used only as a comparison object after the startup transfer.

There are two normalizations. The finite-window normalization is

$$\sigma^{\mathrm{bRR}}_{n,n_0}(u) := \sqrt{\sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u)}, \quad \Xi^{\mathrm{bRR}}_{n,n_0}(u) := \frac{T^{\mathrm{RR}}_{n,n_0}(u)}{\sigma^{\mathrm{bRR}}_{n,n_0}(u)}, \tag{226}$$

where $\sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u)$ is the deterministic variance proxy defined in Eq. 262. The asymptotic normalization is

$$\sigma(u) := \sqrt{u^\top \Sigma_\infty u}, \quad \Xi^{\mathrm{asy,RR}}_{n,n_0}(u) := \frac{T^{\mathrm{RR}}_{n,n_0}(u)}{\sigma(u)}. \tag{227}$$

We first control $\Xi^{\mathrm{bRR}}_{n,n_0}(u)$ and then pass to $\Xi^{\mathrm{asy,RR}}_{n,n_0}(u)$. A final corollary converts the $\sqrt{m}$ statistic to the final $\sqrt{n}$ statistic when the burn-in is at the mixing scale, of order α^{-1} times logarithmic factors.

5.2. Burned-in Deterministic Weights

With the $\sqrt{m}$ normalization, the depth-zero linearized term has the weights

$$Q^{(\alpha)}_{l;n_0,n} := \alpha \sum_{k=\max(n_0,l)}^{n-1} B^{k-l}_\alpha, \quad 1 \le l \le n-1, \tag{228}$$

and $Q^{(\alpha)}_{l;n_0,n} = 0$ for $l \ge n$. The RR weight is

$$Q^{\mathrm{RR}}_{l;n_0,n} := 2Q^{(\alpha)}_{l;n_0,n} - Q^{(2\alpha)}_{l;n_0,n}. \tag{229}$$

Thus the leading burned-in sum is

$$W^{\mathrm{RR}}_{n,n_0} := -\frac{1}{\sqrt{m}} \sum_{l=1}^{n-1} Q^{\mathrm{RR}}_{l;n_0,n} \epsilon(Z_l). \tag{230}$$

For $l \geq n_0$ these weights are full-window weights with horizon $n - l$. For $l < n_0$, the lower summation limit is n_0 rather than l, so the weight comparison, Poisson decomposition, and variance proxy must be restated for $Q^{\mathrm{RR}}_{l;n_0,n}$.

5.3. Closed Forms and Burned-in Weight Bounds

Write $B_w := I - w\overline{A}$ for $w \in \{\alpha, 2\alpha\}$ and recall $m = n - n_0$. The burned-in weights have two closed forms. If $l \geq n_0$, then

$$Q^{(w)}_{l;n_0,n} = w \sum_{k=l}^{n-1} B_w^{k-l} = \overline{A}^{-1}\big(I - B_w^{n-l}\big). \tag{231}$$

If $l < n_0$, then

$$Q^{(w)}_{l;n_0,n} = w \sum_{k=n_0}^{n-1} B_w^{k-l} = \overline{A}^{-1}\big(B_w^{n_0-l} - B_w^{n-l}\big). \tag{232}$$

Thus only the post-burn-in weights are close to the asymptotic kernel $\overline{A}^{-1}$. The pre-burn-in weights are instead exponentially small as one moves backward from n_0. Empty sums below are interpreted as zero.

Lemma 17. Assume $\alpha, 2\alpha \in (0, \alpha_\infty]$ and the Lyapunov contraction Eq. 12. Let

$$\rho_\alpha := \sqrt{1 - \alpha a}, \quad C_Q := \kappa_Q^{1/2} \parallel \overline{A}^{-1} \parallel, \quad \tilde{C}_A := \kappa_Q \parallel \overline{A} \parallel. \tag{233}$$

(i) Post-burn-in comparison. If $l \geq n_0$ and $k := n - l$, then

$$\parallel Q^{\mathrm{RR}}_{l;n_0,n} - \overline{A}^{-1} \parallel \leq 3C_Q \rho_\alpha^k. \tag{234}$$

(ii) Pre-burn-in energy. If $1 \leq l < n_0$ and $r := n_0 - l$, then

$$\parallel Q^{\mathrm{RR}}_{l;n_0,n} \parallel \leq 6C_Q \rho_\alpha^r. \tag{235}$$

Consequently, there are constants $C_{\mathrm{burn,E}}$ and $C_{\mathrm{burn,V}}$, depending only on κ_Q, $\parallel \overline{A}^{-1} \parallel$, and $\parallel \overline{A} \parallel$, such that, uniformly in n and n_0,

$$\sum_{l=1}^{n_0-1} \parallel Q^{\mathrm{RR}}_{l;n_0,n} \parallel^2 + \sum_{l=\max(1,n_0)}^{n-1} \parallel Q^{\mathrm{RR}}_{l;n_0,n} - \overline{A}^{-1} \parallel^2 \leq \frac{C_{\mathrm{burn,E}}}{\alpha a}, \tag{236}$$

and

$$\sum_{l=1}^{n-2} \parallel Q^{\mathrm{RR}}_{l+1;n_0,n} - Q^{\mathrm{RR}}_{l;n_0,n} \parallel \leq \frac{C_{\mathrm{burn,V}}}{a^2}. \tag{237}$$

Proof. For $l \geq n_0$, Eq. 231 gives

$$Q^{\mathrm{RR}}_{l;n_0,n} - \overline{A}^{-1} = -\overline{A}^{-1}\big(2B_\alpha^k - B_{2\alpha}^k\big). \tag{238}$$

The same contraction argument as in the full-window estimate gives Eq. 234.

For $l < n_0$, set $r := n_0 - l$. By Eq. 232,

$$Q^{\mathrm{RR}}_{l;n_0,n} = \overline{A}^{-1}\left(2(B^r_\alpha - B^{r+m}_\alpha) - (B^r_{2\alpha} - B^{r+m}_{2\alpha})\right). \tag{239}$$

Taking norms and using $\| B^j_{2\alpha} \|_Q \le \rho^j_\alpha$ gives Eq. 235. Squaring Eq. 234 and Eq. 235 and summing the resulting geometric series gives Eq. 236.

It remains to bound the total variation. In the post-burn-in region $l, l+1 \ge n_0$, the full-window identity gives

$$Q^{\mathrm{RR}}_{l+1;n_0,n} - Q^{\mathrm{RR}}_{l;n_0,n} = -2\alpha(B^s_\alpha - B^s_{2\alpha}), \quad s := n - l - 1. \tag{240}$$

The elementary identity $X^s - Y^s = (X - Y)\sum_{i=1}^{s} X^{i-1}Y^{s-i}$ with $X = B_\alpha$ and $Y = B_{2\alpha}$ yields, for $s \ge 1$,

$$\| B^s_\alpha - B^s_{2\alpha} \| \le \alpha \tilde{C}_A \, s \, \rho^{s-1}_\alpha. \tag{241}$$

The left-factored form is legitimate because B_α and $B_{2\alpha}$ are polynomials in the same matrix $\overline{A}$, hence commute with each other and with $\overline{A}$. Thus the post-burn-in contribution is bounded by a constant times a^{-2}, as in the full-window proof.

For $l, l+1 < n_0$, put $s := n_0 - l - 1$. The pre-burn-in closed form gives

$$Q^{\mathrm{RR}}_{l+1;n_0,n} - Q^{\mathrm{RR}}_{l;n_0,n} = 2\alpha(B^s_\alpha - B^s_{2\alpha} - B^{s+m}_\alpha + B^{s+m}_{2\alpha}). \tag{242}$$

Here $s \ge 1$. Applying Eq. 241 to the powers s and $s+m$ and summing over $s \ge 1$ again gives a constant times a^{-2}. If the boundary $l = n_0 - 1$ exists, then

$$Q^{\mathrm{RR}}_{n_0;n_0,n} - Q^{\mathrm{RR}}_{n_0-1;n_0,n} = 2\alpha(B^m_{2\alpha} - B^m_\alpha), \tag{243}$$

and Eq. 241 with $s = m$ bounds this term by the same a^{-2} scale. Combining the pre-burn-in, boundary, and post-burn-in contributions proves Eq. 237. □

5.4. Deterministic Transient After Burn-in

For a generic step size $w \in \{\alpha, 2\alpha\}$ define the deterministic transient

$$D^{(w)}_{\mathrm{tr},n,n_0} := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} B^k_w(\theta_0 - \theta^*), \quad B_w := I - w\overline{A}. \tag{244}$$

The Richardson–Romberg transient entering Eq. 225 is

$$D^{\mathrm{RR}}_{\mathrm{tr},n,n_0}(u) := u^\top\left(2D^{(\alpha)}_{\mathrm{tr},n,n_0} - D^{(2\alpha)}_{\mathrm{tr},n,n_0}\right). \tag{245}$$

Lemma 18. Assume $\alpha, 2\alpha \in (0, \alpha_\infty]$ and the Lyapunov contraction Eq. 12. Then, for each $w \in \{\alpha, 2\alpha\}$,

$$\| D^{(w)}_{\mathrm{tr},n,n_0} \| \le \frac{2\sqrt{\kappa_Q}}{wa\sqrt{m}}(1 - wa)^{n_0/2} \| \theta_0 - \theta^* \|. \tag{246}$$

Consequently,

$$| D^{\mathrm{RR}}_{\mathrm{tr},n,n_0}(u) | \le \frac{5\sqrt{\kappa_Q} \| u \| \| \theta_0 - \theta^* \|}{\alpha a\sqrt{m}}(1 - \alpha a)^{n_0/2}. \tag{247}$$

Proof. Let $r_w := \sqrt{1 - wa}$. By Eq. 12 and equivalence of the Euclidean and Q-norms,

$$\| B^k_w(\theta_0 - \theta^*) \| \le \sqrt{\kappa_Q} r^k_w \| \theta_0 - \theta^* \|. \tag{248}$$

Therefore

$$\| D^{(w)}_{\mathrm{tr},n,n_0} \| \le \frac{\sqrt{\kappa_Q}\ \| \theta_0 - \theta^* \|}{\sqrt{m}} \sum_{k=n_0}^{n-1} r_w^k \le \frac{\sqrt{\kappa_Q}\ \| \theta_0 - \theta^* \|}{\sqrt{m}} \frac{r_w^{n_0}}{1 - r_w}. \tag{249}$$

Since $1 - \sqrt{1-x} \ge x/2$ for $0 \le x \le 1$, the last display gives Eq. 246. Applying this estimate at $w = \alpha$ and $w = 2\alpha$, using $1 - 2\alpha a \le 1 - \alpha a$, and taking the triangle inequality in Eq. 245 gives

$$| D^{\mathrm{RR}}_{\mathrm{tr},n,n_0}(u) | \le \| u \| \left(2 \| D^{(\alpha)}_{\mathrm{tr},n,n_0} \| + \| D^{(2\alpha)}_{\mathrm{tr},n,n_0} \|\right) \le \frac{5\sqrt{\kappa_Q}\ \| u \| \| \theta_0 - \theta^* \|}{\alpha a \sqrt{m}} (1 - \alpha a)^{n_0/2}. \tag{250}$$

□

Corollary 8. (Mixing-scale burn-in removes the deterministic transient.) If, for some $\beta > 0$,

$$n_0 \ge \frac{2\beta}{\alpha a} \log n, \tag{251}$$

then

$$| D^{\mathrm{RR}}_{\mathrm{tr},n,n_0}(u) | \le \frac{5\sqrt{\kappa_Q}\ \| u \| \| \theta_0 - \theta^* \|}{\alpha a \sqrt{m}} n^{-\beta}. \tag{252}$$

In particular, at the balanced scale $\alpha = cn^{-1/2}$ and $m \ge n/2$, this is $O\big(n^{-\beta}\big)$. Taking $\beta = 1$ makes the deterministic transient negligible relative to the stationary $n^{-1/4}$ Berry–Esseen rate.

Proof. The elementary inequality $(1 - \alpha a)^{n_0/2} \le \exp(-\alpha a n_0/2)$ and Eq. 251 imply $(1 - \alpha a)^{n_0/2} \le n^{-\beta}$. Substitute this into Eq. 247. If $\alpha = cn^{-1/2}$ and $m \ge n/2$, then $\big(\alpha\sqrt{m}\big)^{-1} \le \sqrt{2}/c$, so the displayed $O\big(n^{-\beta}\big)$ bound follows. □

The deterministic transient is not the full initial-condition contribution: the exact recursion contains the random product $\Gamma^{(w)}_{1:k} := \prod_{j=1}^{k}\big(I - wA\big(Z_j\big)\big)$. The difference between this product and B_w^k is another finite-start term. We use the same random-product stability input that appears in Levin et al. (2025, Appendix D.1, Proposition 9). Let $\alpha_{\mathrm{st}}(p)$ denote the minimum of the Levin depth-two startup ceiling and the product-stability ceiling at moment order $2p$; the startup section below uses the same threshold.

Lemma 19. (Imported random-product stability.) Under the stability and bounded-noise assumptions used in Levin et al. (2025, Appendix D.1, Proposition 9), if $2\alpha \le \alpha_{\mathrm{st}}(p)$, then there exist constants $C_{\mathrm{prod}} < \infty$ and $c_{\mathrm{prod}} > 0$ such that, for every $p \ge 2$, every $w \in \{\alpha, 2\alpha\}$, every $0 \le s < k$, and every $\mathcal{F}_s$-measurable vector V_s,

$$\| \Gamma^{(w)}_{s+1:k} V_s \|_{L_p} \le C_{\mathrm{prod}} \exp\big(-c_{\mathrm{prod}} w a (k - s)/p\big) \| V_s \|_{L_{2p}}. \tag{253}$$

Define the accumulated RR random initial-product discrepancy by

$$\mathcal{J}^{\mathrm{init,RR}}_{n,n_0}(u) := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} u^\top \left[2\left(\Gamma^{(\alpha)}_{1:k} - B^k_\alpha\right) - \left(\Gamma^{(2\alpha)}_{1:k} - B^k_{2\alpha}\right)\right](\theta_0 - \theta^*), \tag{254}$$

where the empty product at $k = 0$ is the identity.

Lemma 20. (Burned-in random initial-product transient.) Assume the hypotheses of Lemma 19, $\alpha, 2\alpha \in (0, \alpha_\infty]$, $\alpha a \le 1/4$, and the Lyapunov contraction Eq. 12. Then, for every $p \ge 2$,

$$\| \mathcal{J}^{\mathrm{init,RR}}_{n,n_0}(u) \|_{L_p} \le \frac{C_{\mathrm{init,RR}}\ \|u\|\ \|\theta_0 - \theta^*\|\ p}{\alpha a \sqrt{m}} \exp(-c_{\mathrm{init}} \alpha a n_0/p), \tag{255}$$

where $C_{\mathrm{init,RR}} < \infty$ and $c_{\mathrm{init}} > 0$ depend only on C_{prod}, c_{prod}, and the Lyapunov norm-equivalence constant.

Proof. Let $e_0 := \theta_0 - \theta^*$. By Lemma 19 with $s = 0$ and $V_0 = e_0$,

$$\| \Gamma^{(w)}_{1:k} e_0 \|_{L_p} \le C_{\mathrm{prod}} \exp\bigl(-c_{\mathrm{prod}} wak/p\bigr) \|e_0\|. \tag{256}$$

The deterministic Lyapunov contraction gives

$$\| B^k_w e_0 \| \le \sqrt{\kappa_Q}(1-wa)^{k/2} \|e_0\| \le \sqrt{\kappa_Q} \exp(-wak/(2p)) \|e_0\|, \tag{257}$$

since $p \ge 1$. Hence, after decreasing the exponential constant,

$$\| \left(\Gamma^{(w)}_{1:k} - B^k_w\right) e_0 \|_{L_p} \le C \exp(-c_{\mathrm{init}} wak/p) \|e_0\|. \tag{258}$$

Apply this estimate at $w = \alpha$ and $w = 2\alpha$, use the triangle inequality in Eq. 254, and extend the geometric sum to infinity:

$$\| \mathcal{J}^{\mathrm{init,RR}}_{n,n_0}(u) \|_{L_p} \le \frac{C \|u\| \|e_0\|}{\sqrt{m}} \sum_{k=n_0}^{\infty} \exp(-c_{\mathrm{init}} \alpha a k/p). \tag{259}$$

Because $\alpha a \le 1/4$ and $p \ge 2$, $1 - \exp(-c_{\mathrm{init}}\alpha a/p) \ge C^{-1}\alpha a/p$, which gives Eq. 255. □

Corollary 9. (Mixing-scale burn-in removes the random initial-product discrepancy.) If, for some $\beta > 0$,

$$n_0 \ge \frac{\beta p}{c_{\mathrm{init}}\alpha a} \log n, \tag{260}$$

then

$$\| \mathcal{J}^{\mathrm{init,RR}}_{n,n_0}(u) \|_{L_p} \le \frac{C_{\mathrm{init,RR}} \|u\| \|\theta_0 - \theta^*\| \, p}{\alpha a \sqrt{m}} n^{-\beta}. \tag{261}$$

At the balanced scale $\alpha = cn^{-1/2}$, with $m \ge n/2$ and p logarithmic in n, this is $\mathrm{polylog}(n)\, n^{-\beta}$.

Proof. Substitute Eq. 260 into Eq. 255. At the balanced scale, $\left(\alpha\sqrt{m}\right)^{-1} = O(1)$, so the remaining factor is logarithmic. □

5.5. Burned-in Variance Proxy

For the stochastic depth-zero term, write $Q^{\mathrm{bRR}}_l := Q^{\mathrm{RR}}_{l;n_0,n}$ and $\Sigma := \Sigma^{(\mathrm{M})}_\epsilon$. We take $n \ge 2$ in this subsection. The martingale part produced by the Poisson decomposition below starts at $l = 2$, so the deterministic variance proxy is

$$\Sigma^{\mathrm{bRR}}_{n,n_0} := \frac{1}{m} \sum_{l=2}^{n-1} Q^{\mathrm{bRR}}_l \, \Sigma \left(Q^{\mathrm{bRR}}_l\right)^\top, \quad \sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u) := u^\top \Sigma^{\mathrm{bRR}}_{n,n_0} u. \tag{262}$$

Lemma 21. Assume the conditions of Lemma 17 and $\| \Sigma \| < \infty$. There exists a constant $C_{\mathrm{burn,3}}$, depending only on κ_Q, $\| \overline{A}^{-1} \|$, $\| \overline{A} \|$, and $\| \Sigma \|$, such that

$$\| \Sigma^{\mathrm{bRR}}_{n,n_0} - \Sigma_\infty \| \le \frac{C_{\mathrm{burn,3}}}{m\,\alpha a}. \tag{263}$$

Consequently,

$$| \sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u) - \sigma^2(u) | \le \frac{C_{\mathrm{burn,3}} \| u \|^2}{m\,\alpha a}. \tag{264}$$

Proof. For post-burn-in indices put $\Delta_l := Q^{\mathrm{bRR}}_l - \overline{A}^{-1}$. Then

$$Q^{\mathrm{bRR}}_l \Sigma \bigl(Q^{\mathrm{bRR}}_l\bigr)^\top - \Sigma_\infty = \Delta_l \Sigma \overline{A}^{-\top} + \overline{A}^{-1} \Sigma \Delta_l^\top + \Delta_l \Sigma \Delta_l^\top. \tag{265}$$

By Eq. 234,

$$\sum_{l\geq \max(2,n_0)} \| \Delta_l \| \leq \frac{C}{\alpha a}, \quad \sum_{l\geq \max(2,n_0)} \| \Delta_l \|^2 \leq \frac{C}{\alpha a}. \tag{266}$$

For pre-burn-in indices there is no comparison with $\overline{A}^{-1}$; instead Eq. 236 gives

$$\sum_{2\leq l<n_0} \| Q_l^{\mathrm{bRR}} \|^2 \leq \frac{C}{\alpha a}. \tag{267}$$

Replacing all post-burn-in weights by $\overline{A}^{-1}$ gives at most m copies of Σ_∞; the martingale index set starts at $l = 2$, so the possible finite-index mismatch is bounded by two copies of Σ_∞. Combining these three estimates and dividing by m yields Eq. 263, after absorbing the harmless $2\,\|\Sigma_\infty\|\frac{}{m}$ term into $\frac{C_{\mathrm{burn},3}}{m\alpha a}$ using $\alpha a \leq 1$. The scalar bound follows from $|u^\top H u| \leq \|H\|\,\|u\|^2$. □

Assume $\sigma^2(u) > 0$ and impose the burned-in variance lower-bound condition

$$m\,\alpha\,a \geq \frac{2C_{\mathrm{burn},3} \| u \|^2}{\sigma^2(u)}. \tag{268}$$

Then Eq. 264 gives $\sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u) \geq \sigma^2(u)/2$, hence $\sigma^{\mathrm{bRR}}_{n,n_0}(u) \geq \sigma(u)/\sqrt{2}$. At the balanced scale $\alpha = c\,n^{-1/2}$ with $m \geq n/2$, this condition holds for all sufficiently large n depending on u, c, and the problem constants.

5.6. Burned-in Poisson Martingale Approximation

Let $\hat{\epsilon} := \sum_{j=0}^{\infty} Q^j \epsilon$ be the Poisson solution, so

$$\hat{\epsilon} - Q\hat{\epsilon} = \epsilon, \quad \| \hat{\epsilon} \|_\infty \leq 3\,t_{\mathrm{mix}} \| \epsilon \|_\infty. \tag{269}$$

The stationary Poisson identity applies because the coefficients Q_l^{bRR} are deterministic.

Lemma 22. Assume **UGE 1** and $\pi(\epsilon) = 0$. Define, for $2 \leq l \leq n-1$,

$$\Delta M_l^{\mathrm{bRR}} := Q_l^{\mathrm{bRR}} \left(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})\right), \tag{270}$$

and set $M_{n,n_0}^{\mathrm{bRR}} := \sum_{l=2}^{n-1} \Delta M_l^{\mathrm{bRR}}$. Then $\left\{\Delta M_l^{\mathrm{bRR}}\right\}_{l=2}^{n-1}$ is a sequence of $\mathcal{F}_l$-martingale differences and

$$W_{n,n_0}^{\mathrm{RR}} = -\frac{1}{\sqrt{m}} M_{n,n_0}^{\mathrm{bRR}} + D_{2,n,n_0}^{\mathrm{bRR}}, \tag{271}$$

where

$$D_{2,n,n_0}^{\mathrm{bRR}} := -\frac{1}{\sqrt{m}} \left[Q_1^{\mathrm{bRR}}\,\hat{\epsilon}(Z_1) + \sum_{l=1}^{n-2} \left(Q_{l+1}^{\mathrm{bRR}} - Q_l^{\mathrm{bRR}}\right) Q\hat{\epsilon}(Z_l) \right]. \tag{272}$$

Moreover, with $C_{\mathrm{burn,Q}} := \| \overline{A}^{-1} \| + 6C_Q$,

$$\| D_{2,n,n_0}^{\mathrm{bRR}} \|_\infty \leq \frac{3\,t_{\mathrm{mix}} \| \epsilon \|_\infty}{\sqrt{m}} \left(C_{\mathrm{burn,Q}} + \frac{C_{\mathrm{burn,V}}}{a^2} \right). \tag{273}$$

Proof. The martingale-difference property follows from the Markov property: $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$.

Substitute the Poisson equation in Eq. 230. The $l = 1$ term is kept as a left boundary term, and for $l \geq 2$ we add and subtract $Q\hat{\epsilon}(Z_{l-1})$. Abel summation of the telescope gives exactly Eq. 271. The right boundary vanishes because $Q^{\mathrm{RR}}_{n-1;n_0,n} = 2\alpha I - 2\alpha I = 0$.

For the sup-norm bound, use $\| Q\hat{\epsilon} \|_\infty \leq \| \hat{\epsilon} \|_\infty$, the uniform weight bound $\| Q_l^{\mathrm{bRR}} \| \leq C_{\mathrm{burn,Q}}$ from Eq. 234 and Eq. 235, and the total-variation estimate Eq. 237. □

5.7. Burned-in Predictable Variance Comparison

For the martingale in Lemma 22 define

$$\mathcal{V}_\epsilon(z) := Q(\hat{\epsilon}\hat{\epsilon}^\top)(z) - (Q\hat{\epsilon})(z)\,(Q\hat{\epsilon})(z)^\top. \tag{274}$$

This is the same matrix-valued conditional covariance function as in the stationary chapter; it is not the vector noise ϵ. As before, $\pi(\mathcal{V}_\epsilon) = \Sigma$ and

$$\| \mathcal{V}_\epsilon \|_\infty \le 18\, t_{\mathrm{mix}}^2 \| \epsilon \|_\infty^2. \tag{275}$$

The predictable quadratic variation is

$$\langle M^{\mathrm{bRR}} \rangle_{n,n_0} := \sum_{l=2}^{n-1} \mathbb{E}\Big[\Delta M_l^{\mathrm{bRR}} \left(\Delta M_l^{\mathrm{bRR}}\right)^\top \mid \mathcal{F}_{l-1}\Big] = \sum_{l=2}^{n-1} Q_l^{\mathrm{bRR}}\, \mathcal{V}_\epsilon(Z_{l-1}) \left(Q_l^{\mathrm{bRR}}\right)^\top. \tag{276}$$

Lemma 23. Assume **UGE 1**, $\pi(\epsilon) = 0$, and $\| \epsilon \|_\infty < \infty$. There exists a universal constant $C_4 > 0$ such that, for every $u \in \mathbb{R}^d$, every $p \ge 2$, every initial distribution ξ, every $n \ge 2$, and every $0 \le n_0 < n$,

$$\mathbb{E}_\xi^{1/p}\Big[|\ u^\top \langle M^{\mathrm{bRR}} \rangle_{n,n_0} u - m\, \sigma_{n,n_0}^{2,\,\mathrm{bRR}}(u)\ |^p\Big] \le C_4\, C_{\mathrm{burn,Q}}^2 \| u \|^2 \| \epsilon \|_\infty^2\, t_{\mathrm{mix}}^{5/2} \sqrt{p\, n}. \tag{277}$$

Proof. Set

$$h_{l(z)} := u^\top Q_l^{\mathrm{bRR}}\, \mathcal{V}_\epsilon(z) \left(Q_l^{\mathrm{bRR}}\right)^\top u, \quad g_{l(z)} := h_{l(z)} - \pi(h_l). \tag{278}$$

By Eq. 275 and $\| Q_l^{\mathrm{bRR}} \| \le C_{\mathrm{burn,Q}}$,

$$\| g_l \|_\infty \le 36\, C_{\mathrm{burn,Q}}^2 \| u \|^2\, t_{\mathrm{mix}}^2 \| \epsilon \|_\infty^2. \tag{279}$$

Moreover, Eq. 276 and Eq. 262 imply

$$u^\top \langle M^{\mathrm{bRR}} \rangle_{n,n_0} u - m\, \sigma_{n,n_0}^{2,\,\mathrm{bRR}}(u) = \sum_{l=2}^{n-1} g_{l(Z_{l-1})}. \tag{280}$$

The scalar Markov concentration lemma used in the stationary chapter is valid for arbitrary initial law by Levin et al. (2025, Lemma 11). Applied to the time-inhomogeneous centered functions g_l, it gives the displayed $\sqrt{pn}$ bound. □

Corollary 10. Under the assumptions of Lemma 23,

$$\mathbb{E}_\xi^{1/p}\Big[|\ u^\top \langle M^{\mathrm{bRR}} \rangle_{n,n_0} u - m\, \sigma^2(u)\ |^p\Big] \le C_4\, C_{\mathrm{burn,Q}}^2 \| u \|^2 \| \epsilon \|_\infty^2\, t_{\mathrm{mix}}^{5/2} \sqrt{p\, n} + \frac{C_{\mathrm{burn},3} \| u \|^2}{\alpha a}. \tag{281}$$

Proof. Use the triangle inequality and Eq. 264:

$$m\, |\sigma_{n,n_0}^{2,\,\mathrm{bRR}}(u) - \sigma^2(u)| \le \frac{C_{\mathrm{burn},3} \| u \|^2}{\alpha a}. \tag{282}$$

The concentration sum still runs over the ambient indices $2, ..., n-1$, including pre-burn-in weights, which is why the display keeps $\sqrt{p\, n}$. In the final theorem this is converted to the effective window scale using $m \ge n/2$. □

5.8. Startup Transfer for Augmented-Chain Remainders

After the deterministic transient is removed, a startup discrepancy remains: the finite-start perturbation variables $J^{(\ell,w)}$ and $H^{(2,w)}$ start from zero, whereas the stationary theorem uses the invariant law of the augmented chain. This discrepancy is summed over the post-burn-in window.

For $w \in \{\alpha, 2\alpha\}$ write

$$Y_k^{(w)} := \left(Z_{k+1}, J_k^{(0,w)}, J_k^{(1,w)}, J_k^{(2,w)}\right). \tag{283}$$

For $y = (z, j_0, j_1, j_2)$ and $y' = (z', j_0', j_1', j_2')$ define the depth-two augmented-chain cost used by Levin et al. (2025, Appendix B.2, Eq. (49)):

$$\begin{aligned} c_{J,2}^{(w)}(y, y') &:= \|j_0 - j_0'\| + \|j_1 - j_1'\| + \|j_2 - j_2'\| \\ &\quad + \big(\|j_0\| + \|j_0'\| + \|j_1\| + \|j_1'\| + \|j_2\| + \|j_2'\| + \sqrt{wa}\, \|\epsilon\|_\infty\big)\, 1_{z \neq z'}. \end{aligned} \tag{284}$$

Levin et al. (2025, Appendix B.2, Proposition 5, with constants defined in their Eq. (55)) prove the Wasserstein contraction for this cost. The componentwise estimates in its proof imply that, under their Proposition-5 step-size restriction, two copies of $Y_k^{(w)}$ started from deterministic states y, y' can be coupled so that, for $\ell = 0, 1, 2$,

$$\| \, J_k^{(\ell,w)}(y) - J_k^{(\ell,w)}(y') \, \|_{L_p} \leq C_W\, p^{7/2}\, t_{\mathrm{mix}}^{5/2} \log^{3/2}(1/(wa)) \exp(-wak/(12p))\, c_{J,2}^{(w)}(y, y'). \tag{285}$$

Their Corollary 4 gives the corresponding invariant law $\Pi_{J,2,w}$ for $Y_k^{(w)}$.

The contraction above does not include $H^{(2,w)}$. Levin et al. (2025, Appendix D.1, Proposition 9) prove the required one-trajectory moment bound for $H^{(2,w)}$, using the representation

$$H_k^{(2,w)} = -w \sum_{l=1}^{k} \Gamma_{l+1:k}^{(w)} \tilde{A}(Z_l) J_{l-1}^{(2,w)}. \tag{286}$$

The invariant law for the full augmented state is obtained by a finite-past construction. On a two-sided stationary version of the driving chain, start the $J^{(0,w)}$, $J^{(1,w)}$, $J^{(2,w)}$, and $H^{(2,w)}$ recursions from zero at time $-m$ and evaluate them at time 0. Levin Proposition 5 gives the L_p Cauchy property for the J coordinates. For the H coordinate, use the finite-past version of Eq. 286,

$$H_{0,m}^{(2,w)} := -w \sum_{l=-m+1}^{0} \Gamma_{l+1:0}^{(w)} \tilde{A}(Z_l) J_{l-1,m}^{(2,w)}. \tag{287}$$

Levin Proposition 9 and the random-product stability estimate below make this sequence Cauchy in L_p for each fixed admissible p; its limit is denoted $H_0^{(2,w)}$. Shifting the construction defines the stationary two-sided process $\left(Z_{k+1}, J_k^{(0,w)}, J_k^{(1,w)}, J_k^{(2,w)}, H_k^{(2,w)}\right)_{k\in\mathbb{Z}}$. This is the full augmented-chain invariant law used below.

Lemma 24. (Conditional product stability at a coupling time.) Assume the hypotheses of Lemma 19 and let T be an exact-coupling time for two copies of the base chain satisfying $\mathbb{P}(T > r) \leq C_T \exp(-c_T r)$. Then, after decreasing the exponential constant, the following two estimates hold uniformly in k and $w \in \{\alpha, 2\alpha\}$.

(i) Random restart. If $(V_s)_{s\geq 0}$ is adapted and $\sup_s \|V_s\|_{L_{2p}} \leq B$, then

$$\| \, \Gamma_{T+1:k}^{(w)} V_T 1_{T\leq k} \, \|_{L_p} \leq C \frac{p}{wa} B \exp(-cwak/p). \tag{288}$$

(ii) Stable convolution. If $(U_l)_{l\geq 0}$ is adapted and $\|U_l\|_{L_{2p}} \leq B \exp(-c_0 wal/p)$ for all l, then

$$w \sum_{l=1}^{k} \| \, \Gamma_{l+1:k}^{(w)} U_l \, \|_{L_p} \leq C \frac{p}{a} B \exp(-cwak/p). \tag{289}$$

Proof. The proof of Levin's product-stability estimate is conditional on the past at the starting time, hence Eq. 253 may be applied on each event $T = s$ with the same constants. For the first estimate, Minkowski's inequality gives

$$\| \Gamma^{(w)}_{T+1:k} V_T 1_{T\le k} \|_{L_p} \le C \sum_{s=0}^{k} \exp(-cwa(k-s)/p) \| V_s 1_{T=s} \|_{L_p}. \tag{290}$$

Holder's inequality and the exponential tail of T give $\|V_s 1_{T=s}\|_{L_p} \le B\,\mathbb{P}(T=s)^{1/(2p)} \le CB\exp(-c_T s/(2p))$. Since wa is bounded above by the admissible small-step constant, the convolution of $\exp(-cwa(k-s)/p)$ and $\exp(-c_T s/(2p))$ is bounded by $C(p/(wa))\exp(-cwak/p)$ after decreasing c. This proves Eq. 288.

For the second estimate, apply Eq. 253 at each deterministic time l and use the assumed decay of U_l:

$$w\sum_{l=1}^{k} \exp(-cwa(k-l)/p) B\exp(-c_0 wal/p) \le C\frac{p}{a} B\exp(-c' wak/p). \tag{291}$$

Renaming c' as c proves Eq. 289. □

For startup transfer, however, one must compare two copies of this display. The following lemma records the resulting full-state contraction. It is a technical extension of the Levin Proposition-5 coupling: the $J^{(0)}$, $J^{(1)}$, and $J^{(2)}$ coordinates use Levin Appendix B.2 directly, while the $H^{(2)}$ coordinate is controlled by applying the random-product stability estimate used inside Levin Appendix D.1, Proposition 9, to the difference of two representations Eq. 286.

The threshold $\alpha_{\rm st}(p)$ was fixed above as the common ceiling for the Levin depth-two startup contraction and the random-product stability estimate. For $w \in \{\alpha, 2\alpha\}$ write

$$R^{(w)}_{k,\,\rm fin} := J^{(1,w)}_{k,\,\rm fin} + J^{(2,w)}_{k,\,\rm fin} + H^{(2,w)}_{k,\,\rm fin} \tag{292}$$

for the finite-start depth-two remainder, and define $R^{(w)}_{k,\,\rm aug}$ analogously for a copy initialized from the invariant law of the full augmented chain $\Big(Z_{k+1}, J^{(0,w)}_k, J^{(1,w)}_k, J^{(2,w)}_k, H^{(2,w)}_k\Big)$.

Lemma 25. (Full-state startup contraction for the depth-two augmented remainder.) Assume **UGE 1**, $\pi(\epsilon)=0$, $\|\epsilon\|_\infty < \infty$, the step-size restrictions of the Levin depth-two moment bounds, and $2\alpha \le \alpha_{\rm st}(p)$. There exist constants $c_{\rm st} > 0$ and $C_{\rm st} < \infty$, depending only on the problem constants and on the constants in Eq. 285 and Levin Proposition 9, such that for every $p \ge 2$, every admissible $q \ge 2$, every initial distribution ξ of the base chain with finite-start perturbation coordinates initialized at zero, every $w \in \{\alpha, 2\alpha\}$, and every $k \ge 0$, the finite-start and stationary augmented remainders can be coupled so that

$$\| R^{(w)}_{k,\,\rm fin} - R^{(w)}_{k,\,\rm aug} \|_{L_p} \le A_{\rm st}(p,q,w)\exp(-c_{\rm st} wak/p), \tag{293}$$

where

$$A_{\rm st}(p,q,w) := C_{\rm st}\big(1+d^{1/q}\big)p^8 t_{\rm mix}^5 \frac{1}{a}\sqrt{w/a}\,\log^3(1/(wa)). \tag{294}$$

Proof. Work on the exact-coupling probability space used in Levin Appendix B.2. Let T be the coupling time of the two base chains. By UGE, $\mathbb{P}(T > r) \le C\rho^r$, and Proposition 5 gives, for $\ell \in \{0,1,2\}$,

$$\| J^{(\ell,w)}_{k,\,\rm fin} - J^{(\ell,w)}_{k,\,\rm aug} \|_{L_p} \le Cp^{7/2} t_{\rm mix}^{5/2} \log^{3/2}(1/(wa)) e^{-cwak/p} \mathbb{E}^{1/p}\Big[c^{(w)}_{J,2}\big(Y_0^{\rm fin}, Y_0^{\rm aug}\big)^p\Big]. \tag{295}$$

The finite-start J coordinates are zero. The invariant copy is the finite-past limit described above, so the elementary $J^{(0,w)}$ and $J^{(1,w)}$ estimates and Levin Proposition 8 for $J^{(2,w)}$ give

$$\mathbb{E}^{1/p}\Big[c^{(w)}_{J,2}\big(Y_0^{\rm fin}, Y_0^{\rm aug}\big)^p\Big] \le C\big(1+d^{1/q}\big)p^{7/2} t_{\rm mix}^{5/2}\sqrt{w/a}\,\log^{3/2}(1/(wa)). \tag{296}$$

Thus the $J^{(1,w)}$ and $J^{(2,w)}$ parts of $R^{(w)}_k$ satisfy the pointwise bound with

$$A_{\mathrm{J}}(p,q,w) \coloneqq C\big(1+d^{1/q}\big)p^7 t_{\mathrm{mix}}^5 \sqrt{w/a}\,\log^3(1/(wa)). \tag{297}$$

Since $A_{\mathrm{J}}(p,q,w) \le A_{\mathrm{st}}(p,q,w)$ after enlarging C_{st}, these parts already satisfy Eq. 293.

It remains to control $H^{(2,w)}$. We shall repeatedly use the one-trajectory bound, valid for the finite-start and finite-past stationary copies,

$$\| \, H_s^{(2,w)} \, \|_{L_{2p}} \le B_{H(p,q,w)} \coloneqq C\big(1+d^{1/q}\big)p^{7/2} t_{\mathrm{mix}}^{5/2} w^{3/2} \log^{3/2}(1/(wa)), \tag{298}$$

where C is allowed to change when p is replaced by $2p$. On the event $T \le k$, the base chains are identical after time T, and subtracting Eq. 286 from time T onward gives

$$\Delta H_k^{(2,w)} = \Gamma_{T+1:k}^{(w)} \Delta H_T^{(2,w)} - w \sum_{l=T+1}^{k} \Gamma_{l+1:k}^{(w)} \tilde{A}(Z_l) \Delta J_{l-1}^{(2,w)}, \tag{299}$$

where Δ denotes finite-start minus stationary.

On the bad event $T > k$, Holder inequality gives the explicit reduction

$$\| \, \Delta H_k^{(2,w)} 1_{T>k} \, \|_{L_p} \le \Big(\| \, H_{k,\,\mathrm{fin}}^{(2,w)} \, \|_{L_{2p}} + \| \, H_{k,\,\mathrm{aug}}^{(2,w)} \, \|_{L_{2p}} \Big) \, \mathbb{P}(T>k)^{1/(2p)}. \tag{300}$$

The bound Eq. 298 controls both one-trajectory terms uniformly in k, and UGE gives $\mathbb{P}(T>k)^{1/(2p)} \le Ce^{-ck/p}$. Therefore

$$\| \, \Delta H_k^{(2,w)} 1_{T>k} \, \|_{L_p} \le C B_{H(p,q,w)} e^{-ck/p}. \tag{301}$$

Since $wa \le 1$, this is bounded by the right-hand side of Eq. 293 after absorbing a fixed power of a^{-1} into C_{st}.

For the first term in Eq. 299, apply Lemma 24 with $V_s = \Delta H_s^{(2,w)}$. The input $\sup_s \|V_s\|_{L_{2p}} \le C B_{H(p,q,w)}$ follows from Eq. 298 for the two coupled copies. Hence

$$\| \, \Gamma_{T+1:k}^{(w)} \Delta H_T^{(2,w)} 1_{T\le k} \, \|_{L_p} \le C \frac{p}{wa} B_{H(p,q,w)} e^{-cwak/p} \le C A_{\mathrm{st}}(p,q,w) e^{-cwak/p}. \tag{302}$$

For the convolution term, set $U_l \coloneqq \tilde{A}(Z_l) \Delta J_{l-1}^{(2,w)}$. Applying Eq. 295 with moment order $2p$ gives

$$\| \, U_l \, \|_{L_{2p}} \le C A_{\mathrm{J}}(2p,q,w) \exp(-cwal/p), \tag{303}$$

where replacing $l-1$ by l only changes the constant. The stable convolution estimate Eq. 289 yields

$$\begin{aligned} &w \sum_{l=1}^{k} \| \, \Gamma_{l+1:k}^{(w)} \tilde{A}(Z_l) \Delta J_{l-1}^{(2,w)} \, \|_{L_p} \\ &\le C \frac{p}{a} A_{\mathrm{J}}(2p,q,w) e^{-cwak/p} \\ &\le C A_{\mathrm{st}}(p,q,w) e^{-cwak/p}. \end{aligned} \tag{304}$$

The additional factor p/a is the reason for the enlarged definition of A_{st} relative to the pure J-coordinate scale A_{J}. Combining Eq. 301, Eq. 302, and Eq. 304 gives the same startup bound for $H^{(2,w)}$. Adding the already controlled $J^{(1,w)}$ and $J^{(2,w)}$ pieces proves Eq. 293. □

Define the RR startup discrepancy accumulated over the burned-in window by

$$\mathcal{U}_{n,n_0}^{\mathrm{start,RR}} \coloneqq \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} \Big[2\Big(R_{k,\,\mathrm{fin}}^{(\alpha)} - R_{k,\,\mathrm{aug}}^{(\alpha)} \Big) - \Big(R_{k,\,\mathrm{fin}}^{(2\alpha)} - R_{k,\,\mathrm{aug}}^{(2\alpha)} \Big) \Big]. \tag{305}$$

Lemma 26. (Accumulated startup transfer.) Under Lemma 25, if $\alpha, 2\alpha \in (0, \alpha_\infty]$ and $\alpha a \leq 1/4$, then for every $p \geq 2$ and admissible $q \geq 2$,

$$\| \mathcal{U}^{\mathrm{start,RR}}_{n,n_0} \|_{L_p} \leq \frac{C_{\mathrm{start,RR}} p \, A_{\mathrm{st}}(p,q,\alpha)}{\alpha a \sqrt{m}} \exp(-c_{\mathrm{st}} \alpha a n_0 / p). \tag{306}$$

Proof. Apply Eq. 293 at $w = \alpha$ and $w = 2\alpha$, then use the triangle inequality:

$$\| \mathcal{U}^{\mathrm{start,RR}}_{n,n_0} \|_{L_p} \leq \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} \big(2 A_{\mathrm{st}}(p,q,\alpha) e^{-c_{\mathrm{st}} \alpha a k / p} + A_{\mathrm{st}}(p,q,2\alpha) e^{-2 c_{\mathrm{st}} \alpha a k / p} \big). \tag{307}$$

Since $\alpha a \leq 1/4$, $A_{\mathrm{st}}(p,q,2\alpha) \leq C A_{\mathrm{st}}(p,q,\alpha)$ and $1 - \exp(-c_{\mathrm{st}} \alpha a / p) \geq C^{-1} \alpha a / p$. Extending the geometric sum to infinity gives Eq. 306. □

Corollary 11. (Mixing-scale burn-in removes the startup discrepancy.) If, for some $\beta > 0$,

$$n_0 \geq \frac{\beta p}{c_{\mathrm{st}} \alpha a} \log n, \tag{308}$$

then

$$\| \mathcal{U}^{\mathrm{start,RR}}_{n,n_0} \|_{L_p} \leq \frac{C_{\mathrm{start,RR}} p \, A_{\mathrm{st}}(p,q,\alpha)}{\alpha a \sqrt{m}} n^{-\beta}. \tag{309}$$

At the balanced scale $\alpha = c n^{-1/2}$, with $m \geq n/2$ and p, q logarithmic in n, this is $\mathrm{polylog}(n) \, n^{-1/4-\beta}$. Thus the Berry–Esseen choice $p \asymp \log n$ requires a mixing-scale burn-in with logarithmic-square factor, n_0 of order $(\alpha a)^{-1} \log^2 n$, under this L_p contraction.

Proof. Substitute Eq. 308 into Eq. 306. For the balanced scale, $A_{\mathrm{st}}(p,q,\alpha) = \mathrm{polylog}(n) \, \alpha^{1/2}$, hence $A_{\mathrm{st}}(p,q,\alpha) / (\alpha \sqrt{m}) = O\big(\mathrm{polylog}(n) n^{-1/4}\big)$. □

5.9. Burned-in Depth-Two Misadjustment Bound

Define the finite-start burned-in RR misadjustment by

$$R^{\mathrm{mis,RR}}_{n,n_0,\,\mathrm{fin}} := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} \Big(2 R^{(\alpha)}_{k,\,\mathrm{fin}} - R^{(2\alpha)}_{k,\,\mathrm{fin}} \Big). \tag{310}$$

For comparison, let

$$R^{\mathrm{mis,RR}}_{m,\,\mathrm{aug}} := \frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} \Big(2 R^{(\alpha)}_{j,\,\mathrm{aug}} - R^{(2\alpha)}_{j,\,\mathrm{aug}} \Big) \tag{311}$$

be the stationary augmented-chain depth-two misadjustment over a window of length m. By stationarity, the same distribution is obtained if the sum in Eq. 311 is taken over $j = n_0, ..., n-1$.

Theorem 4. (Burned-in PR-averaged RR misadjustment bound.) Assume **UGE 1**, $\pi(\epsilon) = 0$, $\| \epsilon \|_\infty < \infty$, $\alpha, 2\alpha \in (0, \alpha_\infty]$, $2\alpha \le \alpha_{\text{inv}}$, $\alpha a \le 1/4$, and the step-size restrictions of the Levin depth-two and startup-contraction bounds, where α_{inv} is defined in Eq. 92. Set

$$\Phi_+(p, \alpha) := 1 + p^{3/2}\, t_{\text{mix}}^{1/2}/a + p^{1/2}\, t_{\text{mix}}^{3/2} \sqrt{\alpha/a}. \tag{312}$$

There exists a constant $C_{\text{burn,mis}}$ depending only on the stationary misadjustment constants, the startup-contraction constants, and the problem constants such that, for every $p \ge 2$, every $q \ge 2$ satisfying $p \le q/2$, every $2\alpha \le \alpha_*(q, t_{\text{mix}})$ and $2\alpha \le \alpha_{\text{st}}(p)$, and every $m \ge 2$,

$$\begin{aligned}\| R^{\text{mis,RR}}_{n,n_0,\,\text{fin}} \|_{L_p} \le C_{\text{burn,mis}} \sqrt{m}\, \alpha^2 & \\ + C_{\text{burn,mis}} (1 + d^{1/q}) p^{7/2} t_{\text{mix}}^{5/2} \sqrt{m}\, \alpha^{3/2} \log^{3/2}(1/(\alpha a)) & \\ + C_{\text{burn,mis}} p^{3/2} \sqrt{\alpha} & \\ + C_{\text{burn,mis}} p^3 (\alpha m)^{-1/2} \log^{1/p}(1/(\alpha a)) & \\ + C_{\text{burn,mis}} \Phi_+(p, \alpha)\, m^{-1/2} & \\ + \frac{C_{\text{burn,mis}} p\, A_{\text{st}}(p, q, \alpha)}{\alpha a \sqrt{m}} \exp(-c_{\text{st}} \alpha a n_0 / p). &\end{aligned} \tag{313}$$

Proof. Couple the finite-start and stationary augmented-chain remainders as in Lemma 25, and define the stationary window on the same indices by

$$\tilde{R}^{\text{mis,RR}}_{n,n_0,\,\text{aug}} := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} \Big(2R^{(\alpha)}_{k,\,\text{aug}} - R^{(2\alpha)}_{k,\,\text{aug}} \Big). \tag{314}$$

Then, with the notation of Eq. 305,

$$R^{\text{mis,RR}}_{n,n_0,\,\text{fin}} = \tilde{R}^{\text{mis,RR}}_{n,n_0,\,\text{aug}} + \mathcal{U}^{\text{start,RR}}_{n,n_0} \tag{315}$$

under this coupling. Therefore

$$\| R^{\text{mis,RR}}_{n,n_0,\,\text{fin}} \|_{L_p} \le \| \tilde{R}^{\text{mis,RR}}_{n,n_0,\,\text{aug}} \|_{L_p} + \| \mathcal{U}^{\text{start,RR}}_{n,n_0} \|_{L_p}. \tag{316}$$

By stationarity, $\| \tilde{R}^{\text{mis,RR}}_{n,n_0,\,\text{aug}} \|_{L_p} = \| R^{\text{mis,RR}}_{m,\,\text{aug}} \|_{L_p}$, and the latter is exactly the stationary augmented-chain misadjustment bound Theorem 2 with n replaced by m. The second term is Lemma 26. Combining the two estimates gives Eq. 313. □

Corollary 12. (Balanced-scale burned-in misadjustment rate.) Assume the hypotheses of Theorem 4. Let $\alpha = c\, n^{-1/2}$, $m \ge n/2$, $p = \max(2, \lceil \log n \rceil)$, and $q = \max(2p, \lceil \log(e\, d) \rceil, 2)$. If, for some fixed $\beta > 0$,

$$n_0 \ge \frac{\beta p}{c_{\text{st}} \alpha a} \log n, \tag{317}$$

and n is large enough that $2\alpha \le \alpha_*(q, t_{\text{mix}})$ and $2\alpha \le \alpha_{\text{st}}(p)$ and $2\alpha \le \alpha_{\text{inv}}$, then

$$\| R^{\text{mis,RR}}_{n,n_0,\,\text{fin}} \|_{L_p} \le C_{\text{burn,mis}}\, \text{polylog}(n)\, n^{-1/4}. \tag{318}$$

Proof. The stationary part of Eq. 313 is the bound of Corollary 4 with n replaced by m, and $m \ge n/2$ keeps the rate unchanged. The startup term is controlled by Corollary 11 and is $\text{polylog}(n)\, n^{-1/4-\beta}$ at this scale. □

5.10. Burned-in Martingale Berry–Esseen

The depth-zero martingale in Lemma 22 has deterministic coefficients, but its normalization is based on the effective sample size $m = n - n_0$. We assume $m \ge n/2$ and impose the mixing-scale burn-in lower bounds with logarithmic factors, so the inhomogeneous concentration term of order $\sqrt{pn}$ can be written on the m scale.

Fix $u \in \mathbb{R}^d$ and put $X_l^{\mathrm{bRR}} := u^\top \Delta M_l^{\mathrm{bRR}}$ for $2 \le l \le n-1$. Then

$$|X_l^{\mathrm{bRR}}| \le \kappa_{\mathrm{burn}}(u), \quad \kappa_{\mathrm{burn}}(u) := 6\, t_{\mathrm{mix}}\, C_{\mathrm{burn,Q}} \parallel \epsilon \parallel_\infty \parallel u \parallel. \tag{319}$$

Indeed, this is the same bounded-increment estimate as in the stationary chapter, with $C_{\mathrm{burn,Q}}$ replacing the full-window weight bound. Set

$$s_{n,n_0}^2(u) := m\, \sigma_{n,n_0}^{2,\,\mathrm{bRR}}(u). \tag{320}$$

Under Eq. 268, $s_{n,n_0}^2(u) \ge m\sigma^2(u)/2$. Also, $m \ge n/2$ and the uniform bound $\|Q_l^{\mathrm{bRR}}\| \le C_{\mathrm{burn,Q}}$ imply the deterministic upper bound

$$s_{n,n_0}^2(u) \le 2m\, C_{\mathrm{burn,Q}}^2 \parallel \Sigma_\epsilon^{(\mathrm{M})} \parallel \parallel u \parallel^2. \tag{321}$$

Theorem 5. (Burned-in martingale Berry–Esseen bound.) Assume **UGE 1**, $\pi(\epsilon) = 0$, $\parallel \epsilon \parallel_\infty < \infty$, $\sigma^2(u) > 0$, $\alpha, 2\alpha \in (0, \alpha_\infty]$, $m \ge n/2$, and the burned-in variance lower-bound condition Eq. 268. There exist constants $C_{\mathrm{bK},1}(u), C_{\mathrm{bK},2}(u) > 0$, depending only on $\|u\|$, $\sigma(u)$, $C_{\mathrm{burn,Q}}$, t_{mix}, $\|\epsilon\|_\infty$, $\|\Sigma_\epsilon^{(\mathrm{M})}\|$, and the universal Bolthausen–Fan constants, such that for every $n \ge 3$,

$$d_K\left(\frac{u^\top M_{n,n_0}^{\mathrm{bRR}}}{\sqrt{m}\,\sigma_{n,n_0}^{\mathrm{bRR}}(u)}, \mathcal{N}(0,1)\right) \le \frac{C_{\mathrm{bK},1}(u)\log^{3/4} n}{m^{1/4}} + \frac{C_{\mathrm{bK},2}(u)\log n}{\sqrt{m}}. \tag{322}$$

Proof. Apply the Bolthausen–Fan inequality used in the stationary chapter to the martingale differences X_l^{bRR}. The bounded-increment input is Eq. 319, and the target variance is $s_{n,n_0}^2(u)$. The first and third Bolthausen–Fan terms are bounded exactly as before, with n replaced by m in the denominator and the harmless factor $n/m \le 2$ absorbed into the constants. For the conditional-variance term use Lemma 23:

$$\mathbb{E}^{1/p}\left[\mid u^\top \langle M^{\mathrm{bRR}}\rangle_{n,n_0} u - s_{n,n_0}^2(u) \mid^p\right] \le C\sqrt{p\,n}. \tag{323}$$

Together with $m \ge n/2$, Eq. 268, and Eq. 321, this gives

$$\text{Term II} \le C(u)\, p^{(3p+1)/(2(2p+1))}\, m^{-p/(2(2p+1))}. \tag{324}$$

Taking $p = \lceil \log n \rceil$ gives the displayed $\log^{3/4}(n) m^{-1/4}$ bound. The classical Bolthausen and Lindeberg terms give the second displayed term. $\square$

5.11. Finite-Window Smoothing Assembly

The finite-start scalar statistic decomposes as

$$T_{n,n_0}^{\mathrm{RR}}(u) = -\frac{u^\top M_{n,n_0}^{\mathrm{bRR}}}{\sqrt{m}} + \mathcal{R}_{n,n_0,\,\mathrm{fin}}^{\mathrm{bRR}}(u), \tag{325}$$

where the composite non-Gaussian remainder is

$$\mathcal{R}_{n,n_0,\,\mathrm{fin}}^{\mathrm{bRR}}(u) := D_{\mathrm{tr},n,n_0}^{\mathrm{RR}}(u) + \mathcal{J}_{n,n_0}^{\mathrm{init,RR}}(u) + u^\top D_{2,n,n_0}^{\mathrm{bRR}} + u^\top R_{n,n_0,\,\mathrm{fin}}^{\mathrm{mis,RR}}. \tag{326}$$

The terms are controlled by Lemma 18, Lemma 20, Lemma 22, and Theorem 4, respectively.

For reference, the finite-window assembly uses the following components:

Component	**Input bound**	**Role in smoothing**

$D^{\mathrm{RR}}_{\mathrm{tr},n,n_0}(u)$	Eq. 247	Deterministic transient from $\theta_0 - \theta^*$.
$\mathcal{J}^{\mathrm{init,RR}}_{n,n_0}(u)$	Eq. 255	Random initial-product discrepancy.
$u^\top D^{\mathrm{bRR}}_{2,n,n_0}$	Eq. 273	Poisson Abel boundary remainder.
$u^\top R^{\mathrm{mis,RR}}_{n,n_0,\mathrm{fin}}$	Eq. 313	Depth-two RR misadjustment plus startup transfer.
$u^\top M^{\mathrm{bRR}}_{n,n_0}$	Theorem 5 and Corollary 10	Leading martingale Berry–Esseen term.

Let $\mathcal{B}_{\mathrm{mis}}(m, n_0, p, q, \alpha)$ denote the right-hand side of Eq. 313.

Lemma 27. Assume the hypotheses of Lemma 18, Lemma 20, Lemma 22, and Theorem 4. Then, for every $p \geq 2$ and admissible $q \geq 2$,

$$
\begin{aligned}
\| \mathcal{R}^{\mathrm{bRR}}_{n,n_0,\mathrm{fin}}(u) \|_{L_p} \leq \; & \frac{5\sqrt{\kappa_Q} \, \| u \| \, \| \theta_0 - \theta^* \|}{\alpha a \sqrt{m}} (1-\alpha a)^{n_0/2} \\
& + \frac{C_{\mathrm{init,RR}} \, \|u\| \, \|\theta_0 - \theta^*\| \, p}{\alpha a \sqrt{m}} \exp(-c_{\mathrm{init}} \alpha a n_0 / p) \\
& + \frac{C_{\mathrm{burn,D2}} \| u \|}{\sqrt{m}} + \| u \| \mathcal{B}_{\mathrm{mis}}(m, n_0, p, q, \alpha),
\end{aligned}
\tag{327}
$$

where

$$
C_{\mathrm{burn,D2}} := 3 \, t_{\mathrm{mix}} \, \| \epsilon \|_\infty \left(C_{\mathrm{burn,Q}} + \frac{C_{\mathrm{burn,V}}}{a^2} \right). \tag{328}
$$

Proof. Apply the triangle inequality in Eq. 326. The deterministic part is Eq. 247. The random initial-product term is Eq. 255. The Poisson boundary term is bounded by

$$
\| u^\top D^{\mathrm{bRR}}_{2,n,n_0} \|_{L_p} \leq \|u\| \, \|D^{\mathrm{bRR}}_{2,n,n_0}\|_\infty \leq \frac{C_{\mathrm{burn,D2}} \, \|u\|}{\sqrt{m}} \tag{329}
$$

using Eq. 273. The last term is exactly Eq. 313. □

Dividing Eq. 325 by the finite-window normalization gives

$$
\Xi^{\mathrm{bRR}}_{n,n_0}(u) = X^{\mathrm{bRR}}_{n,n_0} + Y^{\mathrm{bRR}}_{n,n_0}, \tag{330}
$$

where

$$
X^{\mathrm{bRR}}_{n,n_0} := -\frac{u^\top M^{\mathrm{bRR}}_{n,n_0}}{\sqrt{m} \, \sigma^{\mathrm{bRR}}_{n,n_0}(u)}, \quad Y^{\mathrm{bRR}}_{n,n_0} := \frac{\mathcal{R}^{\mathrm{bRR}}_{n,n_0,\mathrm{fin}}(u)}{\sigma^{\mathrm{bRR}}_{n,n_0}(u)}. \tag{331}
$$

Theorem 6. (Finite-window burned-in PR-averaged RR Berry–Esseen bound.) Assume the hypotheses of Theorem 5, Lemma 20, and Theorem 4. Let $p = \max(2, \lceil \log n \rceil)$ and $q = \max(2p, \lceil \log(e\,d) \rceil, 2)$. If $2\alpha \le \alpha_*(q, t_{\mathrm{mix}})$ and $2\alpha \le \alpha_{\mathrm{st}}(p)$, then

$$d_K\Big(\Xi^{\mathrm{bRR}}_{n,n_0}(u), \mathcal{N}(0,1)\Big) \le \frac{C_{\mathrm{bK},1}(u)\log^{3/4} n}{m^{1/4}} + \frac{C_{\mathrm{bK},2}(u)\log n}{\sqrt{m}} \\ + \frac{e\,\|\,\mathcal{R}^{\mathrm{bRR}}_{n,n_0,\,\mathrm{fin}}(u)\,\|_{L_p}}{\sqrt{2\pi}\,\sigma^{\mathrm{bRR}}_{n,n_0}(u)} + \frac{e}{n}, \tag{332}$$

with the composite remainder bounded by Lemma 27.

Proof. Apply the smoothing inequality Eq. 209 to the split Eq. 330. The martingale Berry–Esseen term is Theorem 5; the minus sign in X^{bRR}_{n,n_0} is handled by applying that theorem to the signed increments $-u^\top \Delta M^{\mathrm{bRR}}_l$. The bounded-increment constant and predictable quadratic variation are unchanged. Since $p = \max(2, \lceil \log n \rceil)$, the smoothing tail e^{-p} is at most e/n. The L_p norm of the perturbation is

$$\|Y^{\mathrm{bRR}}_{n,n_0}\|_{L_p} = \frac{\|\mathcal{R}^{\mathrm{bRR}}_{n,n_0,\,\mathrm{fin}}(u)\|_{L_p}}{\sigma^{\mathrm{bRR}}_{n,n_0}(u)}, \tag{333}$$

which gives Eq. 332. □

5.12. Balanced Burn-in Berry–Esseen Bound

The finite-window bound uses $\sigma^{\mathrm{bRR}}_{n,n_0}(u)$. The final inference statement uses $\sigma(u)$, via the following burned-in analogue of Corollary 7.

Lemma 28. Assume $\sigma^2(u) > 0$ and the burned-in variance lower-bound condition Eq. 268. Put

$$r_{n,n_0}(u) := \frac{\sigma^{\mathrm{bRR}}_{n,n_0}(u)}{\sigma(u)}. \tag{334}$$

Then

$$\frac{1}{\sqrt{2}} \le r_{n,n_0}(u) \le \sqrt{3/2}, \tag{335}$$

and, for any real random variable W,

$$d_K\Big(r_{n,n_0}(u)W, \mathcal{N}(0,1)\Big) \le d_K(W, \mathcal{N}(0,1)) + \frac{C_{\mathrm{norm}} C_{\mathrm{burn},3}\,\|u\|^2}{m\,\alpha\,a\,\sigma^2(u)}, \tag{336}$$

where C_{norm} is a universal constant.

Proof. The variance lower-bound condition and Eq. 264 give

$$|\sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u) - \sigma^2(u)| \le \sigma^2(u)/2. \tag{337}$$

Hence $r_{n,n_0}(u) \in \big[1/\sqrt{2}, \sqrt{3/2}\big]$. For r in this compact interval,

$$\sup_x |\ \Phi(x/r) - \Phi(x)\ | \le C_{\mathrm{norm}}\,|r-1|, \tag{338}$$

because $\sup_x |x\varphi(x)| < \infty$. Therefore

$$d_K(rW, \mathcal{N}(0,1)) \le d_K(W, \mathcal{N}(0,1)) + C_{\mathrm{norm}}\,|r-1|. \tag{339}$$

Finally,

$$|r_{n,n_0}(u)-1| = \frac{|\sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u)-\sigma^2(u)|}{\sigma(u)\left(\sigma^{\mathrm{bRR}}_{n,n_0}(u)+\sigma(u)\right)} \le \frac{C_{\mathrm{burn},3}\,\|u\|^2}{m\,\alpha\,a\,\sigma^2(u)}, \tag{340}$$

using Eq. 264 and $\sigma^{\mathrm{bRR}}_{n,n_0}(u)+\sigma(u)\ge\sigma(u)$. □

Theorem 7. (Balanced-scale burned-in PR-averaged RR Berry–Esseen bound.) Assume Assumptions 1–3 from Section 1.5. Assume also the Lyapunov contraction Eq. 12 for the two step sizes used below, the imported Levin and Samsonov inputs summarized in Section 4.1, and the non-degeneracy condition $\sigma^2(u)>0$. Fix $c>0$ and set

$$\alpha \coloneqq c\,n^{-1/2},\quad m\coloneqq n-n_0,\quad p\coloneqq \max(2,\lceil\log n\rceil),\quad q\coloneqq\max(2p,\lceil\log(e\,d)\rceil,2). \tag{341}$$

Put

$$\alpha_{\mathrm{adm}}(p,q)\coloneqq \min\Big(\alpha_\infty,\alpha_{\mathrm{inv}},\frac{1}{2a},\alpha_*(q,t_{\mathrm{mix}}),\alpha_{\mathrm{st}}(p)\Big). \tag{342}$$

Here α_{inv} is the local inverse ceiling from Eq. 92, $\alpha_*(q,t_{\mathrm{mix}})$ collects the Levin depth-two moment admissibility restrictions, and $\alpha_{\mathrm{st}}(p)$ is the common product-stability and full-state startup threshold in Lemma 19 and Lemma 25. Suppose $n\ge 3$ is such that

$$m\ge n/2,\quad 2\alpha\le\alpha_{\mathrm{adm}}(p,q),\quad m\,\alpha\,a\ge\frac{2C_{\mathrm{burn},3}\ \|\ u\ \|^2}{\sigma^2(u)}, \tag{343}$$

and the burn-in satisfies the explicit mixing-scale conditions with logarithmic factors

$$n_0\ge\frac{2}{\alpha a}\log n,\quad n_0\ge\frac{p}{c_{\mathrm{init}}\alpha a}\log n,\quad n_0\ge\frac{p}{c_{\mathrm{st}}\alpha a}\log n. \tag{344}$$

Then there exists a finite constant $C_{\mathrm{burn,final}}(u,c,\theta_0)$, depending only on u, c, $\|\theta_0-\theta^*\|$, and the problem and universal constants in the preceding bounds, such that

$$d_K\Big(\Xi^{\mathrm{bRR}}_{n,n_0}(u),\mathcal{N}(0,1)\Big)\le\frac{C_{\mathrm{burn,final}}(u,c,\theta_0)\,\mathrm{polylog}(n)}{n^{1/4}}, \tag{345}$$

and

$$d_K\Big(\Xi^{\mathrm{asy,RR}}_{n,n_0}(u),\mathcal{N}(0,1)\Big)\le\frac{C_{\mathrm{burn,final}}(u,c,\theta_0)\,\mathrm{polylog}(n)}{n^{1/4}}. \tag{346}$$

Proof. Apply the finite-window assembly theorem Theorem 6. Since $m\ge n/2$, polynomial prefactors in n may be written on the m scale up to universal constants; for instance $\sqrt{n}/m\le\sqrt{2}/\sqrt{m}$. In particular, the martingale terms satisfy

$$\frac{\log^{3/4}n}{m^{1/4}}+\frac{\log n}{\sqrt{m}}\le C\,\frac{\mathrm{polylog}(n)}{n^{1/4}}. \tag{347}$$

It remains to bound the composite remainder in Lemma 27. The first condition in Eq. 344 is Eq. 251 with $\beta=1$; by Corollary 8 and $\alpha=cn^{-1/2}$, $m\ge n/2$, the deterministic transient is $O(n^{-1})$. The second condition is Eq. 260 with $\beta=1$, so Corollary 9 makes the random initial-product term $O(\mathrm{polylog}(n)n^{-1})$. The Poisson Abel remainder is $O(m^{-1/2})$. The third condition in Eq. 344 is the startup condition Eq. 308 with $\beta=1$, so Corollary 12 gives

$$\|\ R^{\mathrm{mis,RR}}_{n,n_0,\,\mathrm{fin}}\ \|_{L_p}\le C\,\mathrm{polylog}(n)\,n^{-1/4}. \tag{348}$$

Together with $\sigma^{\mathrm{bRR}}_{n,n_0}(u) \geq \sigma(u)/\sqrt{2}$, this makes the smoothing remainder in Eq. 332 of order $\mathrm{polylog}(n) n^{-1/4}$. The smoothing tail e/n is lower order, so Eq. 345 follows.

For the asymptotic normalization, write

$$\Xi^{\mathrm{asy,RR}}_{n,n_0}(u) = r_{n,n_0}(u)\, \Xi^{\mathrm{bRR}}_{n,n_0}(u). \tag{349}$$

By Lemma 28, the additional cost is at most

$$\frac{C\,\|u\|^2}{m\,\alpha\,a\,\sigma^2(u)} = O\big(n^{-1/2}\big), \tag{350}$$

because $m \geq n/2$ and $\alpha = cn^{-1/2}$. This is absorbed into the balanced finite-window rate, proving Eq. 346. □

Condition Eq. 343 collects the finite-n admissibility requirements. The inequality $2\alpha \leq \alpha_{\mathrm{adm}}(p, q)$ enforces the Lyapunov small-step ceiling, the local inverse ceiling, the Levin depth-two admissibility threshold, the random-product stability estimate Lemma 19, and the full-state startup contraction Lemma 25. Since $\alpha = cn^{-1/2}$ and $m \geq n/2$, the elementary step-size constraints and Eq. 268 hold automatically for all sufficiently large n. The remaining non-elementary large-n requirement is

$$2cn^{-1/2} \leq \min(\alpha_*(q, t_{\mathrm{mix}}), \alpha_{\mathrm{st}}(p)), \tag{351}$$

with p, q as in the theorem. Under this Levin/startup admissibility condition, the large-n reading of the theorem keeps only $m \geq n/2$ and the burn-in lower bounds in Eq. 344.

Corollary 13. ($\sqrt{n}$-normalization for the burned-in RR statistic.) Under the assumptions of Theorem 7, define the final scalar statistic

$$T^{\mathrm{RR,n}}_{n,n_0}(u) := \sqrt{n}\, u^\top \Big(\bar{\theta}^{(\mathrm{RR},\alpha)}_{n,n_0} - \theta^*\Big) = \sqrt{n/m}\, T^{\mathrm{RR}}_{n,n_0}(u), \tag{352}$$

and its asymptotic normalization

$$\Xi^{\mathrm{n,RR}}_{n,n_0}(u) := \frac{T^{\mathrm{RR,n}}_{n,n_0}(u)}{\sigma(u)}. \tag{353}$$

For the Berry–Esseen moment choice $p = \max(2, \lceil \log n \rceil)$ in Theorem 7, the lower burn-in conditions Eq. 344 are implied by

$$n_0 \geq C_-\,(\alpha a)^{-1} \log^2 n \tag{354}$$

with any fixed C_- large enough. At the balanced scale $\alpha = cn^{-1/2}$ this is $n_0 = O\big(n^{1/2} \log^2 n\big)$, not a purely logarithmic burn-in in n. If, in addition, the burn-in window stays in the same mixing-scale window with logarithmic-square factor,

$$C_-\,(\alpha a)^{-1} \log^2 n \leq n_0 \leq C_0\,(\alpha a)^{-1} \log^2 n \tag{355}$$

for some finite C_0 and such a fixed C_-, then there exists a finite constant $C_{\mathrm{burn,n}}(u, c, \theta_0, C_0, C_-)$ such that

$$d_K\Big(\Xi^{\mathrm{n,RR}}_{n,n_0}(u), \mathcal{N}(0,1)\Big) \leq \frac{C_{\mathrm{burn,n}}(u, c, \theta_0, C_0, C_-)\,\mathrm{polylog}(n)}{n^{1/4}}. \tag{356}$$

Proof. Put $s_{n,n_0} := \sqrt{n/m}$. Since Eq. 343 gives $m \geq n/2$,

$$0 \leq s_{n,n_0} - 1 = \frac{n_0}{m\,\big(s_{n,n_0} + 1\big)} \leq \frac{2n_0}{n}. \tag{357}$$

For every real random variable W and every $s \in \big[1, \sqrt{2}\big]$,

$$d_K(sW, \mathcal{N}(0,1)) \le d_K(W, \mathcal{N}(0,1)) + C\,|s-1|, \tag{358}$$

by the same scaling argument as in Lemma 28. Applying this with $W = \Xi^{\mathrm{asy,RR}}_{n,n_0}(u)$ and using Theorem 7 gives the already proved $\operatorname{polylog}(n)n^{-1/4}$ term. The upper burn-in bound in Eq. 355 and $\alpha = cn^{-1/2}$ give

$$|s_{n,n_0} - 1| \le \frac{2C_0}{ca}\frac{\log^2 n}{n^{1/2}}, \tag{359}$$

which is lower order and is absorbed into the same balanced-scale rate. $\square$

The lower side of Eq. 355 is the logarithmic-square mixing-scale burn-in needed by the current L_p startup contraction; the upper side keeps the $\sqrt{n/m}$ rescaling lower order. At $\alpha = cn^{-1/2}$ this window has order $n^{1/2}\log^2 n$.

6. References

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